Notes on Probability Theory

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The notes are based on the lecture taught by Prof. David Anderson and Prof. Benedek Valkó at University of Wisconsin-Madison in 2025-2026. The course assumes a certain amount of knowledges in real analysis. For some classic results in real analysis, one can refer to my notes on real analysis.

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1. Probability Space

1.1. Probability Space

Definition 1.1
Let Ω be a set. A collection of subsets $\mathcal{F}$ forms a ***σ-algebra*** *if*

(a) $\varnothing \in \mathcal{F}$.

(b) $A \in \mathcal{F}$ *implies* $A^c \in \mathcal{F}$.

(c) If $A_i \in \mathcal{F}$ *are countably many sets,* $\cup_i A_i \in \mathcal{F}$.

The dual $(\Omega, \mathcal{F})$ *is called a* ***measurable space*** *and the sets falling in* $\mathcal{F}$ *are said to be* ***measurable****.*

Definition 1.2
Let $(\Omega, \mathcal{F})$ *be a measurable space. A set function* $\mu : \mathcal{F} \to [0, \infty]$ *is a* ***measure*** *if*

(a) $\mu(\varnothing) = 0$.

(b) For countably many disjoint $A_i \in \mathcal{F}$, $\mu(\cup_i A_i) = \sum_i \mu(A_i)$.

The triple $(\Omega, \mathcal{F}, \mu)$ *is called a* ***measure space****.*

Definition 1.3
A ***probability space*** *is a measure space* $(\Omega, \mathcal{F}, \mathrm{P})$ *such that* $\mathrm{P}(\Omega) = 1$.

Lemma 1.4
Let $\mathcal{S}$ be a collection of sets. Then there exists the smallest σ-algebra containing $\mathcal{S}$.

Proof. Let $\mathcal{F}$ be the intersection of all σ-algebra containing $\mathcal{S}$. $\mathcal{F}$ is non-empty since the power set is a σ-algebra containing $\mathcal{S}$. Now it is clear that $\varnothing \in \mathcal{F}$ since $\varnothing \in \mathcal{A}$ for every σ-algebra $\mathcal{A}$ contiaining $\mathcal{S}$. If $A \in \mathcal{F}$, $A \in \mathcal{A}$ for all $\mathcal{A}$ containing $\mathcal{S}$ and $A^c \in \mathcal{A}$ for all $\mathcal{A}$. Thus $A^c \in \mathcal{F}$. Finally, if $A_i \in \mathcal{F}$ are countably many sets, then each A_i lies in every $\mathcal{A}$ containing $\mathcal{S}$; so does $\cup_i A_i$ and thus $\cup_i A_i \in \mathcal{F}$. The minimality follows by the construction of $\mathcal{F}$. ∎

Definition 1.5
For any collection of sets $\mathcal{S}$, the smallest σ-algebra is denoted as $\sigma(\mathcal{S})$.

Theorem 1.6
Let $(\Omega, \mathcal{F}, \mathrm{P})$ *be a probability space. Then*

(a) If $A, B \in \mathcal{F}$ *and* $A \subset B$, *then* $\mathrm{P}(A) \leq \mathrm{P}(B)$.

(b) For countably many $A_i \in \mathcal{F}$, $\mathrm{P}(\cup_i A_i) \leq \sum_i \mathrm{P}(A_i)$.

(c) If $A_i \nearrow A$, $\mathrm{P}(A_i) \to \mathrm{P}(A)$.

(d) If $A_i \searrow A$, $\mathrm{P}(A_i) \to \mathrm{P}(A)$.

Proof. (a) and (b) are clear. For (c), write $E_i = A_i - A_{i-1}$ and $A_0 = \varnothing$. Then since E_i are disjoint and $A_n = \cup_{i=1}^n E_i$,

$$\mathrm{P}(A_n) = \mathrm{P}\big(\cup_{i=1}^n E_i\big) = \sum_{i=1}^n \mathrm{P}(E_i) \to \sum_i \mathrm{P}(E_i) = \mathrm{P}(\cup_i E_i) = \mathrm{P}(A)$$

as $n \to \infty$.

For (d), note that $A_i^c \nearrow A^c$. Thus $1 - \mathrm{P}(A_i) = \mathrm{P}(A_i^c) \to \mathrm{P}(A^c) = 1 - \mathrm{P}(A)$. Thus $\mathrm{P}(A_i) \to \mathrm{P}(A)$. ∎

Definition 1.7

*The **Borel σ-algebra** is the σ-algebra generated by all open sets.*

Definition 1.8

Let P *be a probability measure on* $(\mathbb{R}, \mathcal{B})$*. The **distribution function** F is defined as*

$$F(x) = \mathrm{P}((-\infty, x])$$

for $x \in \mathbb{R}$*.*

Proposition 1.9

The distribution function in $(\mathbb{R}, \mathcal{B})$ *satisfies that*

(a) $F(x) \le F(y)$ *for all* $x \le y$*.*

(b) $F(x) \to F(y)$ *as* $x \to y^+$*.*

(c) $F(-\infty) = 0$ *and* $F(\infty) = 1$*.*

Proof. For (a), note that $(-\infty, x] \subset (-\infty, y]$ and

$$F(x) = \mathrm{P}((-\infty, x]) \le \mathrm{P}((-\infty, y]) = F(y).$$

For (b), notice that for $x_n \to y^+$, $(-\infty, x_n] \searrow (-\infty, y]$. Hence

$$F(x_n) = \mathrm{P}((-\infty, x_n]) \to \mathrm{P}((-\infty, y]) = F(y).$$

Similarly, taking $x_n \to \pm\infty$ gives (c). ∎

Definition 1.10

*A collection $\mathcal{S}$ of sets is called an **algebra** if*

(a) $\varnothing \in \mathcal{S}$*.*

(b) If $A \in \mathcal{S}$*, then* $A^c \in \mathcal{S}$*.*

(c) If $A, B \in \mathcal{S}$*, then* $A \cup B \in \mathcal{S}$*.*

Remark

An algebra is closed under finite unions. It is also clear that a σ-algebra is an algebra, while the converse is not true. An example is the collection of all finite unions of intervals in $\mathbb{R}$*.*

Definition 1.11
A collection $\mathcal{S}$ of sets is called a ***semi-algebra*** *if*

(a) If $A, B \in \mathcal{S}$, then $A \cap B \in \mathcal{S}$.

(b) If $A \in \mathcal{S}$, then A^c can be written as a finite disjoint union of sets in $\mathcal{S}$.

Remark
A semi-algebra must contain $\varnothing$ since for any $A \in \mathcal{S}$, $A^c = \cup_i A_i$, where $A_i \in \mathcal{S}$ are disjoint. Then $A \cap A_1 = \varnothing \in \mathcal{S}$.

Remark
An example of being a semi-algebra but not an algebra is the collection of all intervals of the form $(a_i, b_i]$ for $-\infty \leq a_i < b_i \leq \infty$ with the empty set.

Lemma 1.12
If $\mathcal{S}$ is a semi-algebra, then $\overline{S} = \{$finite disjoint unions of sets in $\mathcal{S}\}$ forms an algebra.

Proof. It has been shown that $\varnothing \in \mathcal{S}$. For $A, B \in \overline{\mathcal{S}}$, write $A = \cup_{i=1}^n A_i$ and $B = \cup_{j=1}^m B_j$ for disjoint $A_i, B_j \in \mathcal{S}$, respectively. Then $A \cap B = \cup_{i,j}(A_i \cap B_j) \in \overline{\mathcal{S}}$. Thus $\overline{\mathcal{S}}$ is closed under intersection. Now if $A \in \overline{\mathcal{S}}$, $A = \cup_{i=1}^n A_i$ for disjoint $A_i \in \mathcal{S}$. Then $A^c = \cap_{i=1}^n A_i^c$. By the definition of semi-algebra, A_i^c can be written as a finite disjoint union of sets in $\mathcal{S}$ and thus $A_i^c \in \overline{\mathcal{S}}$. Since $\overline{\mathcal{S}}$ is closed under finite intersection, $A^c = \cap_{i=1}^n A_i^c \in \overline{\mathcal{S}}$. Finally, for $A, B \in \overline{\mathcal{S}}$, $A \cup B = (A^c \cap B^c)^c \in \overline{\mathcal{S}}$. We conclude that $\overline{\mathcal{S}}$ is indeed an algebra. ∎

Definition 1.13
Suppose $\mathcal{S}$ is a semi-algebra. $\overline{\mathcal{S}} = \{$finite disjoint unions of sets in $\mathcal{S}\}$ is called the ***algebra generated by*** *$\mathcal{S}$.*

Definition 1.14
Let $\mathcal{S}$ be an algebra. A set function $\mu_0 : \mathcal{S} \to [0, \infty]$ is called a ***premeasure*** *if*

(a) $\mu_0(\varnothing) = 0$.

(b) For countable disjoint $A_i \in \mathcal{S}$ such that $\cup_i A_i \in \mathcal{S}$,

$$\mu_0(\cup_i A_i) = \sum_i \mu_0(A_i).$$

Theorem 1.15
Let ν be a set function on a semi-algebra $\mathcal{S}$ such that $\nu(\varnothing) = 0$. Suppose that

(a) if $A \in \mathcal{S}$ and $A = \cup_{i=1}^n A_i$ for disjoint $A_i \in \mathcal{S}$, then $\nu(A) = \sum_{i=1}^n \nu(A_i)$;

(b) if $A_i \in \mathcal{S}$ are countably many sets and $A = \cup_i A_i \in \mathcal{S}$, then $\nu(A) \leq \sum_i \nu(A_i)$.

Then ν can be extended to a unique premeasure μ_0 on the algebra generated by $\mathcal{S}$.

Proof. We first show the existence. From lemma 1.12 we know that $\mathcal{S}$ generates an algebra $\mathcal{A} = \{$finite disjoint union of sets in $\mathcal{S}\}$. Define our candidate μ_0 by $\mu_0(A) = \sum_i \nu(A_i)$ for

$A = \cup_i A_i$ where $A_i \in \mathcal{S}$ are disjoint. To see that μ_0 is well-defined, suppose $A = \cup_i B_i$ for disjoint $B_i \in \mathcal{S}$. Observe that

$$A_i = \cup_j (A_i \cap B_j) \quad \text{and} \quad B_j = \cup_i (A_i \cap B_j)$$

are finite disjoint unions. Then

$$\sum_i \nu(A_i) = \sum_i \sum_j \nu(A_i \cap B_j) = \sum_j \sum_i \nu(A_i \cap B_j) = \sum_j \nu(B_j)$$

by (a). Thus μ_0 is well-defined.

Now we check that μ_0 is a premeasure. Clearly $\mu_0(\varnothing) = 0$. For finitely many disjoint $A_i \in \mathcal{A}$ such that $\cup_i A_i \in \mathcal{A}$, we can write $A_i = \cup_j B_{ij}$ for disjoint $B_{ij} \in \mathcal{S}$. Then (a) implies that

$$\mu_0(\cup_i A_i) = \mu_0(\cup_{i,j} B_{ij}) = \sum_{i,j} \nu(B_{ij}) = \sum_i \sum_j \mu_0(B_{ij}) = \sum_i \mu_0(A_i).$$

Next, for countably many disjoint $A_i \in \mathcal{A}$ such that $A = \cup_i A_i \in \mathcal{A}$, write $A_i = \cup_j B_{ij}$, where $B_{ij} \in \mathcal{S}$ are finite disjoint for each i. Then $\mu_0(A_i) = \sum_j \nu(B_{ij})$ and

$$\sum_i \mu_0(A_i) = \sum_i \sum_j \nu(B_{ij}).$$

Without loss of generality, we may choose A_i to be those in $\mathcal{S}$ since otherwise we can replace A_i by B_{ij}. We assume that $A_i \in \mathcal{S}$ from now on. Since $A \in \mathcal{A}$, $A = \cup_i C_i$ for finite disjoint $C_i \in \mathcal{S}$. $C_i = \cup_j (C_i \cap A_j)$. Thus (b) gives that

$$\nu(C_i) \leq \sum_j \nu(C_i \cap A_j).$$

Then

$$\mu_0(A) = \sum_i \nu(C_i) \leq \sum_i \sum_j \nu(C_i \cap A_j) = \sum_j \sum_i \nu(C_i \cap A_j) = \sum_j \nu(A_j) = \sum_j \mu_0(A_j).$$

For the opposite inequality, set $B_n = \cup_{i=1}^n A_i$ and $C_n = A - B_n$. Since $\mathcal{A}$ is an algebra, $C_n \in \mathcal{A}$ and the finite additivity shows that

$$\mu_0(A) = \sum_{i=1}^n \mu_0(A_i) + \mu_0(C_n) \geq \sum_{i=1}^n \mu_0(A_i).$$

Taking $n \to \infty$ gives the desired inequality and thus μ_0 is σ-additive on $\mathcal{A}$.

Finally, if μ_1 is another premeasure on $\mathcal{A}$ extending ν, then for $A = \cup_i A_i$ for disjoint $A_i \in \mathcal{S}$,

$$\mu_1(A) = \sum_i \nu(A_i) = \mu_0(A).$$

■

Definition 1.16
A collection of sets $\mathcal{P}$ is called a ***π-system*** *if $A, B \in \mathcal{P}$ implies $A \cap B \in \mathcal{P}$.*

Definition 1.17
A collection of sets $\mathcal{L}$ is called a ***λ-system*** *if*

(a) $\Omega \in \mathcal{L}$.

(b) If $A, B \in \mathcal{L}$ and $A \subset B$, then $B - A \in \mathcal{L}$.

(c) If $A_i \in \mathcal{L}$ and $A_i \nearrow A$, then $A \in \mathcal{L}$.

Theorem 1.18 (Sierpiński-Dynkin π-λ)
If $\mathcal{P}$ is a π-system and $\mathcal{L}$ is a λ-system containing $\mathcal{P}$, then $\sigma(\mathcal{P}) \subset \mathcal{L}$.

Proof. First we show that a collection $\mathcal{S}$ is a σ-algebra if and only if it is both a π-system and a λ-system. Suppose first that $\mathcal{S}$ is a π-system and a λ-system. $\varnothing = \Omega - \Omega \in \mathcal{S}$. If $A \in \mathcal{S}$, then $A^c = \Omega - A \in \mathcal{S}$. For $A, B \in \mathcal{S}$, $A \cup B = (A^c \cap B^c)^c \in \mathcal{S}$ since we have shown that $\mathcal{S}$ is closed under complement and intersection by being a π-system. Thus $\mathcal{S}$ is also closed under finite unions. If $A_i \in \mathcal{S}$ are countably many sets, let $B_n = \cup_{i=1}^{n} A_i \in \mathcal{S}$. Then $B_n \nearrow \cup_i A_i$ and thus $\cup_i A_i \in \mathcal{S}$.

Conversely, if $\mathcal{S}$ is a σ-algebra, then for $A, B \in \mathcal{S}$, $A \cap B = (A^c \cup B^c)^c \in \mathcal{S}$. Thus $\mathcal{S}$ is a π-system. If $A, B \in \mathcal{S}$ and $A \subset B$, then $B - A = B \cap A^c \in \mathcal{S}$. Finally, if $A_i \in \mathcal{S}$ and $A_i \nearrow A$, then $A = \cup_i (A_i - A_{i-1}) \in \mathcal{S}$ with $A_0 = \varnothing$. Thus $\mathcal{S}$ is a λ-system.

Now set $\mathcal{L}$ to be the smallest λ-system containing $\mathcal{P}$. It suffices to show that $\mathcal{L}$ is also a π-system and thus by the above conclusion, $\mathcal{L}$ is a σ-algebra containing $\mathcal{P}$; hence $\sigma(\mathcal{P}) \subset \mathcal{L}$.

To show that $\mathcal{L}$ is a π-system, let $A, B \in \mathcal{L}$. If $A, B \in \mathcal{P}$, $A \cap B \in \mathcal{P} \subset \mathcal{L}$. To extend the result for general $A, B \in \mathcal{L}$, we first fix $B \in \mathcal{P}$ and define

$$\mathcal{L}_B = \{A \mid A \cap B \in \mathcal{L}\}.$$

We claim that $\mathcal{L}_B$ is a λ-system containing $\mathcal{P}$. For $A \in \mathcal{P}$, $A \cap B \in \mathcal{L}$. Thus $\mathcal{P} \subset \mathcal{L}_B$. Clearly $\Omega \in \mathcal{L}_B$. If $E, F \in \mathcal{L}_B$ and $E \subset F$, then

$$(F - E) \cap B = (F \cap B) - (E \cap B) \in \mathcal{L}.$$

Thus $F - E \in \mathcal{L}_B$. Finally, if $E_i \in \mathcal{L}_B$ and $E_i \nearrow E$, then

$$E \cap B = \cup_i (E_i \cap B) \in \mathcal{L}.$$

Hence $E \in \mathcal{L}_B$ and we conclude that $\mathcal{L}_B$ is a λ-system. Since it is a λ-system containing $\mathcal{P}$, it also contains the smallest λ-system $\mathcal{L}$ with the intersection property. Thus $A \cap B \in \mathcal{L}$ whenever $A \in \mathcal{L}$ and $B \in \mathcal{P}$.

Next, fix $A \in \mathcal{L}$ and define $\mathcal{L}_A = \{B \mid A \cap B \in \mathcal{L}\}$. Clearly $\mathcal{L}_A$ contains $\mathcal{L}$ and $\Omega \in \mathcal{L}_A$. If $E, F \in \mathcal{L}_A$ and $E \subset F$, then

$$(F - E) \cap A = (F \cap A) - (E \cap A) \in \mathcal{L}.$$

Thus $F - E \in \mathcal{L}_A$. Finally, if $E_i \in \mathcal{L}_A$ and $E_i \nearrow E$, then

$$E \cap A = \cup_i (E_i \cap A) \in \mathcal{L}.$$

Hence $E \in \mathcal{L}_A$ and we conclude that $\mathcal{L}_A$ is a λ-system. Since it contains $\mathcal{L}$, $A, B \in \mathcal{L}$ implies $A \cap B \in \mathcal{L}$; in other words, $\mathcal{L}$ is a π-system and the proof is complete. ∎

Corollary 1.19
Let μ and ν be two probability measures agreeing on a π-system $\mathcal{P}$, i.e., $\mu(A) = \nu(A)$ for all $A \in \mathcal{P}$. Then $\mu(A) = \nu(A)$ for all $A \in \sigma(\mathcal{P})$.

Proof. Put

$$\mathcal{L} = \{A \mid \mu(A) = \nu(A)\}.$$

We claim that $\mathcal{L}$ is a λ-system containing $\mathcal{P}$. It is clear that by our assumption, $\mathcal{P} \subset \mathcal{L}$ and $\Omega \in \mathcal{L}$. If $A, B \in \mathcal{L}$ and $A \subset B$, then

$$\mu(B - A) = \mu(B) - \mu(A) = \nu(B) - \nu(A) = \nu(B - A).$$

Thus $B - A \in \mathcal{L}$. Finally, if $A_i \in \mathcal{L}$ and $A_i \nearrow A$, then

$$\mu(A) = \lim_{i \to \infty} \mu(A_i) = \lim_{i \to \infty} \nu(A_i) = \nu(A).$$

Hence $A \in \mathcal{L}$ and we conclude that $\mathcal{L}$ is a λ-system. By the Sierpiński-Dynkin π-λ theorem, $\sigma(\mathcal{P}) \subset \mathcal{L}$; in other words, μ and ν agree on $\sigma(\mathcal{P})$. ∎

Definition 1.20
*A measure μ on a measurable space $(\Omega, \mathcal{F})$ is called σ-**finite** if there exists countable $A_i \in \mathcal{F}$ such that $\cup_i A_i = \Omega$ and $\mu(A_i) < \infty$.*

Definition 1.21
*A set function $\mu^* : 2^\Omega \to [0, \infty]$ is called an **outer measure** if*

(a) $\mu^*(\varnothing) = 0$.

(b) If $A \subset B$, then $\mu^*(A) \leq \mu^*(B)$.

(c) For countably many $A_i \subset \Omega$, $\mu^*(\cup_i A_i) \leq \sum_i \mu^*(A_i)$.

Definition 1.22
*Let μ^* be an outer measure. A set $A \subset \Omega$ is said to be **Carathéodory measurable** or μ^*-*

measurable *if for all* $E \subset \Omega$*,*

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c).$$

Lemma 1.23
Let μ^* *be an outer measure on* Ω*. Then the collection of all* μ^**-measurable sets forms a* σ*-algebra* $\mathcal{F}$ *and* $\mu^*|_{\mathcal{F}}$ *is a measure.*

Proof. Put

$$\mathcal{F} = \{A \subset \Omega \mid \mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \text{ for all } E \subset \Omega\}.$$

We first show that $\mathcal{F}$ is a σ-algebra. Clearly $\varnothing \in \mathcal{F}$ and if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$. For $A, B \in \mathcal{F}$, let $C = A \cup B$. The property of outer measure gives that $\mu^*(E) \leq \mu^*(E \cap C) + \mu^*(E \cap C^c)$. To see the opposite inequality, note that $C = A \cup (B \cap A^c)$ and

$$\begin{aligned}\mu^*(E \cap C) + \mu^*(E \cap C^c) &\leq \mu^*(E \cap A) + \mu^*(E \cap B \cap A^c) + \mu^*(E \cap A^c \cap B^c) \\ &= \mu^*(E \cap A) + \mu^*(E \cap A^c) = \mu^*(E).\end{aligned}$$

Hence $C \in \mathcal{F}$ and $\mathcal{F}$ is closed under finite unions. For countable disjoint $A_i \in \mathcal{F}$ with $A = \cup_i A_i$, let $B_n = \cup_{i=1}^n A_i \in \mathcal{F}$. Then

$$\mu^*(E \cap A) = \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c) \geq \mu^*(E \cap B_n) = \sum_{i=1}^n \mu^*(E \cap A_i).$$

Taking $n \to \infty$ gives that

$$\mu^*(E \cap A) \geq \sum_i \mu^*(E \cap A_i) \geq \mu^*(E \cap A)$$

by the σ-subadditivity of outer measure. Hence $\mu^*(E \cap A) = \sum_i \mu^*(E \cap A_i)$. Note also that $E \cap A^c \subset E \cap B_n^c$ so $\mu^*(E \cap A^c) \leq \mu^*(E \cap B_n^c)$. Thus

$$\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c) \geq \sum_{i=1}^n \mu^*(E \cap A_i) + \mu^*(E \cap A^c) \to \mu^*(E \cap A) + \mu^*(E \cap A^c) \geq \mu^*(E)$$

by the σ-subadditivity of outer measure. We conclude that $\mathcal{F}$ is a σ-algebra.

Finally, denote $\mu^*|_{\mathcal{F}}$ by μ. Clearly $\mu(\varnothing) = 0$. For countably many disjoint $A_i \in \mathcal{F}$ such that $A = \cup_i A_i \in \mathcal{F}$, let $B_n = \cup_{i=1}^n A_i \in \mathcal{F}$. Then

$$\mu(A) = \mu(B_n) + \mu(A \cap B_n^c) \geq \mu(B_n) = \sum_{i=1}^n \mu(A_i) \to \sum_i \mu(A_i) \geq \mu(A).$$

Hence $\mu(A) = \sum_i \mu(A_i)$ and μ is a measure on $\mathcal{F}$. ∎

Theorem 1.24 (Carathéodory Extension)
Let ν be a finitely additive, σ-subadditive set function on a semi-algebra $\mathcal{S}$ such that $\nu(\varnothing) = 0$. Then ν can be extended to a measure on $\sigma(\mathcal{S})$.

Proof. By theorem 1.15, ν can be extended to a premeasure μ_0 on the algebra $\mathcal{A}$ generated by $\mathcal{S}$. Define the outer measure by

$$\mu^*(A) = \inf\left\{\sum_i \mu_0(E_i) \;\middle|\; A \subset \cup_i E_i, E_i \in \mathcal{A}\right\}$$

for all $A \subset \Omega$ with the convention that $\inf \varnothing = \infty$. We check that μ^* is indeed an outer measure. Clearly $\mu^*(\varnothing) = 0$. If $A \subset B$, then any cover of B by sets in $\mathcal{A}$ is also a cover of A and hence $\mu^*(A) \leq \mu^*(B)$. For countably many $A_i \subset \Omega$, we can find $\{E_{ij}\}_j$ covering A_i such that

$$\sum_j \mu_0(E_{ij}) \leq \mu^*(A_i) + 2^{-i}\epsilon$$

for some $\epsilon > 0$. Then $\cup_{i,j} E_{ij}$ covers $\cup_i A_i$ and

$$\mu^*(\cup_i A_i) \leq \sum_i \sum_j \mu_0(E_{ij}) \leq \sum_i \mu^*(A_i) + \epsilon.$$

Since ϵ is arbitrary, we conclude that $\mu^*(\cup_i A_i) \leq \sum_i \mu^*(A_i)$ and μ^* is indeed an outer measure.

It follows from lemma 1.23 that the collection of all μ^*-measurable sets forms a σ-algebra $\mathcal{F}$ and μ^* restricted on $\mathcal{F}$ is a measure. It is clear that $\mathcal{A} \subset \mathcal{F}$ and $\sigma(\mathcal{S}) \subset \sigma(\mathcal{A}) \subset \mathcal{F}$ and $\mu = \mu^*|_{\sigma(\mathcal{S})}$ is also a measure. Finally, for $A, A_i \in \mathcal{S}$ where A_i covers A,

$$\mu(A) = \mu^*(A) \leq \nu(A) \leq \sum_i \nu(A \cap A_i) \leq \sum_i \nu(A_i).$$

Taking the infimum over all such covers, we get $\nu(A) = \mu^*(A)$ and μ is indeed an extension of ν. ∎

Remark
If the measures are probability measures, then we have that the extension is unique by corollary 1.19.

Theorem 1.25
If F is non-decreasing, right-continuous and satisfies that $F(-\infty) = 0$, $F(\infty) = 1$, then there is a unique probability measure such that

$$\mathrm{P}((-\infty, x]) = F(x).$$

Proof. Define

$$\mathcal{S} = \{(a, b] \mid -\infty \leq a < b \leq \infty\} \cup \{\varnothing\}.$$

It is clear that $\mathcal{S}$ is a semi-algebra. Define the set function $\mathrm{P} : \mathcal{S} \to [0,1]$ by

$$\mathrm{P}((a,b]) = F(b) - F(a)$$

and $\mathrm{P}(\varnothing) = 0$. For disjoint, at most countable $(a_i, b_i] \in \mathcal{S}$, we define

$$\mathrm{P}(\cup_i (a_i, b_i]) = \sum_i \mathrm{P}((a_i, b_i]) = \sum_i F(b_i) - F(a_i).$$

It is clear that P is finitely additive. If $(a,b] = \cup_i (a_i, b_i]$ for disjoint $(a_i, b_i] \in \mathcal{S}$, we may assume without loss of generality that $a = a_1 < b_1 < b_2 < \cdots < b_n = b$ and

$$\mathrm{P}((a,b]) = F(b) - F(a) = \sum_i F(b_i) - F(a_i) = \sum_i \mathrm{P}((a_i, b_i]).$$

Hence P is σ-additive. It now follows from the Carathéodory extension theorem that P can be extended uniquely to a probability measure on $\sigma(\mathcal{S}) = \mathcal{B}$. ■

Remark

This theorem shows that the distribution function completely characterizes the probability measure. In other words, the term "distribution function" can refer to either the CDF or the probability measure.

1.2. Random Variable

Definition 1.26

*Let Ω be a probability space. A **random variable** X is a measurable function $X : \Omega \to (S, \mathcal{S})$, where $(S, \mathcal{S})$ is a measurable space.*

Remark

The codomain is often taken to be $(\mathbb{R}, \mathcal{B})$ or $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, but it is also possible to define random functions, i.e., $(S, \mathcal{S})$ is a function space.

Definition 1.27

*Let $X : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ be a random variable. The **distribution** of X is the pushforward measure of P under X, i.e.,*

$$\mu_X(A) = \mathrm{P}(X \in A) = \mathrm{P}(X^{-1}(A)), \quad A \in \mathcal{S}.$$

Definition 1.28

*Let $X : (\Omega, \mathcal{F}) \to (\mathbb{R}^d, \mathcal{B})$ be a random variable. The **cumulative distribution function** of X is defined as*

$$F_X(x) = \mathrm{P}(X \leq x) = \mathrm{P}(X_1 \leq x_1, \ldots, X_d \leq x_d), \quad x = (x_1, \ldots, x_d) \in \mathbb{R}^d.$$

Proposition 1.29

Let $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B})$ be a random variable and F be its cumulative distribution function. Then,

(a) F is non-decreasing, i.e., $x \leq y$ implies $F(x) \leq F(y)$;

(b) $F(-\infty) = 0$ and $F(\infty) = 1$;

(c) F is right-continuous, i.e., $\lim_{y \to x^+} F(y) = F(x)$;

(d) $F(x^-) = \mathrm{P}(X < x)$;

(e) $\mathrm{P}(X = x) = F(x) - F(x^-)$.

Proof. (a) comes from that $\{X \leq x\} \subset \{X \leq y\}$ for $x \leq y$.

Take $a_n \to \infty$. Then $\{X \leq a_n\} \nearrow \Omega$ and $\{X \leq -a_n\} \searrow \emptyset$. By theorem 1.6, we have that

$$F(a_n) = \mathrm{P}(X \leq a_n) \to \mathrm{P}(\Omega) = 1, \quad F(-a_n) = \mathrm{P}(X \leq -a_n) \to \mathrm{P}(\emptyset) = 0.$$

(c) is similar to (b). Take $y_n \to x^+$, then $\{X \leq y_n\} \searrow \{X \leq x\}$. By theorem 1.6, we have that

$$F(y_n) = \mathrm{P}(X \leq y_n) \to \mathrm{P}(X \leq x) = F(x).$$

For (d), take $x_n \to x^-$, then $\{X \leq x_n\} \nearrow \{X < x\}$. By theorem 1.6, we have that

$$F(x_n) = \mathrm{P}(X \leq x_n) \to \mathrm{P}(X < x).$$

For (e), $\mathrm{P}(X = x) = \mathrm{P}(X \leq x) - \mathrm{P}(X < x) = F(x) - F(x^-)$. ■

Theorem 1.30

Let F be a non-decreasing, right-continuous function satisfying that $F(-\infty) = 0$ and $F(\infty) = 1$. Then there is a random variable X such that

$$F(x) = \mu_X((-\infty, x]).$$

Proof. Put $\Omega = (0,1)$, $\mathcal{F} = \mathcal{B}$, P be the Lebesgue measure and $X(\omega) = \sup\{x \mid F(x) < \omega\}$. Notice that

$$\begin{aligned}\{X \leq x\} &= \{\omega \in \Omega \mid \sup\{y \mid F(y) < \omega\} \leq x\} \\ &= \{\omega \in \Omega \mid \text{for all } y > x, F(y) \geq \omega\} \\ &= \{\omega \in \Omega \mid F(x) \geq \omega\}.\end{aligned}$$

Hence $\mathrm{P}(X \leq x) = \mathrm{P}(\{\omega \in \Omega \mid \omega \leq F(x)\}) = F(x)$. ■

Definition 1.31

*If X and Y are random variables mapping to some measurable space $(S, \mathcal{S})$, then X and Y are said to be **equal in distribution** if $\mu_X = \mu_Y$, denoted by $X \overset{d}{=} Y$.*

Definition 1.32

*Let $X : \Omega \to \mathbb{R}$ be a random variable with distribution F. $f : \mathbb{R} \to \mathbb{R}$ is said to be the **density** of X if*

$$F(x) = \int_{-\infty}^{x} f(y)dy$$

for all $x \in \mathbb{R}$.

Remark

If f and g are both densities of X, then $f = g$ a.e.

Remark

If $\mu_X \ll \lambda$, where λ is the Lebesgue measure, then by Radon-Nikodym theorem, there is a density f such that

$$\mu_X(A) = \int_A f(x)d\lambda(x)$$

for all $A \in \mathcal{B}$. Or equivalently, F is absolutely continuous.

Example

Not all random variables have densities, even when its CDF is continuous. Consider the

Cantor function

$$F(x) = \begin{cases} \sum_n \frac{a_n}{2^n}, & x = \sum_n \frac{2a_n}{3^n} \in C \text{ for some } \{a_n\} \in \{0,1\}^{\mathbb{N}} \\ \sup_{y \le x, y \in C} F(y), & x \in [0,1] - C \\ 0, & x < 0 \\ 1, & x > 1 \end{cases}$$

where C is the Cantor set. Then F is a valid CDF, but has no density.

Definition 1.33
A probability measure P *is said to be* ***discrete*** *if there is a countable set S such that* $\mathrm{P}(S^c) = 0$.
A random variable X is said to be ***discrete*** *if its distribution is.*

Theorem 1.34
Suppose $X : (\Omega, \mathcal{F}) \to (S, \sigma(\mathcal{A}))$ and $\mathcal{A}$ is a collection of subsets in S. If $X^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{A}$, then X is a random variable.

Proof. Set $\mathcal{G} = \left\{A \subset S \;\middle|\; X^{-1}(A) \in \mathcal{F}\right\}$. Clearly $\varnothing \in \mathcal{G}$ and if $A \in \mathcal{G}$, $X^{-1}(A^c) = (X^{-1}(A))^c \in \mathcal{F}$, so $A^c \in \mathcal{G}$. If $A_n \in \mathcal{G}$, then $X^{-1}(\cup_n A_n) = \cup_n X^{-1}(A_n) \in \mathcal{F}$, so $\cup_n A_n \in \mathcal{G}$. Hence $\mathcal{G}$ is a σ-algebra containing $\mathcal{A}$, so $\sigma(\mathcal{A}) \subset \mathcal{G}$. It follows that $X^{-1}(B) \in \mathcal{F}$ for all $B \in \sigma(\mathcal{A})$, so X is a random variable. ∎

Corollary 1.35
If X_i are random variables, then

$$\inf_i X_i, \quad \sup_i X_i, \quad \liminf_{i\to\infty} X_i, \quad \limsup_{i\to\infty} X_i$$

are all random variables.

Proof. Since the sets of the form $(-\infty, x]$ generate $\mathcal{B}$, it suffices to check that the inverse images of these sets are in $\mathcal{F}$. For $\inf_i X_i$,

$$\left\{\inf_i X_i \le x\right\} = \cup_i \{X_i \le x\} \in \mathcal{F}.$$

For $\sup_i X_i$, since $\sup_i X_i = -\inf_i(-X_i)$, it is also a random variable. Finally, write

$$\liminf_i X_i = \sup_n \inf_{i \ge n} X_i, \quad \limsup_i X_i = \inf_n \sup_{i \ge n} X_i.$$

The results follow from the measurability of $\inf_i X_i$ and $\sup_i X_i$. ∎

Definition 1.36
Let X be a random variable. $\sigma(X)$ is the smallest σ-algebra such that X is measurable.

Remark
If $X : \Omega \to (S, \mathcal{S})$, then $\sigma(X) = X^{-1}(\mathcal{S})$.

Definition 1.37
*Let X be a random variable. The **expectation** of X is defined as*

$$\mathrm{E}[X] = \int X d\,\mathrm{P}.$$

Theorem 1.38 (Jensen's Inequality)
Let $X : \Omega \to \mathbb{R}^d$ be a random variable such that $\mathrm{E}[\|X\|_1] < \infty$ and $\phi : \mathbb{R}^d \to \mathbb{R}$ be a convex function. Then

$$\phi(\mathrm{E}[X]) \leq \mathrm{E}[\phi(X)].$$

Proof. For any given $y \in \mathbb{R}^d$, note that $\left\{x \in \mathbb{R}^d \,\middle|\, \phi(x) > \phi(y)\right\}$ is a open convex set. By the Hahn-Banach seperation theorem, there is a hyperplane $\{f(x) = a + \langle b, x\rangle\}$ separating $\left\{(x, \phi(x)) \in \mathbb{R}^{d+1} \,\middle|\, \phi(x) > \phi(y)\right\}$ and $\{(y, \phi(y))\}$. Note that $\phi(y) = f(y)$ and $\phi(x) \geq f(x)$ for all $x \in \mathbb{R}^d$. Take $y = \mathrm{E}[X]$, then

$$\phi(\mathrm{E}[X]) = f(\mathrm{E}[X]) = \mathrm{E}[f(X)] \leq \mathrm{E}[\phi(X)].$$

■

Theorem 1.39 (Hölder's Inequality)
Let X, Y be random variables and $p, q \in [1, \infty]$ such that $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$\mathrm{E}[|XY|] \leq \mathrm{E}[|X|^p]^{1/p}\,\mathrm{E}[|Y|^q]^{1/q}.$$

Proof. If $\mathrm{E}[|X|^p]$ and $\mathrm{E}[|Y^q|]$ are zero or infinite, the result is trivial. We assume that $\mathrm{E}[|X|^p] = \mathrm{E}[|Y|^q] = 1$. For fixed $y \geq 0$, set $\phi(x) = x^p/p + y^p/p - xy$ for $x \geq 0$.

$$\phi'(x) = x^{p-1} - y, \quad \phi''(x) = (p-1)x^{p-2} \geq 0.$$

Thus ϕ is convex and minimized at $x = y^{1/(p-1)}$ with minimum $\phi(y^{1/(p-1)}) = 0$. Hence $x^p/p + y^q/q \geq xy$ for all $x, y \geq 0$.

$$\mathrm{E}[|XY|] \leq \mathrm{E}\left[\frac{|X|^p}{p} + \frac{|Y|^q}{q}\right] = \frac{1}{p} + \frac{1}{q} = 1 = \mathrm{E}[|X|^p]^{1/p}\,\mathrm{E}[|Y|^q]^{1/q}.$$

■

Theorem 1.40 (Markov's Inequality)
If $X \geq 0$ is a random variable, then for any $c > 0$,

$$\mathrm{P}(X \geq c) \leq \frac{1}{c}\,\mathrm{E}[X].$$

Proof.

$$\mathrm{P}(X \geq c) = \int \mathbf{1}\{X \geq c\}\, d\,\mathrm{P} \leq \int \frac{X}{c} d\,\mathrm{P} = \frac{1}{c}\,\mathrm{E}[X].$$

■

Example
Suppose $\phi : \mathbb{R} \to \mathbb{R}$ is a non-negative function. Put

$$I_A = \inf_{y \in A} \phi(y),$$

where A is some measurable set. Then for any random variable X,

$$I_A \mathbf{1}\{X \in A\} \leq \phi(x)\mathbf{1}\{X \in A\} \leq \phi(x).$$

Thus

$$I_A \, \mathrm{P}(X \in A) \leq \mathrm{E}\left[\phi(X)\right].$$

Corollary 1.41 (Chebyshev's Inequality)
Let X be a random variable. Then for any $c > 0$ and $\alpha \in \mathbb{R}$,

$$\mathrm{P}(|X - \alpha| \geq c) \leq \frac{1}{c^2} \mathrm{E}\left[(X - \alpha)^2\right].$$

Proof. By the Markov's inequality,

$$\mathrm{P}(|X - \alpha| \geq c) = \mathrm{P}((X - \alpha)^2 \geq c^2) \leq \frac{1}{c^2} \mathrm{E}\left[(X - \alpha)^2\right].$$

■

Theorem 1.42
Suppose X is a random variable of $(S, \mathcal{S})$ with distribution μ and $f : (S, \mathcal{S}) \to (\mathbb{R}, \mathcal{B})$ is measurable. If either

(a) $f \geq 0$, *or*

(b) $\mathrm{E}[|f(X)|] < \infty$,

then

$$\mathrm{E}[f(X)] = \int f(x) d\mu(x).$$

Proof. Suppose first that $f = \mathbf{1}_A$ for some $A \in \mathcal{S}$. Then

$$\mathrm{E}[f(X)] = \mathrm{P}(X \in \mathcal{A}) = \mathrm{P}(X^{-1}(A)) = \mu(A) = \int \mathbf{1}_A d\mu.$$

By linearity we can extend this result to simple functions. Now suppose first that (a) holds. For such f, there is a sequence of simple functions $s_n \nearrow f$ and $s_n \circ X \nearrow f \circ X$. By LMCT,

$$\mathrm{E}[f(X)] = \mathrm{E}\left[\lim_n s_n(X)\right] = \lim_n \mathrm{E}[s_n(X)] = \lim_n \int s_n d\mu = \int f d\mu.$$

Suppose that (b) is the case. Write $f = f^+ - f^-$ and apply the previous result.

$$\mathrm{E}\left[f(X)\right] = \mathrm{E}\left[f^+(X)\right] - \mathrm{E}\left[f^-(X)\right] = \int f^+ d\mu - \int f^- d\mu = \int f d\mu.$$

■

Definition 1.43
*The k-**th moment** of a random variable X is* $\mathrm{E}\left[X^k\right]$.

Definition 1.44
*The **variance** of a random variable X is* $\mathrm{Var}\,\mathrm{E}\left[(X - \mathrm{E}\left[X\right])^2\right]$.

Definition 1.45
*The **covariance** of two integrable random variables X, Y is*

$$\mathrm{Cov}(X, Y) = \mathrm{E}\left[(X - \mathrm{E}\left[X\right])(Y - \mathrm{E}\left[Y\right])\right].$$

Definition 1.46
For $1 \leq p < \infty$, the $\mathcal{L}^p(\Omega, \mathrm{P})$ space is defined as

$$\mathcal{L}^p(\Omega, \mathrm{P}) = \{X : \Omega \to S \mid X \text{ measurable and } \mathrm{E}\left[|X|^p\right] < \infty\}.$$

For $p = \infty$,

$$\mathcal{L}^\infty(\Omega, \mathrm{P}) = \{X : \Omega \to S \mid X \text{ measurable and } \operatorname{ess\,sup}_{\omega \in \Omega} X(\omega) < \infty\}.$$

Proposition 1.47
Let $1 \leq p < q \leq \infty$. Then $\mathcal{L}^q(\mathrm{P}) \subset \mathcal{L}^p(\mathrm{P})$.

Proof. Suppose first that $q < \infty$. If $X \in \mathcal{L}^q(\mathrm{P})$, then

$$\mathrm{E}\left[|X|^p\right] \leq \mathrm{E}\left[|X|^q \mathbf{1}\left\{|X| \geq 1\right\}\right] + \mathrm{E}\left[|X|^p \mathbf{1}\left\{|X| < 1\right\}\right] \leq \mathrm{E}\left[|X|^q\right] + 1 < \infty.$$

Hence $X \in \mathcal{L}^p(\mathrm{P})$. If $q = \infty$, X is essentially bounded, i.e., $X \leq M$ for some $M \in \mathbb{R}$ almost surely. Hence $X \in \mathcal{L}^p$. ■

1.3. Independence

Definition 1.48
*Let $(\Omega, \mathcal{F}, \mathrm{P})$ be a probability space. Suppose $\mathcal{F}_\beta \subset \mathcal{F}$, $\beta \in B$ are a collection of sub-σ-algebras. Then $\{\mathcal{F}_\beta\}$ are **independent** if for all finite $\{\mathcal{F}_i\}_{i=1}^n \subset \{\mathcal{F}_\beta\}$,*

$$\mathrm{P}(\cap_{i=1}^n A_i) = \prod_{i=1}^n \mathrm{P}(A_i)$$

where $A_i \in \mathcal{F}_i$.

Definition 1.49
*A collection of random variables $\{X_\beta \mid \beta \in B\}$ on $(\Omega, \mathcal{F}, \mathrm{P})$ is **independent** if the collection of the generating σ-algebras $\{\sigma(X_\beta) \mid \beta \in B\}$ is.*

Remark
In other words,

$$\mathrm{P}(\cap_i \{X_{\beta_i} \in A_i\}) = \prod_i \mathrm{P}(X_{\beta_i} \in A).$$

Note that these random variables can map into different measurable space.

Definition 1.50
*A collection of events $\mathcal{S}$ is **independent** if $\{\mathbf{1}_A \mid A \in \mathcal{S}\}$ is.*

Proposition 1.51
Let $X_1, \ldots, X_n$ be independent random variables and $g_1, \ldots g_n$ are measurable functions. Then $g_1(X_1), \ldots, g_n(X_n)$ are independent.

Proof. Suppose $g_i : (S_i, \mathcal{S}_i) \to (T_i, \mathcal{T}_i)$. For $A_i \in \mathcal{T}_i$, $g^{-1}(A_i) \in \mathcal{S}_i$ and

$$\mathrm{P}(\cap_i \{g_i(X_i) \in A_i\}) = \mathrm{P}(\cap_i \{X_i \in g^{-1}(A_i)\}) = \prod_i \mathrm{P}(X_i \in g^{-1}(A_i)) = \prod_i \mathrm{P}(g_i(X_i) \in A_i).$$

$g_1(X_1), \ldots, g_n(X_n)$ are independent. ∎

Theorem 1.52
Let $\mathcal{S}_1, \ldots \mathcal{S}_n$ be a collection of π-system. If $\Omega \in \mathcal{S}_i$ for all $i = 1, \ldots, n$ and for all $A_i \in \mathcal{S}_i$,

$$\mathrm{P}(\cap_i A_i) = \prod_i \mathrm{P}(A_i),$$

then $\sigma(\mathcal{S}_1), \ldots, \sigma(\mathcal{S}_n)$ are independent.

Proof. Fix $\mathcal{S}_2, \ldots, \mathcal{S}_n$. Put

$$\mathcal{L} = \left\{ A \in \mathcal{F} \;\middle|\; \mathrm{P}(A \cap (\cap_{i=2}^n A_i)) = \mathrm{P}(A) \prod_{i=2}^n \mathrm{P}(A_i), A_i \in \mathcal{S}_i \text{ for } i = 2, \ldots, n \right\}.$$

We claim that $\mathcal{L}$ forms a λ-system. First, by assumption we can pick $A_i = \Omega$ for $i = 2, \ldots, n$ to see that $\Omega \in \mathcal{L}$. Suppose that $A \subset B$, $A, B \in \mathcal{L}$,

$$\begin{aligned}\mathrm{P}((B - A) \cap (\cap_{i=2}^n A_i)) &= \mathrm{P}((B \cap (\cap_{i=2}^n A_i)) - (A \cap (\cap_{i=2}^n A_i))) \\ &= \mathrm{P}(B) \prod_{i=2}^n \mathrm{P}(A_i) - \mathrm{P}(A) \prod_{i=2}^n \mathrm{P}(A_i) = \mathrm{P}(B - A) \prod_{i=2}^n \mathrm{P}(A_i).\end{aligned}$$

Hence $B - A \in \mathcal{L}$. Let $S_j \nearrow S$, $S_j \in \mathcal{L}$. Then

$$\mathrm{P}(S \cap (\cap_{i=2}^n A_i)) = \lim_{j \to \infty} \mathrm{P}(S_j \cap (\cap_{i=2}^n A_i)) = \lim_{j \to \infty} \mathrm{P}(S_j) \prod_{i=2}^n \mathrm{P}(A_i) = \mathrm{P}(S) \prod_{i=2}^n \mathrm{P}(A_i).$$

Thus $S \in \mathcal{L}$ and $\mathcal{L}$ is a λ-system. By Dynkin's π-λ, $\sigma(\mathcal{S}_1), \mathcal{S}_2, \ldots, \mathcal{S}_n$ satisfies the product property. Repeat the procedure for $\mathcal{S}_2, \ldots, \mathcal{S}_n$. We have that $\sigma(\mathcal{S}_1), \ldots, \sigma(\mathcal{S}_n)$ satisfies the product property. That is, they are independent. ∎

Corollary 1.53
Let $X_1, \ldots, X_n$ be $\mathbb{R}$-valued random variables. Then they are independent if and only if

$$\mathrm{P}(X_1 \leq s_1, \ldots, X_n \leq s_n) = \prod_{i=1}^n \mathrm{P}(X_i \leq s_i)$$

for all $s_i \in \mathbb{R}$, $1 \leq i \leq n$.

Proof. The sufficient part is trivial. For the converse, put $\mathcal{S}_i = \{\{X_i \leq t\} \mid t \in \mathbb{R}\} \cup \{\Omega\}$. Clearly $\mathcal{S}_i$ are π-system and $\Omega \in \mathcal{S}_i$ for all i. $\sigma(\mathcal{S}_i)$ are independent and $\mathcal{S}_i$ generates $\sigma(X_i)$. Applying theorem 1.52 shows that X_i are independent. ∎

Corollary 1.54
If $\mathcal{F}_{ij}$, $1 \leq i \leq n, 1 \leq j \leq m(i)$ are independent σ-algebras, then $\mathcal{G}_i = \sigma(\cup_j \mathcal{F}_{ij})$ are independent.

Proof. Put $\mathcal{H}_i = \left\{\cap_j A_j \;\middle|\; A_j \in \mathcal{F}_{ij}\right\}$. We claim that $\sigma(\mathcal{H}_i) = \mathcal{G}_i$. Indeed, by choosing sets of the form

$$(\Omega, \ldots, \Omega, A_j, \Omega, \ldots, \Omega) \in \mathcal{F}_{i1} \times \cdots \times \mathcal{F}_{im(i)},$$

it is clear that $\cup_j \mathcal{F}_{ij} \subset \mathcal{H}_i$. Also, if $A \in \mathcal{H}_i$, then

$$A = \cap_j A_j = (\cup_j (A_j^c))^c \in \sigma(\cup_j \mathcal{F}_{ij}).$$

Thus $\cup_j \mathcal{F}_{ij} \subset \mathcal{H}_i \subset \sigma(\cup_j \mathcal{F}_{ij})$ and $\sigma(\mathcal{H}_i) = \sigma(\cup_j \mathcal{F}_{ij}) = \mathcal{G}_i$. Also notice that $\mathcal{H}_i$ contain Ω and form π-systems. For $A_i \in \mathcal{H}_i$, write $A_i = \cap_j A_{ij}$. Then

$$\mathrm{P}(\cap_i A_i) = \mathrm{P}(\cap_{ij} A_{ij}) = \prod_{ij} \mathrm{P}(A_{ij}) = \prod_i \mathrm{P}(\cap_j A_{ij}) = \prod_i \mathrm{P}(A_i).$$

From theorem 1.52 we know that $\mathcal{G}_i = \sigma(\mathcal{H}_i)$ are independent. ∎

Corollary 1.55
If $X_{ij}, 1 \le i \le n, 1 \le j \le m(i)$ are independent random variables, then $Y_i = h_i(X_{i1}, \dots, X_{im(i)})$ are independent provided that h_i are measurable.

Proof. Write $\mathcal{F}_{ij} = \sigma(X_{ij})$. We claim that $\sigma(Y_i) \subset \sigma(\cup_j \mathcal{F}_{ij})$. Indeed, if B_i is a measurable set, $h_i^{-1}(B_i)$ is measurable. Write $h_i^{-1}(B_i) = C_{i1} \times \cdots \times C_{im(i)}$ and since each $X_{ij}^{-1}(C_{ij}) \in \mathcal{F}_{ij}$, we see that $\sigma(Y_i) \subset \sigma(\cup_j \mathcal{F}_{ij})$. It then follows from corollary 1.54 that $\sigma(Y_i)$ are independent and Y_i are independent. ∎

Theorem 1.56
If $X_1, \dots X_n$ are independent $\mathbb{R}$-valued random variables and the distribution of X_i is μ_i. Then the joint distribution of $(X_1, \dots, X_n)$ is $\mu_1 \times \cdots \times \mu_n$.

Proof. Let μ be the distribution of $(X_1, \dots, X_n)$. By definition,

$$\begin{aligned}\mu((X_1, \dots) \in A_1 \times \cdots \times A_n) &= \mu(X_1 \in A_1, \dots, X_n \in A_n) \\ &= \prod_{i=1}^{n} \mu_i(X_i \in A_i) = (\mu_1 \times \cdots \times \mu_n)(A_1 \times \cdots \times A_n).\end{aligned}$$

Now the sets of the forms $A = A_1 \times \cdots \times A_n$ is a π-system generating the product σ-algebra. By corollary 1.19, the joint distribution is exactly $\mu_1 \times \cdots \times \mu_n$. ∎

Theorem 1.57
Let X, Y be two independent random variables. If $h(x, y)$ satisfies either

(a) $\mathrm{E}\left[|h(X, Y)|\right] < \infty$, *or*

(b) h *is non-negative,*

then

$$\mathrm{E}\left[h(X, Y)\right] = \int \int h d\mu_X d\mu_Y,$$

where μ_X, μ_Y are the distributions of X and Y, respectively.

Proof. The proof follows directly from Fubini-Tonelli theorem. If one of the assumptions is true, then

$$\mathrm{E}\left[h(X, Y)\right] = \int_{\mathbb{R}^2} h d(\mu_X \times \mu_Y) = \int \int h d\mu_X d\mu_Y.$$

∎

Remark
If $h(x, y) = h_1(x)h_2(y)$, then

$$\mathrm{E}\left[h_1(X)h_2(Y)\right] = \mathrm{E}\left[h(X, Y)\right] = \int \int h_1 h_2 d\mu_X d\mu_Y = \mathrm{E}\left[h_1(X)\right] \mathrm{E}\left[h_2(Y)\right].$$

Corollary 1.58

If $X_1, \dots X_n$ are independent random variables and

(a) $\mathrm{E}\left[|X_1 \cdots X_n|\right] < \infty$ *or*

(b) $X_i \geq 0$ *for all* i,

then

$$\mathrm{E}\left[X_1 \cdots X_n\right] = \prod_{i=1}^{n} \mathrm{E}\left[X_i\right].$$

Proof. Let $h(x, y) = xy$. By assumptions, we have either $\mathrm{E}\left[|h(X_1, X_2)|\right] < \infty$ or $h(X_1, X_2) \geq 0$. By theorem 1.57, $\mathrm{E}\left[X_1 X_2\right] = \mathrm{E}\left[X_1\right] \mathrm{E}\left[X_2\right]$. Substitute X_1 by $X_1 X_2$ and X_2 by X_3, we see that $\mathrm{E}\left[X_1 X_2 X_3\right] = \mathrm{E}\left[X_1\right] \mathrm{E}\left[X_2\right] \mathrm{E}\left[X_3\right]$. Repeat the procedure n times and the result follows. ■

Definition 1.59

Let X, Y be independent random variables with CDF F and G, respectively. The ***convolution*** *of two CDF is defined as*

$$(F * G)(z) = \int F(z - y) dG(y).$$

Remark

If F and G are absolutely continuous with respect to the Lebesgue measure, then they have Radon-Nikodym derivatives f and g. The definition of convolution becomes

$$(F * G)(z) = \int F(z - y) dG(y) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f(x) g(y) dx dy.$$

Then

$$(F * G)'(z) = \int f(z - y) g(y) dy = (f * g)(z),$$

which is exactly the definition of convolution of two functions.

Proposition 1.60

Let X and Y be independent random variables. Then

$$\mathrm{P}(X + Y \leq z) = (F * G)(z).$$

Proof. By theorem 1.57,

$$\begin{aligned} \mathrm{P}(X + Y \leq z) = \mathrm{E}\left[\mathbf{1}\left\{X + Y \leq z\right\}\right] &= \int \int \mathbf{1}\left\{x + y \leq z\right\} dF(x) dG(y) \\ &= \int F(z - y) dG(y) = (F * G)(z). \end{aligned}$$

■

Remark

Note that the convolution is commutative since

$$(F * G)(z) = \mathrm{P}(X + Y \leq z) = \mathrm{P}(Y + X \leq z) = (G * F)(z).$$

Remark

For discrete X and Y, the convolution becomes

$$\mathrm{P}(X+Y=z)=\sum_{y}\mathrm{P}(X=z-y)\,\mathrm{P}(Y=y).$$

Example

Consider $X\sim\Gamma(\alpha_1,\beta)$ and $Y\sim\Gamma(\alpha_2,\beta)$. Then the density for $X+Y$ is

$$\begin{aligned}
f_{X+Y}(z) &= \int f_X(z-y)f_Y(y)dy \\
&= \int_0^z \frac{1}{\Gamma(\alpha_1)}\beta^{\alpha_1}(z-y)^{\alpha_1-1}e^{-\beta(z-y)}\frac{1}{\Gamma(\alpha_2)}\beta^{\alpha_2}y^{\alpha_2-1}e^{-\beta y}dy \\
&= \frac{1}{\Gamma(\alpha_1)\Gamma(\alpha_2)}\beta^{\alpha_1+\alpha_2}e^{-\beta z}\int_0^z (z-y)^{\alpha_1-1}y^{\alpha_2-1}dy \\
&= \frac{1}{\Gamma(\alpha_1)\Gamma(\alpha_2)}\beta^{\alpha_1+\alpha_2}e^{-\beta z}z^{\alpha_1+\alpha_2-1}\int_0^1 (1-t)^{\alpha_1-1}t^{\alpha_2-1}dt \\
&= \frac{B(\alpha_1,\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)}\beta^{\alpha_1+\alpha_2}e^{-\beta z}z^{\alpha_1+\alpha_2-1} = \frac{1}{\Gamma(\alpha_1+\alpha_2)}\beta^{\alpha_1+\alpha_2}e^{-\beta z}z^{\alpha_1+\alpha_2-1}.
\end{aligned}$$

Hence $X+Y\sim\Gamma(\alpha_1+\alpha_2,\beta)$.

1.4. Convergence of Random Variables

Definition 1.61
A sequence of probability measures P_n *on* $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ *are* ***consistent*** *if*

$$\mathrm{P}_{n+1}((a_1, b_1] \times \cdots \times (a_n, b_n] \times \mathbb{R} = \mathrm{P}_n((a_1, b_1] \times \cdots \times (a_n, b_n])$$

for every n.

Theorem 1.62 (Kolmogorov Extension)
Suppose that a sequence of probability measures P_n *on* $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ *are consistent. Then there is a unique probability measure* P *on* $(\mathbb{R}^{\mathbb{N}}, \mathcal{B})$ *satisfying that*

$$\mathrm{P}(\left\{\omega \in \mathbb{R}^{\mathbb{N}} \;\middle|\; \omega_i \in (a_i, b_i], 1 \le i \le n\right\}) = \mathrm{P}_n((a_1, b_1] \times \cdots \times (a_n, b_n]),$$

where $\mathcal{B}$ *is generated by the collection*

$$\left\{\omega \in \mathbb{R}^{\mathbb{N}} \;\middle|\; \omega_i \in (a_i, b_i], 1 \le i \le n, n \in \mathbb{N}\right\}.$$

Proof. Let

$$\mathcal{S} = \{(a_1, b_1] \times \cdots \times (a_n, b_n] \times \mathbb{R} \times \cdots \mid n \in \mathbb{N}\}.$$

Define P on $\mathcal{S}$ to be

$$\mathrm{P}((a_1, b_1] \times \cdots \times (a_n, b_n] \times \mathbb{R} \times \cdots) = \mathrm{P}_n((a_1, b_1] \times \cdots \times (a_n, b_n])$$

Clearly, $\mathcal{S}$ forms a semi-algebra. From the Carathéodory extension theorem, it suffices to show that P is finitely additive, σ-additive on $\mathcal{S}$ and $\mathrm{P}(\varnothing) = 0$. Note that $\mathrm{P}(\varnothing) = \mathrm{P}(\varnothing \times \mathbb{R} \times \cdots) = \mathrm{P}_1(\varnothing) = 0$. We verify the first two conditions.

First, if $A, B \in \mathcal{S}$ are disjoint, $m \le n$,

$$A = \left\{\omega \in \mathbb{R}^{\mathbb{N}} \;\middle|\; \omega_i \in (a_i, b_i], 1 \le i \le m\right\} \quad \text{and} \quad B = \left\{\omega \in \mathbb{R}^{\mathbb{N}} \;\middle|\; \omega_i \in (c_i, d_i], 1 \le i \le n\right\},$$

then

$$\mathrm{P}(A \cup B) = \mathrm{P}_n((\pi_n A) \cup (\pi_n B)) = \mathrm{P}_n(\pi_n A) + \mathrm{P}_n(\pi_n B) = \mathrm{P}(A) + \mathrm{P}(B),$$

where $\pi_n : \omega \to (\omega_1, \ldots, \omega_n)$ is the projection onto the first n components. Hence P is finitely additive.

Next, suppose $A_1, \ldots \in \mathcal{S}$ are countably many disjoint measurable sets. Put $A = \cup_i A_i$. We can consider the algebra $\bar{\mathcal{S}} = \{\text{finite disjoint union of sets in } \mathcal{S}\}$ generated by $\mathcal{S}$. $B_n = \cup_{i>n} A_i \in \bar{\mathcal{S}}$. Thus

$$\mathrm{P}(A) = \mathrm{P}(B_n) + \sum_{i=1}^{n} \mathrm{P}(A_n)$$

by the previous result. It now suffices to show that $\mathrm{P}(B_n) \to 0$ for any $B_n \searrow \varnothing$. Suppose not,

then there is $\delta > 0$ such that $\mathrm{P}(B_n) \to \delta$ as $B_n \to \varnothing$ by the monotonicity of P.

For such $\{B_n\}$, we claim that there is a sequence of compact set K_n such that $K_n \subset B_n$ and $\mathrm{P}(B_n - K_n) < 2^{-(n+1)}\delta$. Now since $B_1 \in \bar{S}$, there are disjoint $E_1^1, \ldots, E_{m_1}^1$ such that $B_1 = \cup_{i=1}^{m_1} E_i^1$. Now since each E_i^1 is of the product of $(\cdot, \cdot]$. We can find a compact subset K_i^1 of the product of $[\cdot, \cdot]$ such that $\mathrm{P}(E_i^1 - K_i^1) < m_1^{-1} 2^{-2}\delta$. Hence $K_1 = \cup_i K_i^1 \subset B_1$ satisfies that

$$\mathrm{P}(B_1 - K_1) = \sum_{i=1}^{m_1} \mathrm{P}(E_i^1 - K_i^1) < 2^{-2}\delta$$

as desired. Repeat the process and find K_n inductively. The claim follows.

Now, $\cap_{n=1}^m K_n \searrow K$ as $m \to \infty$. Also,

$$\mathrm{P}(B_m - (\cap_{n=1}^m K_n)) \leq \sum_{n=1}^m \mathrm{P}(B_n - K_n) \leq \frac{\delta}{2}.$$

Hence $\delta/2 \leq \mathrm{P}(B_m) - \delta/2 \leq \mathrm{P}(\cap_{n=1}^m K_n)$. We see that $\cap_{n=1}^m K_n$ is non-empty for each m. But this implies that $K \subset \cap_n B_n$ is non-empty, a contradiction. Thus $\mathrm{P}(B_n) \to 0$.

Finally, the σ-additivity follows from that we can take $n \to \infty$ so that

$$\mathrm{P}(A) = \lim_{n\to\infty} \mathrm{P}(B_n) + \sum_{i=1}^n \mathrm{P}(A_n) = \sum_i \mathrm{P}(A_n).$$

Applying Carathéodory extension theorem, such P can be extended on $(\mathbb{R}^{\mathbb{N}}, \mathcal{B})$. ■

Remark

With Kolmogorov extension theorem, we can consider a sequence of independent variable X_i on the product probability space with $\mathcal{F} = \mathcal{B}$, $\tilde{X}_i : \omega \mapsto \omega_i$ and $\mathrm{P}(B_1 \times \cdots B_n) = \prod_{i=1}^n \mu_i(B_i)$, where μ_i is the distribution of X_i.

Definition 1.63

*Let X_n be a sequence of random variable. X_n **converges almost surely** to X if*

$$\mathrm{P}\left\{\lim_{n\to\infty} X_n = X\right\} = 1.$$

We denote it as $X_n \overset{a.s.}{\to} X$ or $X_n \to X$ a.s.

Definition 1.64

*Let X_n be a sequence of random variable. X_n **converges in probability** to X if for every $\epsilon > 0$,*

$$\mathrm{P}\{|X_n - X| > \epsilon\} \to 0$$

as $n \to \infty$. We denote it as $X_n \overset{p}{\to} X$.

Definition 1.65

A sequence of random variable $X_n \in \mathcal{L}^p$ is said to ***converge in*** $\mathcal{L}^p$ *to X if*

$$\mathrm{E}\left[|X_n - X|^p\right]^{1/p} \to 0$$

as $n \to \infty$. If $p = \infty$, the definition becomes

$$\operatorname{ess\,sup}_{\omega\in\Omega} |X_n(\omega) - X(\omega)| \to 0.$$

We denote it as $X_n \to X$ in $\mathcal{L}^p$.

Proposition 1.66
Let X_n be a sequence of independent and indentically distributed random variables. Then

(a) If $X_n \to X$ almost surely, then $X_n \xrightarrow{p} X$.

(b) If $X_n \to X$ in $\mathcal{L}^p$, then $X_n \xrightarrow{p} X$.

Proof. For (a), given $\epsilon > 0$, put

$$E_k = \cup_{n\geq k} \{|X_n - X| > \epsilon\}.$$

Note that $E_k \searrow E = \{|X_n - X| > \epsilon \text{ for infinitely many } n\} = \{\lim_{n\to\infty} X_n = X\}^c$. Hence

$$\mathrm{P}\{|X_k - X| > \epsilon\} \leq \mathrm{P}(E_k) \to \mathrm{P}\left\{\lim_{n\to\infty} X_n = X\right\}^c = 0$$

Hence $X_n \to X$ in probability.

For (b), suppose first that $p < \infty$. By Markov inequality,

$$\mathrm{P}\{|X_n - X| > \epsilon\} = \mathrm{P}\{|X_n - X|^p > \epsilon^p\} \leq \frac{1}{\epsilon^p}\mathrm{E}\left[|X_n - X|^p\right] \to 0.$$

Let $p = \infty$. Note that $\operatorname{ess\,sup}|X_n - X| = \inf\{c \mid \mathrm{P}\{|X_n - X| > c\} = 0\}$. Convergence in $\mathcal{L}^\infty$ implies that for $\epsilon > 0$, there is N such that if $n \geq N$, $\inf\{c \mid \mathrm{P}\{|X_n - X| > c\} = 0\} < \epsilon$. That is, $\mathrm{P}\{|X_n - X| > \epsilon\} = 0$ for $n \geq N$. Hence $X_n \xrightarrow{p} X$. ∎

2. Asymptotic Theory

2.1. Law of Large Number

Definition 2.1
*Let X_i be random variables with $\mathrm{E}\left[X_i^2\right] < \infty$. They are called **uncorrelated** if*

$$\mathrm{E}\left[X_i X_j\right] = \mathrm{E}\left[X_i\right]\mathrm{E}\left[X_j\right].$$

Theorem 2.2 (Weak Law of Large Number I)
Suppose that X_n are uncorrelated random variables with $\mathrm{Var}\left[X_n\right] \leq C\infty$ and $\mathrm{E}\left[X_n\right] = \mu$ for all n. Let $S_n = \sum_{i=1}^n X_i$. Then

$$\frac{1}{n}S_n \to \mu$$

in $\mathcal{L}^2$ and hence in probability.

Proof. Compute that

$$\mathrm{E}\left[\left(\frac{1}{n}S_n - \mu\right)^2\right] = \frac{1}{n^2}\sum_{i=1}^n \mathrm{Var}(X_i) \leq \frac{C}{n} \to 0.$$

Hence $\frac{1}{n}S_n \to \mu$ in $\mathcal{L}^2$ and thus in probability. ∎

Theorem 2.3 (Weak Law of Large Number II, Khinchin)
Suppose that X_i is a sequence of independent and identically distributed random variables with $\mathrm{E}\left[|X_1|\right] < \infty$. Let $S_n = \sum_{i=1}^n X_i$ and $\mu = \mathrm{E}\left[X_1\right]$. Then

$$\frac{1}{n}S_n \to \mu$$

in $\mathcal{L}^1$ and hence in probability.

Proof. By replacing X_i with $X_i - \mu$, we may assume without loss of generality that $\mu = 0$. Now, for $C > 0$,

$$0 = \mathrm{E}\left[X_i\right] = \mathrm{E}\left[X_i \mathbf{1}\left\{|X_i| > C\right\}\right] + \mathrm{E}\left[X_i \mathbf{1}\left\{|X_i| \leq C\right\}\right].$$

Also,

$$\begin{aligned}\frac{1}{n}S_n &= \frac{1}{n}\sum_{i=1}^n X_i \mathbf{1}\left\{|X_i| > C\right\} + \frac{1}{n}\sum_{i=1}^n X_i \mathbf{1}\left\{|X_i| \leq C\right\} \\ &= \frac{1}{n}\sum_{i=1}^n \left(X_i \mathbf{1}\left\{|X_i| > C\right\} - \mathrm{E}\left[X_i \mathbf{1}\left\{|X_i| > C\right\}\right]\right) + \frac{1}{n}\sum_{i=1}^n \left(X_i \mathbf{1}\left\{|X_i| \leq C\right\} - \mathrm{E}\left[X_i \mathbf{1}\left\{|X_i| \leq C\right\}\right]\right).\end{aligned}$$

Notice that by LDCT,

$$\mathrm{E}\left[\left|\frac{1}{n}\sum_{i=1}^{n}(X_i\mathbf{1}\{|X_i|>C\}-\mathrm{E}\left[X_i\mathbf{1}\{|X_i|>C\}\right])\right|\right]\leq 2\,\mathrm{E}\left[|X_1|\,\mathbf{1}\{|X_1|>C\}\right]\to 0$$

as $C\to\infty$ since $|X_1|\,\mathbf{1}\{|X_1|>C\}\leq|X_1|$ and $\mathrm{E}\left[|X_1|\right]<\infty$. Also, by Hölder inequality and the independence,

$$\mathrm{E}\left[\left|\frac{1}{n}\sum_{i=1}^{n}(X_i\mathbf{1}\{|X_i|\leq C\}-\mathrm{E}\left[X_i\mathbf{1}\{|X_i|\leq C\}\right])\right|\right]\leq\sqrt{\frac{1}{n}\operatorname{Var}(X_i\mathbf{1}\{|X_i|\leq C\})}\leq\frac{C}{\sqrt{n}}$$

For any given $\epsilon>0$, there is C such that $2\,\mathrm{E}\left[|X_1|\,\mathbf{1}\{|X_1|>C\}\right]<\epsilon$ and

$$\begin{aligned}\mathrm{E}\left[\left|\frac{1}{n}S_n\right|\right]&\leq\mathrm{E}\left[\left|\frac{1}{n}\sum_{i=1}^{n}(X_i\mathbf{1}\{|X_i|>C\}-\mathrm{E}\left[X_i\mathbf{1}\{|X_i|>C\}\right])\right|\right]\\&\quad+\mathrm{E}\left[\left|\frac{1}{n}\sum_{i=1}^{n}(X_i\mathbf{1}\{|X_i|\leq C\}-\mathrm{E}\left[X_i\mathbf{1}\{|X_i|\leq C\}\right])\right|\right]\\&\leq\epsilon+\frac{C}{\sqrt{n}}\to\epsilon\end{aligned}$$

as $n\to\infty$. Since ϵ can be arbitrarily small, we conclude that $\frac{1}{n}S_n\to 0$ in $\mathcal{L}^1$ and hence in probability. ■

Definition 2.4
Let A_n be a sequence of events.

$$\limsup_{n\to\infty}A_n=\cap_{m=1}^{\infty}\cup_{n=m}^{\infty}A_n$$

and

$$\liminf_{n\to\infty}A_n=\cup_{m=1}^{\infty}\cap_{n=m}^{\infty}A_n.$$

Remark
Observe that

$$\limsup_{n\to\infty}A_n=\{\omega\in\Omega\mid\omega\in A_n\text{ for infinitely many }n\}$$

and

$$\liminf_{n\to\infty}A_n=\{\omega\in\Omega\mid\omega\in A_n\text{ for all but finitely many }n\}.$$

Theorem 2.5 (Borel-Cantelli I)
Let A_n be a sequence of events. If $\sum_n\mathrm{P}(A_n)<\infty$, then

$$\mathrm{P}\left(\limsup_{n\to\infty}A_n\right)=0.$$

Proof. Let $\epsilon > 0$ be given. By assumption, there is n_0 such that $\sum_{n \geq n_0} \mathrm{P}(A_n) < \epsilon$. Then

$$\mathrm{P}\left(\limsup_{n\to\infty} A_n\right) = \mathrm{P}(\cap_{m=1}^{\infty} \cup_{n=m}^{\infty} A_n) \leq \mathrm{P}(\cup_{n=n_0}^{\infty} A_n) \leq \sum_{n=n_0}^{\infty} \mathrm{P}(A_n) < \epsilon.$$

Since ϵ can be arbitrarily small, $\mathrm{P}(\limsup_{n\to\infty} A_n) = 0$. ∎

Corollary 2.6

Suppose for $\epsilon > 0$, $\sum_n \mathrm{P}(|X_n - X| > \epsilon) < \infty$. Then $X_n \to X$ almost surely.

Proof. Let $E_k = \left\{|X_n - X| > k^{-1} \text{ for finitely many } n\right\}$. Note that $E_{k+1} \subset E_k$ and $E_k \searrow E = \{X_n \to X\}$. Now we claim that $\mathrm{P}(E_k) = 1$. Consider $E_k^n = \left\{|X_n - X| > k^{-1}\right\}$. For fixed k, by assumption we have $\sum_n \mathrm{P}(E_k^n) < \infty$. By Borel-Cantelli, $\mathrm{P}(\limsup_{n\to\infty} E_k^n) = 0$. Hence

$$\mathrm{P}(E_k) = \mathrm{P}(\left\{|X_n - X| > k^{-1} \text{ for infinitely many } n\right\}^c) = 1 - \mathrm{P}(\limsup_{n\to\infty} E_k^n) = 1.$$

It now follows by the monotone convergence of measures that $\mathrm{P}(E) = 1$. ∎

Remark

Intuitively, if the convergence is sufficiently fast, the convergence in probability may recover almost sure convergence.

Theorem 2.7 (Strong Law of Large Number I)

Let X_i be independent and identically distributed with $\mu = \mathrm{E}[X_1]$ and $\mathrm{E}\left[X_1^4\right] < \infty$. Let $S_n = \sum_{i=1}^n X_i$. Then

$$\frac{1}{n} S_n \to \mu$$

almost surely.

Proof. Note that

$$\begin{aligned}
\mathrm{E}\left[\left(\frac{1}{n}S_n - \mu\right)^4\right] &= \frac{1}{n^4}\left(\sum_i \mathrm{E}\left[(X_i - \mu)^4\right] + \sum_{i\neq j} \mathrm{E}\left[(X_i - \mu)^2 (X_j - \mu)^2\right]\right) \\
&\leq \frac{1}{n^3}\mathrm{E}\left[(X_1 - \mu)^4\right] + \frac{1}{n^4}\binom{n}{2}\binom{4}{2}\mathrm{E}\left[(X_1 - \mu)^2\right]^2 \leq \frac{C}{n^2}
\end{aligned}$$

for some constant C. By Checyshev's inequality, for $\epsilon > 0$,

$$\mathrm{P}\left\{\left|\frac{1}{n}S_n - \mu\right| > \epsilon\right\} \leq \frac{1}{\epsilon^4}\mathrm{E}\left[\left(\frac{1}{n}S_n - \mu\right)^4\right] \leq \frac{C}{\epsilon^2 n^2}$$

is absolute summable. Hence by corollary 2.6,

$$\frac{1}{n} S_n \to \mu$$

almost surely. ∎

Theorem 2.8
$X_n \xrightarrow{p} X$ if and only if every subsequence of X_n has a further subsequence converging almost surely.

Proof. Suppose first that $X_n \xrightarrow{p} X$. Given a subsequence $X_{n(k)}$, we can choose $n(k_1) < n(k_2) < \cdots$ such that

$$\mathrm{P}\left(\left|X_{n(k_i)} - X\right| > 2^{-i}\right) < 2^{-i}.$$

Since 2^{-i} is summable, by Borel-Cantelli we have

$$\mathrm{P}\left(\left|X_{n(k_i)} - X\right| > 2^{-i} \text{ for infinitely many } i\right) = 0.$$

In other words,

$$\mathrm{P}\left\{X_{n(k_i)} \to X\right\} = \mathrm{P}\left\{\left|X_{n(k_i)} - X\right| > 2^{-i} \text{ for infinitely many } i\right\}^c = 1.$$

For the converse, suppose that $X_n \not\to X$ in probability. Then there exist $\epsilon, \delta > 0$ and

$$\mathrm{P}\left\{\left|X_{n(k)} - X\right| > \epsilon\right\} \geq \delta.$$

By assumption there is a further subsequence converging almost surely and thus in probability, i.e.,

$$\mathrm{P}\left\{\left|X_{n(k_j)} - X\right| > \epsilon\right\} \to 0.$$

This is a contradiction. Hence $X_n \to X$ in probability. ∎

Corollary 2.9
Suppose $X_n \xrightarrow{p} X$. Then the followings are true:

(a) If f is continuous, then $f(X_n) \xrightarrow{p} f(X)$.

(b) If $|X_n| \leq Y$ for some $Y \in \mathcal{L}^1$, then $\mathrm{E}[X_n] \to \mathrm{E}[X]$.

(c) If $X_n \in \mathcal{L}^1$ with $X_n \geq 0$, then $\mathrm{E}[X] \leq \liminf_{n\to\infty} \mathrm{E}[X_n]$.

Proof. For (a), by theorem 2.8, every subsequence has a further subsequence $X_{n(k_j)} \to X$ almost surely and hence $f(X_{n(k_j)}) \to f(X)$ almost surely. Then by theorem 2.8 again we see that $f(X_n) \xrightarrow{p} f(X)$.

For (b), by theorem 2.8, every subsequence has a furhter subsequence $X_{n(k_j)} \to X$ almost surely and LDCT gives $\mathrm{E}\left[X_{n(k_j)}\right] \to \mathrm{E}[X]$. This implies that $\mathrm{E}[X_n] \to \mathrm{E}[X]$ as well.

For (c), note that there is a subsequence X_{n_k} such that $\mathrm{E}\left[X_{n_k}\right] \to \liminf_{n\to\infty} \mathrm{E}[X_n]$. Now, by theorem 2.8, there is a further subsequence of X_{n_k} such that $X_{n_{k(j)}} \to X$ almost surely. By the Fatou's lemma,

$$\mathrm{E}[X] = \mathrm{E}\left[\liminf_{j\to\infty} X_{n_{k(j)}}\right] \leq \liminf_{j\to\infty} \mathrm{E}\left[X_{n_{k(j)}}\right] = \lim_{k\to\infty} \mathrm{E}\left[X_{n_k}\right] = \liminf_{n\to\infty} \mathrm{E}[X_n].$$

∎

Definition 2.10
Let $(\Omega, \mathcal{F}, \mathrm{P})$ *be a probability space.*

$$\mathcal{L}^0(\Omega) = \{X : \Omega \to \mathbb{R} \mid X \text{ is } \mathcal{F}\text{-measurable}\}.$$

Remark
In general, the almost convergence notion on $\mathcal{L}^0$ *is not metrizable, i.e., there is no metric* d *on* $\mathcal{L}^0$ *such that*

$$d(X_n, X) \to 0 \quad \Leftrightarrow \quad X_n \to X \quad a.s.$$

To see this, suppose that the almost sure convergence is metrizable. If $X_n \xrightarrow{p} X$*, any subsequence* $X_{n(k)}$ *converges to* X *in probability as well. By theorem 2.8, we can find a further subsequence converging almost surely and hence in metric* d*, but this implies that* $d(X_n, X) \to 0$*. Then* $X_n \to X$ *almost surely, which is absurd since convergence in probability does not imply almost sure convergence in general.*

However, convergence in probability on $\mathcal{L}^0$ *can be metrized. For instance,*

$$d(X, Y) = \mathrm{E}\left[\max\{|X - Y|, 1\}\right].$$

Theorem 2.11 (Borel-Cantelli II)
Let A_n *be independent events and* $\sum_{n=1}^{\infty} \mathrm{P}(A_n) = \infty$*. Then*

$$\mathrm{P}\left(\limsup_{n\to\infty} A_n\right) = 1.$$

Proof. By assumption we have that $\sum_{n\geq m} \mathrm{P}(A_n) = \infty$ for every $m \in \mathbb{N}$. Notice that $1 + x \leq e^x$. Then

$$\begin{aligned}
\mathrm{P}(\limsup_{n\to\infty} A_n) &= \lim_{m\to\infty} \mathrm{P}(\cup_{n\geq m} A_n) = 1 - \lim_{m\to\infty} \mathrm{P}(\cap_{n\geq m} A_n^c) \\
&= 1 - \lim_{m\to\infty} \lim_{N\to\infty} \mathrm{P}(\cap_{n=m}^{N} A_n^c) = 1 - \lim_{m\to\infty} \lim_{N\to\infty} \prod_{n=m}^{N} \mathrm{P}(A_n^c) \\
&= 1 - \lim_{m\to\infty} \prod_{n=m}^{\infty} (1 - \mathrm{P}(A_n)) \geq 1 - \lim_{m\to\infty} \exp\left(-\sum_{n=m}^{\infty} \mathrm{P}(A_n)\right) = 1.
\end{aligned}$$

Hence $\mathrm{P}(\limsup_{n\to\infty} A_n) = 1$. ∎

Lemma 2.12
Let X *be a non-negative random variable and* $h : \mathbb{R} \to \mathbb{R}$ *be a differentiable function with* $h(0) = 0$ *and* $h' \geq 0$*. Then*

$$\mathrm{E}\left[h(X)\right] = \int_0^{\infty} h'(t)\, \mathrm{P}(X > t) dt.$$

Proof. By Fubini-Tonelli theorem,

$$\begin{aligned}\mathrm{E}\left[h(X)\right] &= \mathrm{E}\left[\int_0^X h'(t)dt\right] = \mathrm{E}\left[\int_0^\infty \mathbf{1}\left\{t < X\right\} h'(t)dt\right] \\ &= \int_0^\infty h'(t)\,\mathrm{E}\left[\mathbf{1}\left\{t < X\right\}\right] dt = \int_0^\infty h'(t)\,\mathrm{P}(X > t)dt.\end{aligned}$$

■

Proposition 2.13

Suppose that X_i are independent and identically distributed random variables with $\mathrm{E}\left[|X_i|\right] = \infty$. Let $S_n = \sum_{i=1}^n X_i$. Then

(a) $\mathrm{P}\left\{|X_n| > n \text{ for infinitely many } n\right\} = 1$.

(b) $\mathrm{P}\left\{\frac{1}{n}S_n \text{ has finite limit}\right\} = 0$.

Proof. For (a), using lemma 2.12 with h being identity,

$$\begin{aligned}\infty = \mathrm{E}\left[|X_1|\right] &= \int_0^\infty \mathrm{P}(|X_1| > t)dt \le \sum_{n=0}^\infty \int_n^{n+1} \mathrm{P}(|X_1| > t)dt \\ &\le \sum_{n=0}^\infty \int_n^{n+1} \mathrm{P}(|X_1| > n)dt = \sum_{n=0}^\infty \mathrm{P}(|X_n| > n).\end{aligned}$$

Now by the second Borel-Cantelli, $\mathrm{P}\left\{|X_n| > n \text{ for infinitely many } n\right\} = 1$.

For (b), consider ω with $\frac{S_n(\omega)}{n} \to Y(\omega) \in \mathbb{R}$. Then for such ω,

$$\frac{X_n}{n} = \frac{S_n - S_{n-1}}{n} = \frac{S_n}{n} - \frac{n-1}{n}\frac{S_{n-1}}{n-1} \to 0$$

as $n \to \infty$. Thus

$$\begin{aligned}\mathrm{P}\left\{\frac{1}{n}S_n \text{ has finite limit}\right\} &\le \mathrm{P}\left\{|X_n| > n \text{ for finitely many } n\right\} \\ &= 1 - \mathrm{P}\left\{|X_n| > n \text{ for infinitely many } n\right\} = 0.\end{aligned}$$

(b) follows. ■

Definition 2.14

*A collection of σ-algebra $\{\mathcal{H}_k\}$ is **pairwise independent** if for any $\mathcal{H}_1, \mathcal{H}_2 \in \{\mathcal{H}_k\}$,*

$$\mathrm{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$$

for any $A \in \mathcal{H}_1$ and $B \in \mathcal{H}_2$.

Remark

As before, a sequence of random variables $\{X_k\}$ is pairwise independent if $\{\sigma(X_k)\}$ is.

Theorem 2.15 (Strong Law of Large Number II, Kolmogorov)
Let X_i be pairwise independent, identically distributed random variables with $\mathrm{E}\left[|X_1|\right] < \infty$ and $S_n = \sum_{i=1}^n X_i$. Then

$$\frac{1}{n} S_n \to \mathrm{E}\left[X_1\right] = \mu$$

almost surely.

Proof. Since we can always decompose $X_i = X_i^+ - X_i^-$ and X_i^+, X_i^- satisfy the assumption of the theorem, we may assume without loss of generality that $X_i \geq 0$. Let $Y_i = X_i \mathbf{1}\left\{X_i \leq i\right\}$ and $T_n = \sum_{i=1}^n Y_i$. Let $\alpha > 1$ and put $k_n = \lfloor \alpha^n \rfloor$. By Chebyshev inequality, for any given $\epsilon > 0$ we have

$$\begin{aligned}
\sum_{n=1}^{\infty} \mathrm{P}\left(\left|\frac{T_{k_n} - \mathrm{E}\left[T_{k_n}\right]}{k_n}\right| > \epsilon\right) &\leq \frac{1}{\epsilon^2} \sum_{n=1}^{\infty} \frac{1}{k_n^2} \mathrm{Var}(T_{k_n}) \\
&= \frac{1}{\epsilon^2} \sum_{n=1}^{\infty} \frac{1}{k_n^2} \sum_{i=1}^{k_n} \mathrm{Var}(Y_i) = \frac{1}{\epsilon^2} \sum_{i=1}^{\infty} \mathrm{Var}(Y_i) \sum_{n:k_n \geq i} \frac{1}{k_n^2}.
\end{aligned}$$

Since $1/k^2$ is summable and k_n repeat at most m_α times, where m_α is an integer such that $\alpha^{m_\alpha + 1} \geq \alpha^{m_\alpha} + 1$, we can find a constant $c_\alpha > 0$ such that

$$\sum_{n:k_n \geq i} \frac{1}{k_n^2} \leq \frac{c_\alpha}{i^2}.$$

Let F be the distribution of X. We have

$$\begin{aligned}
\sum_{n=1}^{\infty} \mathrm{P}\left(\left|\frac{T_{k_n} - \mathrm{E}\left[T_{k_n}\right]}{k_n}\right| > \epsilon\right) &\leq \frac{c_\alpha}{\epsilon^2} \sum_{i=1}^{\infty} \frac{\mathrm{Var}(Y_i)}{i^2} \leq \frac{c_\alpha}{\epsilon^2} \sum_{i=1}^{\infty} \frac{\mathrm{E}\left[Y_i^2\right]}{i^2} \\
&= \frac{c_\alpha}{\epsilon^2} \sum_{i=1}^{\infty} \frac{1}{i^2} \int_0^i x^2 dF(x) = \frac{c_\alpha}{\epsilon^2} \sum_{i=1}^{\infty} \sum_{k=0}^{i-1} \frac{1}{i^2} \int_k^{k+1} x^2 dF(x) \\
&= \frac{c_\alpha}{\epsilon^2} \sum_{k=0}^{\infty} \left(\sum_{i=k+1}^{\infty} \frac{1}{i^2} \right) \int_k^{k+1} x^2 dF(x)
\end{aligned}$$

Also, notice that there is a constant C such that

$$\sum_{i=k+1}^{\infty} \frac{1}{i^2} \leq \frac{C}{k+1}.$$

Hence,

$$\begin{aligned}
\sum_{n=1}^{\infty} \mathrm{P}\left(\left|\frac{T_{k_n} - \mathrm{E}\left[T_{k_n}\right]}{k_n}\right| > \epsilon\right) &\leq \frac{c_\alpha C}{\epsilon^2} \sum_{k=0}^{\infty} \frac{1}{k+1} \int_k^{k+1} x^2 dF(x) \\
&\leq \frac{c_\alpha C}{\epsilon^2} \sum_{k=0}^{\infty} \int_k^{k+1} x dF(x) = \frac{c_\alpha C}{\epsilon^2} \mathrm{E}\left[X_1\right] < \infty.
\end{aligned}$$

Note that for $\delta > 0$ there is an integer M such that $\mathrm{E}\left[X_1 \mathbf{1}\left\{X_1 > M\right\}\right] \leq \delta \leq \mathrm{E}\left[X_1\right]$.

$$\frac{\mathrm{E}\left[T_{k_n}\right]}{k_n} = \frac{1}{k_n}\sum_{i=1}^{k_n} \mathrm{E}\left[Y_i\right] \geq \frac{1}{k_n}\sum_{i=1}^{M} \mathrm{E}\left[Y_i\right] + \frac{1}{k_n}\sum_{i=M+1}^{k_n} \mathrm{E}\left[X_1\right] - \delta.$$

Also,

$$\frac{1}{k_n}\sum_{i=1}^{k_n} \mathrm{E}\left[Y_i\right] \leq \frac{1}{k_n}\sum_{i=1}^{k_n} \mathrm{E}\left[X_1\right] = \mathrm{E}\left[X_1\right].$$

Taking $n \to \infty$ and since δ is arbitrary, we conclude that

$$\frac{\mathrm{E}\left[T_{k_n}\right]}{k_n} \to \mathrm{E}\left[X_1\right].$$

Thus, by the Borel-Cantelli lemma,

$$\mathrm{P}\left\{\frac{T_{k_n}}{k_n} \not\to \mathrm{E}\left[X_1\right]\right\} = \mathrm{P}\left\{\left|\frac{T_{k_n} - \mathrm{E}\left[T_{k_n}\right]}{k_n}\right| > \epsilon \text{ for infinitely many } n\right\} = 0.$$

In other words, $T_{k_n}/k_n \to \mathrm{E}\left[X_1\right]$ almost surely. Also,

$$\begin{aligned}\sum_{k=1}^{\infty} \mathrm{P}\left\{X_k \neq Y_k\right\} = \sum_{k=1}^{\infty} \mathrm{P}\left\{X_k > k\right\} &\leq \sum_{k=1}^{\infty} \int_{k-1}^{k} \mathrm{P}(X_1 > t)dt \\ &= \int_0^{\infty} \mathrm{P}(X_1 > t)dt = \mathrm{E}\left[X_1\right] < \infty\end{aligned}$$

by lemma 2.12. Hence by Borel-Cantelli lemma, $X_k \neq Y_k$ for finitely many k almost surely. This implies that

$$\lim_{n\to\infty} \frac{1}{k_n} S_{k_n} = \lim_{n\to\infty} \frac{T_{k_n}}{k_n} = \mathrm{E}\left[X_1\right]$$

almost surely. Note that S_m is monotone and for each m, we may find $k(n_m) \leq m \leq k(n_{m+1})$ so that

$$\frac{S_{k(n_m)}}{k(n_{m+1})} \leq \frac{S_m}{m} \leq \frac{S_{k(n_{m+1})}}{k(n_m)}.$$

Take $m \to \infty$, we conclude that

$$\frac{1}{\alpha}\mu \leq \liminf_{m\to\infty} \frac{S_m}{m} \leq \limsup_{m\to\infty} \frac{S_m}{m} \leq \alpha\mu$$

almost surely. Taking $\alpha \to 1^+$ gives the desired result. ∎

Theorem 2.16

Let X_i be independent and identically distributed with $\mathrm{E}\left[X_1^+\right] = \infty$ and $\mathrm{E}\left[X_1^-\right] < \infty$. Let $S_n = \sum_{i=1}^n X_i$. Then

$$\frac{1}{n} S_n \to \infty$$

almost surely.

Proof. Write $X_i = X_i^+ - X_i^-$. For X_i^+, consider $Y_i^M = \min\left\{X_i^+, M\right\}$ for some $M > 0$. Note that Y_i^M is independent and identically distributed with finite mean. By the strong law of large number,

$$\frac{1}{n}\sum_{i=1}^{n} Y_i^M \to \mathrm{E}\left[Y_1^M\right]$$

almost surely. Hence,

$$\liminf_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} X_i^+ \geq \lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} Y_i^M = \mathrm{E}\left[Y_1^M\right].$$

Notice that $Y_1^M \nearrow X_1^+$ as $M \to \infty$. By LMCT,

$$\lim_{M\to\infty} \mathrm{E}\left[Y_1^M\right] = \mathrm{E}\left[X_1^+\right] = \infty.$$

We conclude that $\liminf_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} X_i^+ = \infty$. On the other hand, by the strong law of large number,

$$\frac{1}{n}\sum_{i=1}^{n} X_i^- \to \mathrm{E}\left[X_1^-\right]$$

almost surely. We end up with

$$\lim_{n\to\infty} \frac{1}{n} S_n = \lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} X_i^+ - \lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} X_i^- = \infty$$

almost surely. ∎

Example

Let Y_i be independent and identically distributed with density

$$f(y) = \mathbf{1}\{y \geq 1\} \frac{1}{c}\frac{1}{y^2},$$

where c is some normalizing constant. Let $H_i \sim Ber(2^{-i})$. Put $X_i = Y_i H_i$. Then $\mathrm{E}[X_i] = \infty$ for all i, but since

$$\sum_i \mathrm{P}(X_i > 0) = \sum_i 2^{-i} < \infty,$$

by the Borel-Cantelli lemma, $X_i \to 0$ almost surely and

$$\frac{1}{n}\sum_{i=1}^{n} X_i \to 0$$

almost surely.

Example

$Y \geq 0$ is a random variable with $\mathrm{E}[Y] = \infty$. Put $X_i = Y$ for all i. Then X_i are identically distributed with $\mathrm{E}[X_i] = \infty$. But

$$\frac{1}{n}\sum_{i=1}^{n} X_i = Y \not\to \infty$$

almost surely.

Example (Event Streaks)
$X_i \overset{iid}{\sim} Ber(2^{-1})$. Let L_n be the longest streaks of 1 in the first n trials. We have the following:

$$\lim_{n\to\infty} \frac{L_n}{\log_2(n)} = 1.$$

To see this, let ℓ_n be the length of the current streaks. For instance, the following sequence

$$1, 0, 1, 1, 1, 1, 0, \ldots$$

generates $\ell_1 = 1, \ell_2 = 0, \ell_6 = 4$. Observe that $L_n = \max_{m\leq n} \ell_m$. Now,

$$\mathrm{P}(\ell_n \geq k) = \sum_{m=k}^{n} \mathrm{P}(\ell_n = k) = \sum_{m=k}^{n} 2^{-k-1} \leq 2^{-k}$$

as $n \to \infty$. For $\epsilon > 0$,

$$\mathrm{P}(\ell_n \geq (1+\epsilon)\log_2(n)) = \mathrm{P}(\ell_n \geq \lceil (1+\epsilon)\log_2(n) \rceil) \leq 2^{-\lceil (1+\epsilon)\log_2(n)\rceil} \leq 2^{-(1+\epsilon)\log_2(n)} = \frac{1}{n^{1+\epsilon}}$$

is summable. By the Borel-Cantelli lemma,

$$\mathrm{P}\{\ell_n \geq (1+\epsilon)\log_2(n) \text{ for infinitely many } n\} = 0.$$

Hence

$$\mathrm{P}\{\ell_n < (1+\epsilon)\log_2(n) \text{ for all but finitely many } n\} = 1.$$

That is, for almost every ω, there is $N(\omega)$ such that $\ell_n < (1+\epsilon)\log_2(n)$ for $n \geq N(\omega)$. For such ω, we have

$$L_n(\omega) = \max_{m\leq n} \ell_n(\omega) \leq \max_{m\leq n} (1+\epsilon)\log_2(n) = (1+\epsilon)\log_2(n)$$

as $n > N(\omega)$. Thus

$$\limsup_{n\to\infty} \frac{L_n}{\log_2(n)} \leq 1+\epsilon$$

almost surely. Note that

$$\left\{\limsup_{n\to\infty} \frac{L_n}{\log_2(n)} \leq 1+\epsilon\right\} \searrow \left\{\limsup_{n\to\infty} \frac{L_n}{\log_2(n)} \leq 1\right\}$$

as $\epsilon \to 0^+$ *and by the monotone convergence of the measures,*

$$\limsup_{n\to\infty} \frac{L_n}{\log_2(n)} \leq 1$$

almost surely.

For the other side, note that for large n, we may split the sequence into blocks of size $\lceil (1-\epsilon)\log_2(n) \rceil$ *and*

$$\frac{n}{\lceil (1-\epsilon)\log_2(n) \rceil} \geq \frac{n}{\log_2(n)}$$

for large n.

$$\begin{aligned} \mathrm{P}(L_n \leq (1-\epsilon)\log_2(n)) &\leq \mathrm{P}(\textit{each block did not have all 1s}) \\ &\leq (1 - 2^{-\lceil (1-\epsilon)\log_2(n)/2 \rceil})^{n/\lceil (1-\epsilon)\log_2(n)/2 \rceil} \\ &\leq \left(1 - \frac{1}{n^{1-\epsilon}}\right)^{n^{1-\epsilon}\frac{n^\epsilon}{\log_2(n)}} \leq \exp\left(-\frac{n^\epsilon}{\log_2(n)}\right), \end{aligned}$$

whcih is summable, so by the Borel Cantelli lemma,

$$\mathrm{P}\left\{L_n \leq (1-\epsilon)\log_2(n) \textit{ for infinitely many } n\right\} = 0.$$

By a similar argument as above,

$$\liminf_{n\to\infty} \frac{L_n}{\log_2(n)} \geq 1 - \epsilon$$

almost surely and by the monotone convergence of the measures

$$\liminf_{n\to\infty} \frac{L_n}{\log_2(n)} \geq 1$$

almost surely. We conclude that

$$1 \leq \liminf_{n\to\infty} \frac{L_n}{\log_2(n)} \leq \limsup_{n\to\infty} \frac{L_n}{\log_2(n)} \leq 1$$

and the claim follows.

Example (Counting Process)
Let $X_i \in (0,\infty)$ *be independent and identically distributed random variable. Put* $\mu = \mathrm{E}[X_1]$, $T_n = \sum_{i=1}^n X_i$ *and* $N_t = \sup\{n \mid T_n \leq t\}$. *Then we have the following claim:*

$$\lim_{t\to\infty} \frac{N_t}{t} = \frac{1}{\mu}$$

almost surely. To see this, note that since $X_i < \infty$ for all i,

$$\lim_{t\to\infty} N_t = \lim_{t\to\infty} \sup\{n \mid T_n \le t\} = \infty.$$

Now, observe that $T_{N_t} \le t \le T_{N_t+1}$ and hence

$$\frac{T_{N_t}}{N_t} \le \frac{t}{N_t} \le \frac{T_{N_t+1}}{N_t+1}\frac{N_t+1}{N_t}.$$

By the strong law of large number, $T_{N_t}/N_t \to \mu$ almost surely. Thus

$$\lim_{t\to\infty} \frac{N_t}{t} = \frac{1}{\mu}.$$

Theorem 2.17 (Glivenko-Cantelli)
Suppose that $X_i \overset{iid}{\sim} F$ with $X_i \in (-\infty, \infty)$ and

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i \le x\}$$

is the empirical CDF. Then

$$\|F_n - F\|_\infty \to 0$$

almost surely when $n \to \infty$.

Proof. We first claim that for $\epsilon > 0$, we may find a finite partition $\{t_j\}$ such that $-\infty = t_0 < \cdots < t_j = \infty$ and $F(t_{j+1}^-) - F(t_j) \le \epsilon$ for all j. To see the existence of such partition, put $t_0 = -\infty$ and let

$$t_{j+1} = \sup\left\{t \in \mathbb{R} \,\middle|\, F(t) \le F(t_j) + \epsilon\right\}.$$

Observe that $F(t_{j+1}) \ge F(t_j) + \epsilon$. If not, then $F(t_{j+1}) < F(t_j) + \epsilon$. By the right-continuity of F, there is $\delta > 0$ such that $F(t_{j+1} + \delta) \le F(t_j) + \epsilon$, contradicting to the definition of t_{j+1}. It now also follows from the definition that

$$F(t_{j+1}^-) \le F(t_j) + \epsilon.$$

Finally, since F is of finite total variation, the jumps of sizes greater than ϵ can occur only finitely many times and we conclude the existence of such partition.

Next, by the strong law of large number, for almost every ω there is $N(\omega)$ uniform in j such that

$$\left|F_n(t_j) - F(t_j)\right| \le \epsilon$$

for all $n > N(\omega)$. For any $t \in [t_j, t_{j+1})$, we have

$$F(t) - F(t_j) \le F(t_{j+1}^{-1}) - F(t_j) \le \epsilon.$$

Again, by the strong law of the large number,

$$F_n(t_{j+1}^-) - F_n(t_j) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\left\{t_j < X_i < t_{j+1}\right\} \to \mathrm{E}\left[\mathbf{1}\left\{t_j < X_i < t_{j+1}\right\}\right] = F(t_{j+1}^-) - F(t_j)$$

almost surely. That is, for almost every ω, there is $N'(\omega) > N(\omega)$ such that for all j,

$$F_n(t_{j+1}^-) - F_n(t_j) \le F(t_{j+1}^-) - F(t_j) + \epsilon$$

if $n \ge N'(\omega)$. Combining the above estimates, if $n \ge N'(\omega)$,

$$\begin{aligned} |F_n(t) - F(t)| &\le \left|F_n(t) - F_n(t_j)\right| + \left|F_n(t_j) - F(t_j)\right| + \left|F(t_j) - F(t)\right| \\ &\le \left|F_n(t_{j+1}^-) - F_n(t_j)\right| + 2\epsilon \\ &\le F(t_{j+1}^-) - F(t_j) + 3\epsilon \le 4\epsilon. \end{aligned}$$

Since ϵ is arbitrary, we conclude that $F_n \to F$ uniformly for almost every ω and the proof is complete. ∎

Theorem 2.18 (Kolmogorov Maximal Inequality)
Suppose that X_i are independent with $\mathrm{E}[X_i] = 0$ and $\mathrm{Var}[X_i] < \infty$. Let $S_n = \sum_{i=1}^n X_i$. Then

$$\mathrm{P}\left(\max_{1\le k\le n} |S_k| \ge x\right) \le \frac{1}{x^2}\mathrm{Var}[S_n].$$

Proof. Let $A_k = \left\{|S_k| \ge x \text{ and } \left|S_j\right| < x \text{ for } 1 \le j \le k-1\right\}$. Note that

$$\sum_{k=1}^{n} \mathbf{1}_{A_k} = \mathbf{1}\left\{\max_{1\le k\le n} |S_k| \ge x\right\}.$$

$$\begin{aligned} \mathrm{E}\left[S_n^2\right] &\ge \mathrm{E}\left[S_n^2 \sum_{k=1}^{n} \mathbf{1}_{A_k}\right] = \sum_{k=1}^{n} \mathrm{E}\left[S_n^2 \mathbf{1}_{A_k}\right] = \sum_{k=1}^{n} \mathrm{E}\left[(S_n - S_k + S_k)^2 \mathbf{1}_{A_k}\right] \\ &= \sum_{k=1}^{n} \mathrm{E}\left[(S_n - S_k)^2 \mathbf{1}_{A_k}\right] + 2\,\mathrm{E}\left[(S_n - S_k) S_k \mathbf{1}_{A_k}\right] + \mathrm{E}\left[S_k^2 \mathbf{1}_{A_k}\right] \\ &\ge \sum_{k=1}^{n} \mathrm{E}\left[S_k^2 \mathbf{1}_{A_k}\right] + 2\,\mathrm{E}\left[(S_n - S_k) S_k \mathbf{1}_{A_k}\right]. \end{aligned}$$

Notice that $S_n - S_k \in \sigma(X_{k+1}, \ldots, X_n)$ and $S_k \mathbf{1}_{A_k} \in \sigma(X_1, \ldots, X_k)$ are independent. Thus

$$\begin{aligned} \mathrm{E}\left[S_n^2\right] &\ge \sum_{k=1}^{n} \mathrm{E}\left[S_k^2 \mathbf{1}_{A_k}\right] + 2\,\mathrm{E}\left[(S_n - S_k) S_k \mathbf{1}_{A_k}\right] \\ &= \sum_{k=1}^{n} \mathrm{E}\left[S_k^2 \mathbf{1}_{A_k}\right] \ge x^2 \sum_{k=1}^{n} \mathrm{E}\left[\mathbf{1}_{A_k}\right] = x^2\,\mathrm{P}\left(\max_{1\le k\le n} |S_k| \ge x\right). \end{aligned}$$

Rearranging the inequality gives the desired result. ∎

Definition 2.19
Let $(\Omega, \mathcal{F}, \mathrm{P})$ *be a probability space. A sub-*σ*-algebra* $\mathcal{G}$ *is* P-***trivial*** *if for all* $A \in \mathcal{G}$, $\mathrm{P}(A) \in \{0, 1\}$.

Definition 2.20
Let $\mathcal{F}_n$ *be a sequence of* σ*-algebras. The* ***tail*** σ***-alegra*** $\mathcal{T}$ *is defined as* $\mathcal{T} = \cap_n \sigma(\mathcal{F}_n, \ldots)$.

Theorem 2.21 (Kolmogorov Zero-One Law)
Let $\mathcal{F}_i$ *be independent* σ*-algebras,* $\mathcal{G}_n = \sigma(\mathcal{F}_n, \ldots)$ *and the tail* σ*-algebra* $\mathcal{G}_\infty = \cap_{n=1}^{\infty} \mathcal{G}_n$. *Then* $\mathcal{G}_\infty$ *is* P*-trivial.*

Proof. Observe that a σ-algebra $\mathcal{G}$ satisfies that for $A \in \mathcal{G}$, $\mathrm{P}(A) \in \{0, 1\}$ if $\mathcal{G}$ is independent of itself. Indeed, if $\mathcal{G}$ is independent of itself, then for any $A \in \mathcal{G}$,

$$\mathrm{P}(A) = \mathrm{P}(A \cap A) = \mathrm{P}(A)^2$$

implies that $\mathrm{P}(A) = 0$ or 1. Now, for any given n, $\sigma(\mathcal{F}_1, \ldots, \mathcal{F}_{n-1})$ is independent with $\mathcal{G}_n$ and $\mathcal{G}_\infty \subset \mathcal{G}_n$. Hence $\mathcal{G}_\infty$ is independent of $\sigma(\mathcal{F}_1, \ldots, \mathcal{F}_{n-1})$ for all n.

In particular, $\mathcal{G}_\infty$ is independent of $\sigma(\cup_n \mathcal{F}_n)$. To see this, note that $\cup_n \sigma(\cup_{k=1}^n \mathcal{F}_k)$ is a π system that generates $\sigma(\cup_n \mathcal{F}_n)$ and for $A \in \cup_n \sigma(\cup_{k=1}^n \mathcal{F}_k)$, $A \in \sigma(\cup_{k=1}^n \mathcal{F}_k)$ for some n. For $B \in \mathcal{G}_\infty$,

$$\mathrm{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$$

since $\sigma(\cup_{k=1}^n \mathcal{F}_k)$ and $\mathcal{G}_\infty$ is independent. Now it follows from theorem 1.52 that $\mathcal{G}_\infty$ and $\sigma(\cup_n \mathcal{F}_n)$ are independent. Notice that $\mathcal{G}_\infty \subset \sigma(\cup_n \mathcal{F}_n)$ and hence $\mathcal{G}_\infty$ is independent of itself. The proof is complete. ∎

Corollary 2.22
Let X_i *be independent and identically distributed and put* $S_n = \sum_{i=1}^n X_i$. *Then*

(a) S_n *is either almost surely convergent or almost surely divergent.*

(b) If $\frac{1}{n} S_n$ *converges almost surely, its limit is almost surely a constant.*

Proof. Define $\mathcal{F}_i = \sigma(X_i)$ and note that $\{S_n \text{ converges}\} \in \mathcal{G}_\infty = \cap_n \sigma(\cup_{i \geq n} \mathcal{F}_i)$ since

$$\{S_n \text{ converges}\} = \left\{ \lim_{n \to \infty} \sum_{i \geq n} X_i = 0 \right\}$$

is $\mathcal{G}_\infty$-measurable. By the Kolmogorov zero-one law,

$$\mathrm{P}\{S_n \text{ converges}\} \in \{0, 1\}.$$

This proves (a).

For (b), note that by a similar argument, we have

$$\left\{\frac{1}{n}S_n \text{ converges}\right\} \in \mathcal{G}_\infty,$$

where $\mathcal{G}_\infty$ is P-trivial. Also, $\lim_{n\to\infty}\frac{1}{n}S_n$ is $\mathcal{G}_\infty$-measurable. Since $\mathcal{G}_\infty$ is P-trivial,

$$F(t) \coloneqq \mathrm{P}\left\{\lim_{n\to\infty}\frac{1}{n}S_n \le t\right\} \in \{0,1\}.$$

Thus

$$\lim_{n\to\infty}\frac{1}{n}S_n = \sup\{t \mid F(t) = 0\}$$

almost surely, proving that the limit is almost surely a constant. ∎

Theorem 2.23 (Uniform Law of Large Number)
Let $Z_i \overset{iid}{\sim} \mu_Z$ and $\Theta \subset \mathbb{R}^n$ be compact. Suppose that $g(Z,\theta)$ is continuous in $\theta \in \Theta$ Z-almost surely and there is $F(z) \in \mathcal{L}^1(\mu_Z)$ such that $|g(Z,\theta)| \le F(Z)$ for all $\theta \in \Theta$. Then

(a) The mapping $\theta \mapsto \mathrm{E}[g(Z,\theta)]$ is continuous.

(b)

$$\sup_{\theta\in\Theta}\left|\frac{1}{n}\sum_{i=1}^{n} g(Z_i,\theta) - \mathrm{E}[g(Z,\theta)]\right| \to 0$$

almost surely.

Proof. For almost every Z, $g(Z,\theta)$ is continuous in θ. Hence for every $\epsilon > 0$, there is a δ such that $|g(Z,\theta') - g(Z,\theta)| < \epsilon$ if $|\theta' - \theta| < \delta$. Now

$$|\mathrm{E}[g(Z,\theta')] - \mathrm{E}[g(Z,\theta)]| \le \mathrm{E}[|g(Z,\theta') - g(Z,\theta)|] < \epsilon.$$

This proves (a).

For (b), we claim that $\frac{1}{n}\sum_{i=1}^{n} F(Z_i) < C(\{Z_i\})$ for almost every $\{Z_i\}$. Suppose not, then there is a set of $\{Z_i\}$ with positive probability such that there is a subsequence n_j with

$$\frac{1}{n_j}\sum_{i=1}^{n_j} F(Z_i) \to \infty.$$

Then this implies that

$$\mathrm{E}[F(Z)] = \mathrm{E}\left[\frac{1}{n_j}\sum_{i=1}^{n_j} F(Z_i)\right] \to \infty,$$

a contradiction. Also, for almost every $\{Z_i\}$,

$$\left|\frac{1}{n}\sum_{i=1}^{n} g(Z_i,\theta)\right| \le \frac{1}{n}\sum_{i=1}^{n} F(Z_i) < C(\{Z_i\})$$

is bounded uniformly in θ.

Next, we claim that for almost every $\{Z_i\}$, $n^{-1}\sum_{i=1}^n g(Z_i,\theta)$ is equicontinuous. To see this, notice that $g(Z_i,\theta)$ is uniformly continuous. Thus the modulus of continuity

$$\omega(Z_i,\delta) = \sup_{|\theta-\theta'|\le\delta} |g(Z_i,\theta) - g(Z_i,\theta')| \to 0$$

as $\delta \to 0$. Also, $\omega(Z_i,\delta) \le 2F(Z_i)$. By the LDCT,

$$\lim_{\delta\to 0} \mathrm{E}\left[\omega(Z_i,\delta)\right] = \mathrm{E}\left[\lim_{\delta\to 0}\omega(Z_i)\right] = 0.$$

Now

$$\sup_{|\theta-\theta'|\le\delta}\left|\frac{1}{n}\sum_{i=1}^n g(Z_i,\theta) - \frac{1}{n}\sum_{i=1}^n g(Z_i,\theta')\right| \le \frac{1}{n}\sum_{i=1}^n \omega(Z_i,\delta) \to \mathrm{E}\left[\omega(Z_i,\delta)\right]$$

by the strong law of large number. Hence for all $\epsilon > 0$, we can select δ such that $\mathrm{E}\left[\omega(Z_i,\delta)\right] < \epsilon$ and N such that for all $n \ge N$,

$$\left|\sup_{|\theta-\theta'|\le\delta}\left|\frac{1}{n}\sum_{i=1}^n g(Z_i,\theta) - \frac{1}{n}\sum_{i=1}^n g(Z_i,\theta')\right| - \mathrm{E}\left[\omega(Z_i,\delta)\right]\right| < \epsilon.$$

The δ is fixed for all $n \ge N$ and since $g(Z_i,\theta)$ are unifomly continuous for $i = 1,\dots,N$, we see that $\frac{1}{n}\sum_{i=1}^n g(Z_i,\theta)$ is equicontinuous for almost every $\{Z_i\}$.

Suppose that $n^{-1}\sum_{i=1}^n g(Z_i,\theta)$ does not converges uniformly to $\mathrm{E}\left[g(Z,\theta)\right]$. Then there is a subsequence n_j such that

$$\sup_{\theta\in\Theta}\left|n_j^{-1}\sum_{i=1}^{n_j} g(Z_i,\theta) - \mathrm{E}\left[g(Z,\theta)\right]\right| > \epsilon.$$

for all n_j. By the Arzela-Ascoli theorem, there is a further subsequence that converges uniformly. By the strong law of large number, for given θ,

$$n^{-1}\sum_{i=1}^n g(Z_i,\theta) \to \mathrm{E}\left[g(Z,\theta)\right]$$

almost surely. Hence the further subsequence converges to $\mathrm{E}\left[g(Z,\theta)\right]$, posing a contradiction. Thus

$$n^{-1}\sum_{i=1}^n g(Z_i,\theta) \to \mathrm{E}\left[g(Z,\theta)\right]$$

almost surely. ∎

2.2. Convergence in Distribution

Definition 2.24

*Let μ_n and μ be probability measures on (S, d). We say that $\mu_n \to \mu$ **in distribution** or **weakly** if for all $f \in C_b(S, \mathbb{R})$,*

$$\int f d\mu_n \to \int f d\mu.$$

We denote the convergence as $\mu_n \xrightarrow{d} \mu$.

Remark

The notion of convergence is the smallest topology such that the linear functional of the form $\ell_f : \mu \to \int f d\mu$ where $f \in C_b(S)$ is continuous.

Definition 2.25

Let X_n and X be random variables. $X_n \xrightarrow{d} X$ if the corresponding distributions $\mu_n \xrightarrow{d} \mu$.

Remark

The convergence in distribution of the random variables does not depend on the space where the random variables are defined; in fact, they can be defined on different spaces. The definition can also be written as

$$\mathrm{E}\left[f(X_n)\right] \to \mathrm{E}\left[f(X)\right]$$

for all $f \in C_b(S)$.

Theorem 2.26 (Scheffé)

If $f_n : S \to \mathbb{R}$ are density functions such that $f_n \to f$ almost everywhere, where f is a density function, then

$$\sup_{B \in \mathcal{B}} \left| \int_B f_n dx - \int_B f dx \right| \to 0$$

as $n \to 0$. In particular, when $S = \mathbb{R}$, taking $B = [-\infty, x]$ gives the uniform convergence of the CDFs.

Proof. Since

$$\sup_{B \in \mathcal{B}} \left| \int_B f_n dx - \int_B f dx \right| \leq \sup_{B \in \mathcal{B}} \int_B |f_n - f| \, dx \leq \int |f_n - f| \, dx,$$

the theorem follows once we prove that $f_n \to f$ in $\mathcal{L}^1$. Now, since $|f_n - f| \to 0$ almost everywhere and

$$|f_n - f| \leq |f_n| + |f| \quad \Rightarrow \quad 0 \leq |f_n| + |f| - |f_n - f| \, .$$

By the assumptions that f_n and f are density functions,

$$\int f_n dx = 1 = \int f dx.$$

By the Fatou's lemma,

$$\begin{aligned} 2\int |f|\,dx &= \int \liminf_{n\to\infty} |f_n| + |f| - |f_n - f| \le \liminf_{n\to\infty} \int f_n dx + \int f dx - \int |f_n - f|\,dx \\ &= 2\int f dx - \limsup_{n\to\infty} \int |f_n - f|\,dx. \end{aligned}$$

Hence

$$\limsup_{n\to\infty} \int |f_n - f|\,dx \le 0 \quad \Rightarrow \quad \int |f_n - f|\,dx \to 0.$$

Hence $f_n \to f$ in $\mathcal{L}^1$ and the proof is complete. ∎

Proposition 2.27
Let X_n, X be random variables on a separable metric space (S, d). Then $X_n \xrightarrow{p} X$ implies $X_n \xrightarrow{d} X$.

Proof. Let $X_n \xrightarrow{p} X$. To show that $X_n \xrightarrow{d} X$, we need to show that $\mathrm{E}[f(X_n)] \to \mathrm{E}[f(X)]$. Suppose that this does not hold. There is a subsequence X_{n_k} such that $X_{n_k} \to X$ almost surely by theorem 2.8. By continuity we have $f(X_{n_k}) \to f(X)$ almost surely. Since f is bounded, the bounded convergence theorem implies that $\mathrm{E}\left[f(X_{n_k})\right] \to \mathrm{E}[f(X)]$, contradicting to our hypothesis. Hence $\mathrm{E}[f(X_n)] \to \mathrm{E}[f(X)]$ and $X_n \xrightarrow{d} X$. ∎

Theorem 2.28 (Skorokhod Representation)
Suppose $\mu_n \xrightarrow{d} \mu$ on $\mathbb{R}$. Then there are corresponding random variables X_n, X for μ_n and μ such that $X_n \sim \mu_n$, $X \sim \mu$ and $X_n \to X$ almost surely.

Proof. Let F_n and F be the CDFs for μ_n and μ, respectively. Take $\Omega = [0,1]$, $\mathcal{F} = \mathcal{B}$ and P be the Lebesgue measure on $[0,1]$. Put

$$X_n(\omega) = \sup\{x \in \mathbb{R} \mid F_n(x) < \omega\} \quad \text{and} \quad X(\omega) = \sup\{x \in \mathbb{R} \mid F(x) < \omega\}$$

with the convention that $\sup \varnothing = -\infty$. Then

$$\mathrm{P}\{X_n \le x\} = \mathrm{P}\{\omega \mid \omega \le F_n(x)\} = F_n(x) \quad \text{and} \quad \mathrm{P}\{X \le x\} = \mathrm{P}\{\omega \mid \omega \le F(x)\} = F(x).$$

It now suffices to show that $X_n \to X$ almost surely. Indeed, since F_n, F are CDFs, there are only at most countable discontinuities. Let ω be a point of continuity of X. We may find another continuity point such that $F(y) < \omega$. The convergence in distribution implies that $F_n(y) \to F(y)$. Hence for n large enough, we have $F_n(y) < \omega$ and hence $X_n(\omega) > y$. Thus

$$\liminf_{n\to\infty} X_n(\omega) \ge y$$

for all $y \le X(\omega)$. Thus

$$\liminf_{n\to\infty} X_n(\omega) \ge X(\omega).$$

Similarly, pick a continuity point y such that $F(y) \geq \omega$ would give

$$\limsup_{n\to\infty} X_n(\omega) \leq y$$

for all $y \geq X(\omega)$ and thus

$$\limsup_{n\to\infty} X_n(\omega) \leq X(\omega).$$

Combining the above results gives that $X_n \to X$ almost surely, since X is continuous almost surely. ■

Corollary 2.29
Let $g \geq 0$ be a continuous measurable function on $\mathbb{R}$ and $X_n \xrightarrow{d} X$. Then

$$\mathrm{E}\left[g(X)\right] \leq \liminf_{n\to\infty} \mathrm{E}\left[g(X_n)\right].$$

Proof. Let Y_n and Y be the Skorokhod representations for X_n and X, respectively. Since now $g(Y_n) \to g(Y)$ almost surely, the Fatou's lemma shows that

$$\mathrm{E}\left[g(X)\right] = \mathrm{E}\left[g(Y)\right] \leq \liminf_{n\to\infty} \mathrm{E}\left[g(Y_n)\right] = \liminf_{n\to\infty} \mathrm{E}\left[g(X_n)\right].$$

■

Theorem 2.30 (Helly-Bray)
Suppose that X_n and X are $\mathbb{R}$-valued random variables. Then $X_n \xrightarrow{d} X$ if and only if

$$\mathrm{E}\left[g(X_n)\right] \to \mathrm{E}\left[g(X)\right]$$

for all $g \in C_b(\mathbb{R})$.

Proof. Assume first that $X_n \xrightarrow{d} X$. By the Skorokhod representation theorem, we may assume that X_n and X are defined on the same probability space and $X_n \to X$ almost surely. Now, since for all $g \in C_b(\mathbb{R})$, $g(X_n) \to g(X)$ almost surely and are uniformly bounded, the bounded convergence theorem implies that

$$\mathrm{E}\left[g(X_n)\right] \to \mathrm{E}\left[g(X)\right].$$

Conversely, suppose that $\mathrm{E}\left[g(X_n)\right] \to \mathrm{E}\left[g(X)\right]$ for all $g \in C_b(\mathbb{R})$. Let F_n and F be the distribution functions for X_n and X respectively and x be a continuity point of F. For $\epsilon > 0$, consider

$$g_\epsilon(y) = \begin{cases} 1 & y \leq x \\ 1 - \frac{y-x}{\epsilon} & x < y \leq x + \epsilon \\ 0 & y \geq x + \epsilon. \end{cases}$$

Clearly $g_\epsilon \in C_b(\mathbb{R})$. Let $g(y) = \mathbf{1}\{y \le x\}$.

$$\limsup_{n\to\infty} F_n(x) = \limsup_{n\to\infty} \mathrm{E}\left[g(X_n)\right] \le \limsup_{n\to\infty} \mathrm{E}\left[g_\epsilon(X_n)\right] = \mathrm{E}\left[g_\epsilon(X)\right] \le F(x+\epsilon).$$

Since ϵ is arbitrary and F is continuous at x, we have

$$\limsup_{n\to\infty} F_n(x) \le F(x).$$

On the other hand,

$$\liminf_{n\to\infty} F_n(x) = \liminf_{n\to\infty} \mathrm{E}\left[g(X_n)\right] \ge \liminf_{n\to\infty} \mathrm{E}\left[g_\epsilon(X_n+\epsilon)\right] = \mathrm{E}\left[g_\epsilon(X+\epsilon)\right] \ge F(x+\epsilon) \ge F(x).$$

Hence $F_n(x) \to F(x)$ and the proof is complete. ■

Remark
The theorem gives an equivalent definition for the convergence in distribution in $\mathbb{R}$*-valued case.*

Theorem 2.31 (Continuous Mapping Theorem)
Let $X_n \xrightarrow{d} X$ *in* $\mathbb{R}$ *and* g *be a measurable function continuous* μ_X*-almost surely. Then* $g(X_n) \xrightarrow{d} g(X)$.

Proof. By the Skorokhod representation theorem, we may assume that X_n and X are on the same space and $X_n \to X$ almost surely. By the continuity of g, we have $g(X_n) \to g(X)$ almost surely. For all $f \in C_b(\mathbb{R})$, $f(g(X_n)) \to f(g(X))$ almost surely as well. Since $f \circ g$ is bounded, the bounded convergence theorem gives $\mathrm{E}\left[f(g(X_n))\right] \to \mathrm{E}\left[f(g(X))\right]$. By the Helly-Bray theorem, this implies that $g(X_n) \xrightarrow{d} g(X)$. ■

Remark
If g *is bounded, then* $\mathrm{E}\left[g(X_n)\right] \to \mathrm{E}\left[g(X)\right]$ *directly by applying bounded convergence theorem on* $g(X_n)$ *and* $g(X)$.

Example
$X_n \sim U[-\frac{1}{n}, \frac{1}{n}] \xrightarrow{d} \delta_0$. *Let* $g(x) = \mathbf{1}\{x \ge 0\}$. *Then* $g(X_n) \sim Ber(\frac{1}{2}) \not\xrightarrow{d} g(X) \sim \delta_1$.

Definition 2.32
A metric space (S, d) *is* ***Polish*** *if it is complete and separable.*

Theorem 2.33 (Portmanteau)
Suppose that (S, d) *is a Polish space. Let* $X_n, X : \Omega \to S$ *be random variables. Then the followings are equivalent:*

(a) $X_n \xrightarrow{d} X$.

(b) If $G \subset S$ *is an open set, then* $\liminf_{n\to\infty} \mathrm{P}(X_n \in G) \ge \mathrm{P}(X \in G)$.

(c) If $F \subset S$ is a closed set, then $\limsup_{n\to\infty} \mathrm{P}(X_n \in F) \le \mathrm{P}(X \in F)$.

(d) If $A \subset S$ satisfies $\mathrm{P}(X \in \partial A) = 0$*, then* $\lim_{n\to\infty} \mathrm{P}(X_n \in A) = \mathrm{P}(X \in A)$.

Proof. We start from proving (a) implies (b). Let $G \subset S$ be an open set. For any continuous function with $0 \le f \le \mathbf{1}_G$, we have $\mathrm{E}[f(X_n)] \to \mathrm{E}[f(X)]$ and $\mathrm{E}[f(X_n)] \le \mathrm{P}(X_n \in G)$. Hence

$$\mathrm{E}[f(X)] \le \liminf_{n\to\infty} \mathrm{P}(X_n \in G).$$

Take $f \nearrow \mathbf{1}_G$ and by LMCT, since $f(X) \nearrow \mathbf{1}_G(X)$, $\mathrm{E}[\mathbf{1}_G(X)] \le \liminf_{n\to\infty} \mathrm{P}(X_n \in G)$.

To see that (b) and (c) are equivalent, note that if $\liminf_{n\to\infty} \mathrm{P}(X_n \in G) \ge \mathrm{P}(X \in G)$ for all open G, for every closed F,

$$\begin{aligned}\limsup_{n\to\infty} \mathrm{P}(X_n \in F) &= -\liminf_{n\to\infty} -\mathrm{P}(X_n \in F) = 1 - \liminf_{n\to\infty} \mathrm{P}(X_n \in F^c) \\ &\le 1 - \mathrm{P}(X \in F^c) = \mathrm{P}(X \in F).\end{aligned}$$

Conversely, if $\limsup_{n\to\infty} \mathrm{P}(X_n \in F) \le \mathrm{P}(X \in F)$ for every closed F, then

$$\liminf_{n\to\infty} \mathrm{P}(X_n \in G) = 1 - \limsup_{n\to\infty} \mathrm{P}(X_n \in G^c) \ge 1 - \mathrm{P}(X \in G^c) = \mathrm{P}(X \in G).$$

Now, assume that (b) and (c) holds. If $A \subset S$ satisfies that $\mathrm{P}(X \in \partial A) = 0$, then we have $\mathrm{P}(X \in \overline{A}) = \mathrm{P}(X \in A^\circ)$. Now,

$$\begin{aligned}\mathrm{P}(X \in A^\circ) &\le \liminf_{n\to\infty} \mathrm{P}(X_n \in A^\circ) \le \liminf_{n\to\infty} \mathrm{P}(X_n \in A) \\ &\le \limsup_{n\to\infty} \mathrm{P}(X_n \in A) \le \limsup_{n\to\infty} \mathrm{P}(X_n \in \overline{A}) \le \mathrm{P}(X \in \overline{A}) = \mathrm{P}(X \in A^\circ).\end{aligned}$$

Hence $\lim_{n\to\infty} \mathrm{P}(X_n \in A) = \mathrm{P}(X \in A)$.

Finally, assume (d) holds. We prove that (d) implies (c) and (b) implies (a). For any closed F, consider the closed ϵ-neighborhood

$$F_\epsilon = \{s \in S \mid d(F, s) \le \epsilon\}.$$

Now $\partial F_\epsilon = \{s \in S \mid d(F, s) = \epsilon\}$ are disjoint for distinct $\epsilon > 0$. Since X can accumulate mass on only countably many such sets, $\mathrm{P}(X \in \partial F_\epsilon) = 0$ for almost every ϵ. Picking $\epsilon_k \to 0$ such that $\mathrm{P}(X \in \partial F_{\epsilon_k}) = 0$, we have $\mathrm{P}(X_n \in F) \le \mathrm{P}(X_n \in F_\epsilon)$. Taking $n \to \infty$,

$$\limsup_{n\to\infty} \mathrm{P}(X_n \in F) \le \limsup_{n\to\infty} \mathrm{P}(X_n \in F_{\epsilon_k}) = \mathrm{P}(X \in F_{\epsilon_k}).$$

Now taking $\epsilon_k \to 0$,

$$\limsup_{n\to\infty} \mathrm{P}(X_n \in F) \le \mathrm{P}(X \in F).$$

To see that (b) implies (a), let $f \in C_b(S)$ with $f \geq 0$. By lemma 2.12, (b), and Fatou's lemma,

$$\begin{aligned}\mathrm{E}\left[f(X)\right] &= \int_0^\infty \mathrm{P}(f(X) > t)dt \leq \int_0^\infty \liminf_{n\to\infty} \mathrm{P}(f(X_n) > t)dt \\ &\leq \liminf_{n\to\infty} \int_0^\infty \mathrm{P}(f(X_n) > t)dt = \liminf_{n\to\infty} \mathrm{E}\left[f(X_n)\right].\end{aligned}$$

Suppose $f \leq C$, replacing f with $C - f$ gives

$$C - \mathrm{E}\left[f(X)\right] \leq \liminf_{n\to\infty} C - \mathrm{E}\left[f(X_n)\right] = C - \limsup_{n\to\infty} \mathrm{E}\left[f(X_n)\right].$$

Hence $\mathrm{E}\left[f(X)\right] \geq \limsup_{n\to\infty} \mathrm{E}\left[f(X_n)\right]$. We see that $\mathrm{E}\left[f(X)\right] = \lim_{n\to\infty} \mathrm{E}\left[f(X_n)\right]$ for $f \geq 0$, $f \in C_b(S)$. For general f, write $f = f^+ - f^-$ and the conclusion follows by applying above arguments to f^+ and f^-. ∎

Example

For a random variable X on $\mathbb{R}^d$, consider the perturbated version $X_n = X + \sigma_n Z$ with $\sigma_n \searrow 0$ and Z being independent of X, $\mu_Z \ll \lambda$ where λ is the Lebesgue measure. We claim that $X_n \xrightarrow{d} X$. For any open G and $\epsilon > 0$, define

$$G_{-\epsilon} = \left\{x \in \mathbb{R}^d \;\middle|\; \overline{B_\epsilon(x)} \subset G\right\}.$$

Then

$$\mathrm{P}(X_n \in G) \geq \mathrm{P}(X \in G_{-\epsilon}, \|\sigma_n Z\| \leq \epsilon) = \mathrm{P}(X \in G_{-\epsilon})\,\mathrm{P}(\|\sigma_n Z\| \leq \epsilon).$$

Taking $n \to \infty$, $\liminf_{n\to\infty} \mathrm{P}(X_n \in G) \geq \mathrm{P}(X \in G_{-\epsilon})$. Then taking $\epsilon \to 0$ gives

$$\liminf_{n\to\infty} \mathrm{P}(X_n \in G) \geq \mathrm{P}(X \in G).$$

By the portmanteau theorem, $X_n \xrightarrow{d} X$.

Theorem 2.34 (Helly's Selection)

For every sequence of distribution F_n on $\mathbb{R}$, there is a subsequence $F_{n_k} \to F$ pointwise at continuities where F is a non-decreasing, right-continuous function.

Proof. Since $\mathbb{R}$ is separable, there is a countable dense subset D. For $x_1 \in D$, pick a subsequence F_{n_1} such that F_{n_1} converges at x_1. This is possible due to the Bolzano-Weierstrass theorem. Proceed with $x_2 \in D$ and extract subsequence from F_{n_1}. Continue this process and take the diagonal. We obtain a subsequence F_{n_k} such that for every $x_m \in D$, $F_{n_k}(x_m)$ converges. For general $x \in \mathbb{R}$, define

$$F(x) = \inf_{y\in D, y>x} \lim_{k\to\infty} F_{n_k}(y)$$

where $x_m \in D$ is such that $x_m \nearrow x$. It is clear that F is non-decreasing.

Next, for each $x \in \mathbb{R}$ and $\epsilon > 0$, there is $y \in D$ with $y > x$ such that

$$F(y) = \lim_{k\to\infty} F_{n_k}(y) \leq F(x) + \epsilon.$$

Hence F is right-continuous.

If x is a continuity of F, then for $\epsilon > 0$ we can also choose $z \in D$ with $z < x$ such that $F(z) \geq F(x) - \epsilon$. Now

$$F_{n_k}(z) \leq F_{n_k}(x) \leq F_{n_k}(y).$$

Taking $k \to \infty$,

$$F(x) - \epsilon \leq F(z) \leq \liminf_{k\to\infty} F_{n_k}(x) \leq \limsup_{k\to\infty} F_{n_k}(x) \leq F(y) \leq F(x) + \epsilon.$$

Since ϵ is arbitrary, we conclude that $F_{n_k}(x) \to F(x)$. Hence F is the desired function. ■

Remark

A sequence of distribution function F_n converges pointwise at continuities to F does not imply that F is a distribution function. One may consider $F_n(x) = \mathbf{1}\{x \geq n\}$. As $n \to \infty$, F_n converges pointwise at continuities to $F = 0$.

Definition 2.35

*A collection of probability measures $\{\mu_\alpha\}$ on S is **uniformly tight** or simply **tight** if for every $\epsilon > 0$, there is a compact set $K \subset S$ such that $\mu_\alpha(K^c) < \epsilon$ for every α.*

Remark

An alternative definition is that

$$\liminf_{n\to\infty} \mu_n(B_r) \to 1$$

as $r \to \infty$ for $\{\mu_n\}$ with μ_n being defined on normed space.

Definition 2.36

*A sequence of probability measures $\{\mu_n\}$ on $\mathbb{R}$ is said to converge **vaguely** if*

$$\int f d\mu_n \to \int f d\mu$$

for all $f \in C_c(\mathbb{R})$.

Remark

Since $C_c(\mathbb{R}) \subset C_b(\mathbb{R})$, converging weakly implies converging vaguely. Also, the μ in the definition of the vague convergence is not necessarily a probability distribution. This is exactly what we see in the remark of Helly's selecetion theroem.

Theorem 2.37

Let μ_n be probability measures on $\mathbb{R}^d$ such that $\mu_n \xrightarrow{v} \mu$. Then the followings are equivalent:

(a) $\{\mu_n\}$ *is tight.*

(b) μ *is a probability measure, i.e.,* $\mu(\mathbb{R}^d) = 1$.

(c) $\mu_n \xrightarrow{d} \mu$.

Proof. Assuming (c), then

$$\int f d\mu_n \to \int f d\mu$$

for all $f \in C_b(\mathbb{R}^d)$. Take $f = 1$ shows (b).

Suppose that (b) holds. For any open ball $B_r \subset \mathbb{R}^d$, consider $f \in C_c(\mathbb{R}^d)$ with $0 \le f \le \mathbf{1}_{B_r}$. For every n, we have

$$\int f d\mu_n \le \mu_n(B_r).$$

Also,

$$\int f d\mu_n \to \int f d\mu \le \mu(B_r)$$

by the vague convergence. Thus

$$\int f d\mu = \liminf_{n\to\infty} \int f d\mu \le \liminf_{n\to\infty} \mu_n(B_r).$$

Taking $f \nearrow \mathbf{1}_{B_r}$ gives $\mu(B_r) \le \liminf_{n\to\infty} \mu_n(B_r)$. Now taking $r \to \infty$ gives $\liminf_{n\to\infty} \mu_n(B_r) \to 1$ as $r \to \infty$ and hence $\{\mu_n\}$ is tight.

Suppose (a) is true. Fix any $f \in C_b(\mathbb{R}^d)$. For every open ball B_r, consider $g_r \in C_c(\mathbb{R}^d)$ with $f|_{B_r} \le g_r \le f$. Then

$$\left|\int f d\mu_n - \int f d\mu\right| \le \left|\int g_r d\mu_n - \int g_r d\mu\right| + \int |f - g_r| \, d\mu_n + \int |f - g_r| \, d\mu.$$

Since $\{\mu_n\}$ is tight, for every $\epsilon > 0$ there is some M such that $r > M$ implies $\mu_n(B_r^c) < \epsilon$ and

$$\left|\int f d\mu_n - \int f d\mu\right| \le \left|\int g_r d\mu_n - \int g_r d\mu\right| + \|f\|_\infty \epsilon + \int |f - g_r| \, d\mu.$$

Taking $n \to \infty$ yields

$$\limsup_{n\to\infty} \left|\int f d\mu_n - \int f d\mu\right| \le \|f\|_\infty \epsilon + \int |f - g_r| \, d\mu.$$

Taking $r \to \infty$ gives $g_r \to f$ and

$$\limsup_{n\to\infty} \left|\int f d\mu_n - \int f d\mu\right| \le \|f\|_\infty \epsilon.$$

Since ϵ is arbitrary, we conclude that $\int f d\mu_n \to \int f d\mu$ and $\mu_n \xrightarrow{d} \mu$. ∎

Theorem 2.38 (Prokhorov)

Suppose μ_n is a sequence of probability measures on $\mathbb{R}$. $\{\mu_n\}$ is tight if and only if every subsequence of μ_n has a further subsequence converging weakly.

Proof. Suppose that $\{\mu_n\}$ is tight. By Helly's selection theorem, for every subsequence of μ_n we may extract a further subsequence converging vaguely to a measure μ. By the theorem 2.37, μ is a probability measure.

Conversely, suppose that $\{\mu_n\}$ is not tight. For every $\epsilon > 0$ and $k \in \mathbb{N}$, we can pick μ_{n_k} such that $\mu_{n_k}(B_k^c) \geq \epsilon$. For this sequence, we may extract a further subsequence $\mu_{n(k_j)}$ such that it converges weakly to μ and it follows from theorem 2.37 that $\left\{\mu_{n(k_j)}\right\}$ is tight. However $\mu_{n(k_j)}(B_{k_j}^c) \geq \epsilon$ for every k_j, posing a contradiction. Hence $\{\mu_n\}$ is tight. ∎

Proposition 2.39
Let X_n be a sequence of random variables. If there is $\phi(x) \geq 0$ such that $\phi(x) \to \infty$ as $\|x\| \to \infty$ and $\mathrm{E}\left[\phi(X_n)\right] \leq C$ for all n, then $\{X_n\}$ is tight.

Proof. Notice that

$$C \geq \mathrm{E}\left[\phi(X_n)\right] \geq \mathrm{E}\left[\phi(X_n)\mathbf{1}\left\{\|X_n\| \geq r\right\}\right] \geq \left(\inf_{\|x\| \geq r} \phi(x)\right) \mathrm{P}(\|X_n\| \geq r).$$

Hence

$$\mathrm{P}(\|X_n\| \geq r) \leq \frac{C}{\inf_{\|x\| \geq r} \phi(x)} \to 0$$

as $r \to \infty$. Thus $\{X_n\}$ tight. ∎

2.3. Characteristic Functions

Definition 2.40

Let X be an $\mathbb{R}$-valued random variable. The characteristic function of X is defined as

$$\varphi_X(t) = \mathrm{E}\left[e^{itX}\right] = \mathrm{E}\left[\cos(tX)\right] + i\,\mathrm{E}\left[\sin(tX)\right].$$

Remark

The characteristic function always exists for every $t \in \mathbb{R}$ since $x \mapsto \cos(tx)$ and $x \mapsto \sin(tx)$ are bounded functions.

Remark

*If X has distribution μ, we also write $\hat{\mu} = \varphi_X$. It is somtimes called the **Fourier transform** of the probability measure μ.*

Example

If $X \sim U[a,b]$, then

$$\varphi_X(t) = \mathrm{E}\left[\cos(tX)\right] + i\,\mathrm{E}\left[\sin(tX)\right] = \int_a^b \frac{\cos(tx)}{b-a}dx + i\int_a^b \frac{\sin(tx)}{b-a}dx = \frac{e^{ibt}-e^{iat}}{(b-a)it}.$$

Example

If

$$X = \begin{cases} n & \text{with prob. } \frac{1}{2} \\ -n & \text{with prob. } \frac{1}{2}, \end{cases}$$

then

$$\varphi_X(t) = \mathrm{E}\left[e^{itX}\right] = \frac{1}{2}e^{int} + \frac{1}{2}e^{-int} = \cos(nt).$$

Example

If $X \sim Poisson(\lambda)$, then

$$\varphi_X(t) = \sum_{k=0}^{\infty} e^{itk}e^{-\lambda}\frac{\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(e^{it}\lambda)^k}{k!} = \exp\big(\lambda\big(e^{it}-1\big)\big).$$

Proposition 2.41

Let X be an $\mathbb{R}$-valued random variable. Then the followings are true:

(a) $\varphi_X(0) = 1$.

(b) $\varphi_X(-t) = \overline{\varphi_X(t)}$.

(c) $|\varphi_X(t)| \le 1$.

(d) $\varphi_{aX+b}(t) = e^{ibt}\varphi_X(at)$.

(e) If X and Y are independent, then $\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$.

Proof. (a) is trivial. For (b), $\varphi_X(-t) = \mathrm{E}\left[e^{-itX}\right] = \mathrm{E}\left[\cos(tX)\right] - i\,\mathrm{E}\left[\sin(tX)\right] = \overline{\varphi_X(t)}$.

For (c), $|\varphi_X(t)| \leq \mathrm{E}\left[\left|e^{itX}\right|\right] \leq \mathrm{E}[1] = 1$.
For (d), $\varphi_{aX+b}(t) = \mathrm{E}\left[e^{iatX+ibt}\right] = e^{ibt}\,\mathrm{E}\left[e^{iatX}\right] = e^{ibt}\varphi_X(at)$.
For (e), $\varphi_{X+Y}(t) = \mathrm{E}\left[e^{itX+itY}\right] = \mathrm{E}\left[e^{itX}\right]\mathrm{E}\left[e^{itY}\right] = \varphi_X(t)\varphi_Y(t)$. ■

Remark

For (e), inductively we have that for independent variables X_n,

$$\varphi_{X_1+\cdots+X_n}(t) = \prod_{i=1}^{n} \varphi_{X_i}(t).$$

Example

$Z \sim N(0,1)$. Then

$$\begin{aligned}\varphi_Z(t) &= \mathrm{E}[\cos(tZ)] + i\,\mathrm{E}[\sin(tZ)] \\ &= \int \cos(tx)\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right)dx + i\int \sin(tx)\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right)dx \\ &= \int \cos(tx)\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right)dx.\end{aligned}$$

Now

$$\frac{\varphi_Z(t)-\varphi_Z(s)}{t-s} = \int \frac{\cos(tx)-\cos(sx)}{t-s}\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right)dx.$$

By the mean value theorem, $|\cos(tx)-\cos(sx)| \leq |\sin(c)|\,|tx-sx| \leq |x|\,|t-s|$ for some constant lying between tx and sx. Hence

$$\left|\frac{\cos(tx)-\cos(sx)}{t-s}\right|\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right) \leq |x|\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right)$$

which is integrable. It follows by LDCT that taking $t \to s$

$$\begin{aligned}\varphi_Z'(s) &= -\int \sin(sx)x\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right)dx = \int \sin(sx)d\left(\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right)\right) \\ &= -s\int \cos(sx)\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{x^2}{2}\right)dx = -s\varphi_Z(s).\end{aligned}$$

Solving the ODE with initial condition $\varphi_Z(0) = 1$ gives

$$\varphi_Z(t) = \exp\left(-\frac{t^2}{2}\right).$$

In general, if $X \sim N(\mu,\sigma^2)$, $X = \mu + \sigma Z$ and

$$\varphi_X(t) = \exp\left(i\mu t - \frac{\sigma^2 t^2}{2}\right).$$

Theorem 2.42 (Inversion Formula)
Let μ be a probability measure on $\mathbb{R}$ with characteristic function φ. Then

$$\lim_{T\to\infty} \frac{1}{2\pi} \int_{-T}^{T} \frac{e^{-ita} - e^{-itb}}{it} \varphi(t)dt = \mu(a,b) + \frac{1}{2}\mu\{a,b\}.$$

Proof. Put

$$g(T,\lambda) = \int_{-T}^{T} \frac{\sin(\lambda t)}{t} dt = \int_{0}^{T} \frac{\sin(\lambda t)}{t} dt.$$

Note that $g(T,\lambda) \to 2\,\mathrm{sgn}(\lambda)\frac{\pi}{2} = \pi\,\mathrm{sgn}(\lambda)$. Now

$$\begin{aligned}
f(T) &= \frac{1}{2\pi} \int_{-T}^{T} \frac{e^{-ita} - e^{-itb}}{it} \varphi(t)dt \\
&= \frac{1}{2\pi} \int_{-T}^{T} \int_{a}^{b} e^{-ity} \varphi(t)dydt \\
&= \frac{1}{2\pi} \int_{-T}^{T} \int_{a}^{b} \int e^{-it(y-x)} d\mu(x)dydt \\
&= \frac{1}{2\pi} \int \int_{-T}^{T} \int_{a}^{b} e^{-it(y-x)} dydtd\mu(x) \\
&= \frac{1}{2\pi} \int \int_{-T}^{T} \frac{\sin(t(b-x)) - \sin(t(a-x))}{t} dtd\mu(x) \\
&\quad + \frac{1}{2\pi} \int \int_{-T}^{T} \frac{[\cos(t(b-x)) - \cos(t(a-x))]}{t} dtd\mu(x),
\end{aligned}$$

where the fourth equality uses Fubini's theorem, which is valid since $[-T,T]\times[a,b]\times\mathbb{R}$ is of finite measure ($\mu(\mathbb{R}) = 1$) and $\left|e^{-it(y-x)}\right| \le 1$. Notice that

$$t \mapsto \frac{[\cos(t(b-x)) - \cos(t(a-x))]}{t}$$

is a bounded odd function. Hence

$$\begin{aligned}
f(T) &= \frac{1}{2\pi} \int \int_{-T}^{T} \frac{\sin(t(b-x)) - \sin(t(a-x))}{t} dtd\mu(x) \\
&= \frac{1}{2\pi} \int g(T, b-x) - g(T, a-x) d\mu(x).
\end{aligned}$$

Since $g(T,\lambda) \to \pi\,\mathrm{sgn}(\lambda)$,

$$g(T,b-x) - g(T,a-x) \to \begin{cases} 2\pi & \text{if } x \in (a,b) \\ \pi & \text{if } x \in \{a,b\} \\ 0 & \text{otherwise} \end{cases}$$

as $T \to \infty$. By the bounded convergence theorem,

$$f(T) \to \frac{1}{2\pi}(2\pi\mu(a,b) + \pi\mu\{a,b\}) = \mu(a,b) + \frac{1}{2}\mu\{a,b\}$$

as $T \to \infty$. ∎

Remark
The theorem establishes that the characteristic function is unique, i.e., if $\varphi_X = \varphi_Y$, then $X \overset{d}{=} Y$.

Corollary 2.43
Let X be a random variable on $\mathbb{R}$ with characteristic function φ_X. Then X is symmetric if and only if φ_X is real.

Proof. Suppose that X is symmetric. Then $\varphi_X(t) = \varphi_{-X}(t) = \varphi_X(-t) = \overline{\varphi_X(t)}$. Hence $\varphi_X(t)$ is real. Conversely, if φ_X is real, $\varphi_X(t) = \overline{\varphi_X(t)} = \varphi_X(-t) = \varphi_{-X}(t)$. By the uniqueness of characteristic functions, $X \overset{d}{=} -X$ and X is symmetric. ∎

Lemma 2.44
Let μ be a probability measure on $\mathbb{R}$ with characteristic function φ. Then for $r > 0$, we have

$$\mu\{x \mid |x| \geq r\} \leq \frac{r}{2}\int_{-2/r}^{2/r} 1 - \varphi(t)dt$$

Proof. By Fubini's theorem, since $(t,x) \mapsto \left|1 - e^{itx}\right|$ is integrable on $[-2/r, 2/r] \times \mathbb{R}$,

$$\begin{aligned}\int_{-2/r}^{2/r} 1 - \varphi(t)dt &= \int_{-2/r}^{2/r}\int 1 - e^{itx}d\mu(x)dt = \int\int_{-2/r}^{2/r} 1 - e^{itx}dtd\mu(x) \\ &= \frac{4}{r}\int 1 - \frac{\sin(2x/r)}{2x/r}d\mu(x) \geq \frac{2}{r}\mu\left\{x \;\middle|\; \left|\frac{2x}{r}\right| \geq 2\right\}\end{aligned}$$

since $\sin(y) \leq y/2$ for all $y \geq 2$. Rearrange the inequality

$$\mu\{x \mid |x| \geq r\} \leq \frac{r}{2}\int_{-2/r}^{2/r} 1 - \varphi(t)dt.$$

∎

Proposition 2.45
Let $\{\mu_\alpha\}$ be a set of probability measures on $\mathbb{R}$ and $\{\varphi_\alpha\}$ be their characteristic function. Then $\{\mu_\alpha\}$ is tight if and only if $\{\varphi_\alpha\}$ is equicontinuous at zero.

Proof. Suppose that $\{\varphi_\alpha\}$ is equicontinuous at zero. Note that $\varphi_\alpha(0) = 1$ for all α. For every $\epsilon > 0$, there is $r > 0$ such that $1 - \varphi_\alpha(t) < \epsilon$ for all $t \in [-2/r, 2/r]$. Then lemma 2.44 gives

$$\mu_\alpha\{x \mid |x| \geq r\} \leq \frac{r}{2}\int_{-2/r}^{2/r} 1 - \varphi_\alpha(t)dt \leq \frac{r}{2} \cdot 2 \cdot \frac{2}{r}\epsilon = 2\epsilon.$$

Since ϵ is arbitrary, $\{\mu_\alpha\}$ is tight.

Conversely, assume that $\{\mu_\alpha\}$ is tight. Let $X_\alpha \sim \mu_\alpha$ be the corresponding random variables. For any $t \in \mathbb{R}$,

$$|\mu_\alpha(t) - \mu_\alpha(0)| = \left|\mathrm{E}\left[e^{itX_\alpha} - 1\right]\right| \leq \mathrm{E}\left[\left|1 - e^{itX_\alpha}\right|\right] \leq \mathrm{E}\left[\min\{2, |tX_\alpha|\}\right],$$

where the last inequality is due to $\left|1 - e^{itX_\alpha}\right| \leq 2$ and

$$\left|1 - e^{itX_\alpha}\right| = \left|-i\int_0^{tX_\alpha} e^{is} ds\right| \leq \int_0^{|tX_\alpha|} \left|e^{is}\right| ds = |tX_\alpha|.$$

Since $\{X_\alpha\}$ is tight, for every $\epsilon > 0$ there is $M > 0$ such that $\mathrm{P}(|X_\alpha| \geq M) < \epsilon$. Hence

$$|\mu_\alpha(t) - \mu_\alpha(0)| \leq \mathrm{E}\left[\min\{2, |tX_\alpha|\}\right] \leq 2\,\mathrm{P}(|X_\alpha| \geq M) + 2M|t| \leq 2(\epsilon + M|t|).$$

Take $|t| \to 0$ and since ϵ is arbitrary, the equicontinuity follows. ∎

Theorem 2.46 (Levy's Continuity Theorem)
Suppose μ_n are random variables on $\mathbb{R}$ with characteristic function $\varphi_n(t)$. Then

(a) If $\mu_n \xrightarrow{d} \mu$, then $\varphi_n(t) \to \varphi(t)$ for all $t \in \mathbb{R}$ where φ is the characteristic function of μ.

(b) If $\varphi_n(t) \to \varphi(t)$ and $\varphi(t)$ is continuous at 0, then $\{\mu_n\}$ is tight and $\mu_n \xrightarrow{d} \mu$ where μ has characteristic function φ.

Proof. For (a), suppose that $\mu_n \xrightarrow{d} \mu$. Then for every $f \in C_b(\mathbb{R})$,

$$\int f d\mu_n \to \int f d\mu.$$

In particular, taking $f : x \mapsto \cos(tx)$ and $f : x \mapsto \sin(tx)$ shows that

$$\varphi_n(t) = \int \cos(tx) d\mu_n + i\int \sin(tx) d\mu_n \to \int \cos(tx) d\mu + i\int \sin(tx) d\mu = \varphi(t).$$

For (b), by lemma 2.44 and bounded convergence theorem,

$$\limsup_{n\to\infty} \mu_n\{x \mid |x| \geq r\} \leq \lim_{n\to\infty} \frac{r}{2}\int_{-2/r}^{2/r} 1 - \varphi_n(t) dt = \frac{r}{2}\int_{-2/r}^{2/r} 1 - \varphi(t) dt.$$

Since $\varphi(t)$ is continuous at 0,

$$\frac{r}{2}\int_{-2/r}^{2/r} 1 - \varphi(t) dt \to 0$$

as $r \to \infty$. This implies that $\{\mu_n\}$ is tight.

Now, suppose that μ_n does not converge weakly. Then we can find $\epsilon \geq 0$, $f \in C_b$ and a subsequence μ_{n_k} such that

$$\left|\int f d\mu_{n_k} - \int f d\mu\right| \geq \epsilon$$

for every k. By Prokhorov's theorem, we can extract a further subsequence converging weakly, which is a contradiction. Hence $\mu_n \xrightarrow{d} \mu$. ■

Theorem 2.47

Suppose that X is a random variable on $\mathbb{R}$ with $\mathrm{E}[|X|^n] < \infty$ and has characteristic function $\varphi(t)$. Then $\varphi \in C^n(\mathbb{R})$ and

$$\varphi^{(n)}(t) = \mathrm{E}\left[(iX)^n e^{itX}\right].$$

Proof. We prove the result for $n = 1$. The other orders follow inductively. For $t \in \mathbb{R}$,

$$\frac{\varphi(t+h) - \varphi(t)}{h} = \mathrm{E}\left[e^{itX}\frac{e^{ihX} - 1}{h}\right].$$

Note that

$$\left|e^{itx}\frac{e^{ihx} - 1}{h}\right| \leq \left|\frac{\cos(hx) - 1}{h}\right| + \left|\frac{\sin(hx)}{h}\right| \leq 2|x|.$$

Since $\mathrm{E}[2|X|] < \infty$, LDCT gives

$$\varphi'(t) = \lim_{h\to 0} \mathrm{E}\left[e^{itX}\frac{e^{ihX} - 1}{h}\right] = \mathrm{E}\left[e^{itX}(iX)\right].$$

■

Remark

The converse does not hold in general. We provide an example where $\varphi(x)$ is differentiable at 0 but $\mathrm{E}[|X|] = \infty$. Consider the random variable defined by $\mathrm{P}(X = \pm k) = \frac{c}{k^2 \log(k)}$ for $k \geq 2$ where c is some normalizing constant. Then

$$\mathrm{E}[|X|] = 2c\sum_{k=2}^{\infty}\frac{1}{k\log(k)}.$$

Since the integral

$$\int_2^{\infty}\frac{1}{x\log(x)}dx = \log(\log(x))|_2^{\infty} = \infty,$$

the first moment is not finite. However,

$$\varphi(t) = 2c\sum_{k=2}^{\infty}\frac{\cos(kt)}{k^2\log(k)}, \quad \textit{and} \quad \varphi'(0) = 2c\sum_{k=2}^{\infty}\frac{0}{k\log(k)} = 0.$$

Corollary 2.48 (Taylor Expansion)

Let φ be a characteristic function of a random variable X with $\mathrm{E}[|X|^n] < \infty$. Then as $t \to 0$,

$$\varphi(t) = \sum_{k=0}^{n}\frac{(it)^k\,\mathrm{E}\left[X^k\right]}{k!} + o(t^n).$$

Proof. By theorem 2.47, we can consider the Taylor expansion of φ up to order n at $t = 0$.

$$\varphi(t) = \sum_{k=0}^{n} \frac{\varphi^{(k)}(0) t^k}{k!} + o(t^n) = \sum_{k=0}^{n} \frac{\mathrm{E}\left[(iX)^k\right] t^k}{k!} + o(t^n) = \sum_{k=0}^{n} \frac{(it)^k \, \mathrm{E}\left[X^k\right]}{k!} + o(t^n)$$

as $t \to 0$. ∎

2.4. Central Limit Theorems

Theorem 2.49 (Central Limit Theorem I, Lindeberg-Levy)
Let X_i be independent and identically distributed random variables with $\mathrm{E}[X_i] = 0$ and $\mathrm{E}\left[X_i^2\right] = 1$. Let $X \sim N(0,1)$ and $S_n = \sum_{i \le n} X_i$. Then

$$\frac{1}{\sqrt{n}} S_n \xrightarrow{d} X$$

as $n \to \infty$.

Proof. Notice that $\varphi_X(t) = \exp(-t^2/2)$ and by corollary 2.48,

$$\varphi_{\frac{1}{\sqrt{n}}S_n}(t) = \prod_{i=1}^{n} \varphi_{X_i}\left(\frac{t}{\sqrt{n}}\right) = \left(1 - \frac{t^2}{2n} + o(1/n)\right)^n \to \exp(-t^2/2)$$

as $n \to \infty$ for fixed t. Hence $\frac{1}{\sqrt{n}} S_n \xrightarrow{d} X$ by the Levy's continuity theorem. ∎

Theorem 2.50 (Central Limit Theorem II, Lindeberg-Feller)
For each n, let $X_{n,i}$, $1 \le i \le n$ be random variables with mean 0 and $S_n = \sum_{i \le n} X_{n,i}$. Suppose

(a) For given n, $X_{n,i}$ are independent.

(b) $\mathrm{Var}\left[\sum_{i=1}^n X_{n,i}\right] = \sum_{i=1}^n \mathrm{E}\left[X_{n,i}^2\right] \to \sigma^2$. as $n \to \infty$.

(c) For all $\epsilon > 0$, $\lim_{n\to\infty} \sum_{i=1}^n \mathrm{E}\left[X_{n,i}^2 \mathbf{1}\left\{\left|X_{n,i}\right| > \epsilon\right\}\right] = 0$.

Then $S_n \xrightarrow{d} N(0, \sigma^2)$.

Proof. We begin the proof of the theorem by the following claim. For $t \in \mathbb{R}$, we have

$$\left| e^{it} - \sum_{k=0}^{n} \frac{(it)^k}{k!} \right| \le \min\left\{ \frac{2\,|t|^n}{n!}, \frac{|t|^{n+1}}{(n+1)!} \right\}.$$

To see this, let $g_n(t)$ be the difference on the left hand side. For $n = 0$, the inequality holds trivially. Now suppose that the inequality holds for n.

$$g_{n+1}(t) = e^{it} - \sum_{k=0}^{n+1} \frac{(it)^k}{k!} = i \int_0^t e^{is} - \sum_{k=0}^{n} \frac{(is)^k}{k!} ds = i \int_0^t g_n(s) ds.$$

Now

$$|g_{n+1}(t)| \le \int_0^t \frac{2\,|s|^n}{n!} ds = \frac{2\,|t|^{n+1}}{(n+1)!} \quad \text{and} \quad |g_{n+1}(t)| \le \int_0^t \frac{|s|^{n+1}}{(n+1)!} ds = \frac{|t|^{n+2}}{(n+2)!}.$$

The claim follows by induction.

Now notice that for given n, if $Z_{n,i} \sim N(0, \sigma_{n,i}^2)$ are independent with $\sigma_{n,i}^2 = \mathrm{E}\left[X_{n,i}^2\right]$, then $\sum_{i \le n} Z_{n,i} \xrightarrow{d} Z \sim N(0, \sigma^2)$. Let the characteristic functions for $Z_{n,i}$ be $\psi_{n,i}$. Note that for

complex numbers z_i and w_i with $|z_i|, |w_i| \leq 1$, we have

$$\left| \prod_{i \leq n} z_i - \prod_{i \leq n} w_i \right| \leq \sum_{i \leq n} |z_i - w_i| .$$

If $n = 2$, $|z_1 z_2 - w_1 w_2| \leq |z_1 z_2 - z_1 w_2| + |z_1 w_2 - w_1 w_2| \leq |z_2 - w_2| + |z_1 - w_1|$. The general case follows by induction. It follows that if $\varphi_{n,i}$ are characteristic functions for $X_{n,i}$,

$$\begin{aligned}
\left| \prod_{i \leq n} \varphi_{n,i}(t) - \prod_{i \leq n} \psi_{n,i}(t) \right| &\leq \sum_{i \leq n} \left| \varphi_{n,i}(t) - \psi_{n,i}(t) \right| \\
&\leq \sum_{i \leq n} \left| \varphi_{n,i}(t) - 1 + \frac{\sigma_{n,i}^2 t^2}{2} \right| + \sum_{i \leq n} \left| \psi_{n,i}(t) - 1 + \frac{\sigma_{n,i}^2 t^2}{2} \right| \\
&\leq \sum_{i \leq n} \mathrm{E} \left[\left| e^{itX_{n,i}} - 1 + \frac{X_{n,i}^2 t^2}{2} \right| \right] + \sum_{i \leq n} \mathrm{E} \left[\left| e^{itZ_{n,i}} - 1 + \frac{Z_{n,i}^2 t^2}{2} \right| \right] .
\end{aligned}$$

Now by the claim we have for all $\epsilon > 0$,

$$\begin{aligned}
\mathrm{E} \left[\left| e^{itX_{n,i}} - 1 + \frac{X_{n,i}^2 t^2}{2} \right| \right] &\leq \mathrm{E} \left[\min \left\{ \frac{2 \left| tX_{n,i} \right|^2}{2!}, \frac{\left| tX_{n,i} \right|^3}{3!} \right\} \right] \\
&\leq |t|^2 \, \mathrm{E} \left[\left| X_{n,i} \right|^2 \mathbf{1} \left\{ \left| X_{n,i} \right| > \epsilon \right\} \right] + \frac{|t|^3}{6} \mathrm{E} \left[\left| X_{n,i} \right|^3 \mathbf{1} \left\{ \left| X_{n,i} \right| \leq \epsilon \right\} \right] \\
&\leq |t|^2 \, \mathrm{E} \left[\left| X_{n,i} \right|^2 \mathbf{1} \left\{ \left| X_{n,i} \right| > \epsilon \right\} \right] + \frac{\epsilon \, |t|^3}{6} \mathrm{E} \left[\left| X_{n,i} \right|^2 \mathbf{1} \left\{ \left| X_{n,i} \right| \leq \epsilon \right\} \right] \\
&\leq |t|^2 \, \mathrm{E} \left[\left| X_{n,i} \right|^2 \mathbf{1} \left\{ \left| X_{n,i} \right| > \epsilon \right\} \right] + \frac{\epsilon \, |t|^3 \, \sigma_{n,i}^2}{6}
\end{aligned}$$

Let $\sigma_n^2 = \sum_{i \leq n} \sigma_{n,i}^2$. The claim applies to $Z_{n,i}$ gives

$$\begin{aligned}
\sum_{i \leq n} \mathrm{E} \left[\left| e^{itZ_{n,i}} - 1 + \frac{Z_{n,i}^2 t^2}{2} \right| \right] &\leq \frac{|t|^3}{6} \sum_{i \leq n} \mathrm{E} \left[\left| Z_{n,i} \right|^3 \right] = \frac{|t|^3}{6} \sum_{i \leq n} \sigma_{n,i}^3 \, \mathrm{E} \left[|Y|^3 \right] \\
&\leq \frac{|t|^3}{6} \, \mathrm{E} \left[|Y|^3 \right] \sigma_n^2 \sup_{i \leq n} \sigma_{n,i}
\end{aligned}$$

Hence

$$\begin{aligned}
\left| \prod_{i \leq n} \varphi_{n,i}(t) - \prod_{i \leq n} \psi_{n,i}(t) \right| &\leq \sum_{i \leq n} |t|^2 \, \mathrm{E} \left[\left| X_{n,i} \right|^2 \mathbf{1} \left\{ \left| X_{n,i} \right| > \epsilon \right\} \right] + \frac{\epsilon \, |t|^3 \, \sigma_{n,i}^2}{6} + \frac{|t|^3}{6} \, \mathrm{E} \left[|Y|^3 \right] \sigma_n^2 \sup_{i \leq n} \sigma_{n,i} \\
&\to \frac{\epsilon \, |t|^3}{6} \sigma^2 + \frac{|t|^3}{6} \, \mathrm{E} \left[|Y|^3 \right] \sigma^2 \epsilon
\end{aligned}$$

as $n \to \infty$ since

$$\sup_{i \le n} \sigma_{n,i} = \left(\sup_{i \le n} \sigma_{n,i}^2 \right)^{1/2} = \left(\sup_{i \le n} \mathrm{E}\left[X_{n,i}^2 \mathbf{1}\left\{ \left| X_{n,i} \right| > \epsilon \right\} \right] + \mathrm{E}\left[X_{n,i}^2 \mathbf{1}\left\{ \left| X_{n,i} \right| \le \epsilon \right\} \right] \right)^{1/2}$$
$$\le \left(\epsilon^2 + \sum_{i \le n} \mathrm{E}\left[X_{n,i}^2 \mathbf{1}\left\{ \left| X_{n,i} \right| > \epsilon \right\} \right] \right)^{1/2} \to \epsilon$$

as $n \to \infty$. Since ϵ is arbitrary, we conclude that $\left| \prod_{i \le n} \varphi_{n,i}(t) - \prod_{i \le n} \psi_{n,i}(t) \right| \to 0$ and $S_n \xrightarrow{d} N(0, \sigma^2)$. ∎

Corollary 2.51 (Central Limit Theorem III, Lyapunov)
Let X_j be independent with $\mathrm{E}\left[X_j\right] = 0$ and $\alpha_n^2 = \sum_{j \le n} \mathrm{Var}(X_j)$. Suppose that there is $\delta > 0$ such that $\mathrm{E}\left[\left| X_j \right|^{2+\delta} \right] < \infty$ and

$$\lim_{n \to \infty} \frac{1}{\alpha_n^{2+\delta}} \sum_{j \le n} \mathrm{E}\left[\left| X_j \right|^{2+\delta} \right] = 0.$$

Then

$$\frac{1}{\alpha_n} \sum_{j \le n} X_j \xrightarrow{d} N(0,1).$$

Proof. Put $Y_{n,j} = \alpha_n^{-1} X_j$. Note that

$$\mathrm{Var}\left[\sum_{j \le n} Y_{n,j} \right] = \frac{1}{\alpha_n^2} \cdot \alpha_n^2 = 1$$

and for $\epsilon > 0$,

$$\sum_{j \le n} \mathrm{E}\left[Y_{n,j}^2 \mathbf{1}\left\{ \left| Y_{n,j} \right| > \epsilon \right\} \right] = \frac{1}{\alpha_n^2} \sum_{j \le n} \mathrm{E}\left[X_{n,j}^2 \mathbf{1}\left\{ \left| X_{n,j} \right|^{\delta} > (\alpha_n \epsilon)^{\delta} \right\} \right] \le \frac{1}{\alpha_n^{2+\delta} \epsilon^{\delta}} \sum_{j \le n} \mathrm{E}\left[\left| X_{n,j} \right|^{2+\delta} \right] \to 0$$

as $n \to \infty$. Hence by the Lindeberg-Feller theorem, $\alpha_n^{-1} \sum_{j \le n} X_j \xrightarrow{d} N(0,1)$. ∎

Example
Let $X_j \sim Ber(p_j)$ be independent random variables and $S_n = \sum_{j \le n} X_j$. Then

$$\frac{S_n - \mathrm{E}\left[S_n\right]}{\sqrt{\mathrm{Var}(S_n)}} \xrightarrow{d} N(0,1)$$

if $\mathrm{Var}(S_n) \to \infty$. To see this, write $Y_j = X_j - p_j$ and $\sigma_j^2 = \mathrm{Var}(Y_j) = \mathrm{E}\left[Y_j^2\right] = p_j(1 - p_j)$. Let $\delta > 0$ be given. Since $\left| Y_j \right| \le 1$, $\mathrm{E}\left[\left| Y_j \right|^{2+\delta} \right] \le \mathrm{E}\left[Y_j^2 \right] = \sigma_j^2$ and

$$\frac{1}{\mathrm{Var}(S_n)^{2+\delta}} \sum_{j \le n} \mathrm{E}\left[\left| Y_j \right|^{2+\delta} \right] \le \frac{1}{\mathrm{Var}(S_n)^{2+\delta}} \sum_{j \le n} \sigma_j^2 = \mathrm{Var}(S_n)^{-\delta} \to 0.$$

By the Lyapunov theorem, we conclude that

$$\frac{S_n - \mathrm{E}[S_n]}{\sqrt{\mathrm{Var}(S_n)}} \xrightarrow{d} N(0,1).$$

Theorem 2.52

For given n, $X_{n,j} \sim Ber(p_{n,j})$ with $1 \le j \le n$ are independent. Let $S_n = \sum_{j\le n} X_{n,j}$. If

(a) $\sum_{j\le n} p_{n,j} \to \lambda \in (0,\infty)$

(b) $\max_{j\le n} p_{n,j} \to 0$,

then $S_n \xrightarrow{d} Poisson(\lambda)$.

Proof. Let $\varphi_{n,j}(t) = 1 + p_{n,j}(e^{it} - 1)$ be the characteristic functions for $X_{n,j}$. Then

$$\begin{aligned}\log(\varphi_{S_n}(t)) &= \sum_{j=1}^{n} \log(\varphi_{n,j}(t)) = \sum_{j=1}^{n} \log(1 + p_{n,j}(e^{it} - 1)) \\ &= \sum_{j=1}^{n} p_{n,j}(e^{it} - 1) + o(p_{n,j}) \to \lambda(e^{it} - 1),\end{aligned}$$

where we use the fact that $\log(1+x) = x + o(x)$ as $x \to 0$ and condition (b) implies that the Taylor expansion is valid. Hence $\varphi_{S_n}(t) \to \exp(\lambda(e^{it} - 1))$ and $S_n \xrightarrow{d} \mathrm{Poisson}(\lambda)$. ∎

Example

The independence assumption of the random variables can sometimes be relaxed. Consider the random permutations of $\{1,\dots,n\}$. We are interested in finding the distribution of the number of fixed points for the random permutations. Define $X_{n,j} = \mathbf{1}\{\pi(j) = j\}$ where π is a random permutation and $S_n = \sum_{j\le n} X_{n,j}$. Put $A_{n,j} = \{X_{n,j} = 1\}$. Then

$$\mathrm{P}(S_n > 0) = \mathrm{P}(\cup_{j=1}^{n} A_{n,j}) = \sum_j \mathrm{P}(A_{n,j}) - \sum_{k\le j} \mathrm{P}(A_{n,j} \cap \mathrm{P}(A_{n,k})) + \cdots + (-1)^{n+1}\,\mathrm{P}(A_{n,1} \cap \cdots \cap A_{n,n}).$$

Note that

$$\mathrm{P}(A_{n,i_1} \cap \cdots \cap A_{n,i_r}) = \frac{(n-r)!}{n!}.$$

Hence

$$\mathrm{P}(S_n > 0) = n \cdot \frac{(n-1)!}{n!} - \binom{n}{2}\frac{(n-2)!}{n!} + \cdots + (-1)^{n+1}\frac{1}{n!} = 1 - \frac{1}{2!} + \frac{1}{3!} - \cdots + (-1)^{n+1}\frac{1}{n!}.$$

Thus $\mathrm{P}(S_n = 0) = 1 - 1 + \frac{1}{2!} \cdots + (-1)^n \frac{1}{n!} \to e^{-1}$ *as* $n \to \infty$. *Fix k entries. The remaining $n-k$ entries with no fixed points has combinatorics a_{n-k}. Thus*

$$\mathrm{P}(S_n = k) = \binom{n}{k}\frac{a_{n-k}}{n!}, \quad \textit{and} \quad \mathrm{P}(S_{n-k} = 0) = \frac{a_{n-k}}{(n-k)!}.$$

So

$$\mathrm{P}(S_n = k) = \binom{n}{k} \frac{(n-k)!\,\mathrm{P}(S_{n-k} = 0)}{n!} \to \frac{1}{k!} e^{-1}$$

for given k. *We see that* $S_n \xrightarrow{d}$ *Poisson*(1).

3. Martingale

3.1. Conditional Expectation

Definition 3.1
Let $(\Omega, \mathcal{F}, \mathrm{P})$ *be a probability space and* $X \in \mathcal{L}^1(\Omega)$. *The* ***conditional expectation*** *of* X *with respect to a* σ*-algebra* $\mathcal{G} \subset \mathcal{F}$ *is a* $\mathcal{G}$*-measurable random variable* Z *such that*

$$\int_A Z d\,\mathrm{P} = \int_A X d\,\mathrm{P}$$

for every $A \in \mathcal{G}$.

Remark
Define

$$\nu(A) = \int_A X d\,\mathrm{P}\,.$$

It is clear that $\nu \ll \mathrm{P}$ *— in particular on* $\mathcal{G}$ *— and hence the Radon-Nikodym theorem implies that there is a almost surely unique random variable* $Z \in \mathcal{L}^1(\Omega, \mathcal{G}, \mathrm{P})$ *such that* $\nu(A) = \int_A Z d\,\mathrm{P}$. *Hence the conditional expectation is well-defined. We may denote the conditional expectation of* X *with respect to* $\mathcal{G}$ *as* $\mathrm{E}[X|\mathcal{G}]$. *For random variable* Y, *we write* $\mathrm{E}[X|Y] = \mathrm{E}[X|\sigma(Y)]$.

Proposition 3.2
Let $(\Omega, \mathcal{F}, P)$ *be a probability space and* $\mathcal{G} \subset \mathcal{F}$ *is a* σ*-algebra. The followings are true:*

(a) $\mathrm{E}[X|\mathcal{F}] = X$.

(b) If $\sigma(X)$ *is independent of* $\mathcal{G} \subset \mathcal{F}$, *then* $\mathrm{E}[X|\mathcal{G}] = \mathrm{E}[X]$.

(c) For $c \in \mathbb{R}$, $\mathrm{E}[cX + Y|\mathcal{G}] = c\,\mathrm{E}[X|\mathcal{G}] + \mathrm{E}[Y|\mathcal{G}]$.

(d) If $X \le Y$ *almost surely, then* $\mathrm{E}[X|\mathcal{G}] \le \mathrm{E}[Y|\mathcal{G}]$ *almot surely.*

(e) $\mathrm{E}[|\mathrm{E}[X|\mathcal{G}]|] \le \mathrm{E}[|X|]$.

(f) If f *is convex,* $f(\mathrm{E}[X|\mathcal{G}]) \le \mathrm{E}[f(X)|\mathcal{G}]$.

Proof. For (a), note that X is $\mathcal{F}$-measurable and for every $A \in \mathcal{F}$, $\mathrm{E}[X\mathbf{1}_A] = \mathrm{E}[X\mathbf{1}_A]$. Hence X satisfies the required condition for it to be the conditional expectation. $\mathrm{E}[X|\mathcal{F}] = X$ by the uniqueness of the conditional expectation.

For (b), since the constant is $\mathcal{G}$-measurable and $\mathrm{E}[X\mathbf{1}_A] = \mathrm{E}[X]\,\mathrm{E}[\mathbf{1}_A] = \mathrm{E}[\mathrm{E}[X]\,\mathbf{1}_A]$ for every $A \in \mathcal{G}$, we have $\mathrm{E}[X|\mathcal{G}] = \mathrm{E}[X]$.

For (c), it is clear that $c\,\mathrm{E}[X|\mathcal{G}] + \mathrm{E}[Y|\mathcal{G}]$ is measurable. Also, for every $A \in \mathcal{G}$,

$$\begin{aligned}\mathrm{E}[(cX + Y)\mathbf{1}_A] &= c\,\mathrm{E}[X\mathbf{1}_A] + \mathrm{E}[Y\mathbf{1}_A] = c\,\mathrm{E}[\mathrm{E}[X|\mathcal{G}]\,\mathbf{1}_A] + \mathrm{E}[\mathrm{E}[Y|\mathcal{G}]\,\mathbf{1}_A] \\ &= \mathrm{E}[(c\,\mathrm{E}[X|\mathcal{G}] + \mathrm{E}[Y|\mathcal{G}])\mathbf{1}_A]\,.\end{aligned}$$

Hence $\mathrm{E}[cX+Y|\mathcal{G}] = c\,\mathrm{E}[X|\mathcal{G}] + \mathrm{E}[Y|\mathcal{G}]$.

For (d), let $A = \{\mathrm{E}[Y|\mathcal{G}] - \mathrm{E}[X|\mathcal{G}] > 0\} = \{\mathrm{E}[Y-X|\mathcal{G}] > 0\}$. By the definition of conditional expectation, since $A \in \mathcal{G}$,

$$\mathrm{E}[(\mathrm{E}[Y|\mathcal{G}] - \mathrm{E}[X|\mathcal{G}])\mathbf{1}_A] = \mathrm{E}[\mathrm{E}[Y-X|\mathcal{G}]\,\mathbf{1}_A] = \mathrm{E}[(Y-X)\mathbf{1}_A] \geq 0$$

due to the assumption that $X \leq Y$ almost surely. If $\mathrm{P}(A) > 0$, we also have

$$\mathrm{E}[(\mathrm{E}[Y|\mathcal{G}] - \mathrm{E}[X|\mathcal{G}])\mathbf{1}_A] < 0,$$

which is a contradiction. Thus $\mathrm{P}(A) = 0$ and we conclude that $\mathrm{E}[X|\mathcal{G}] \leq \mathrm{E}[Y|\mathcal{G}]$ almost surely.

For (e), we may take $A = \{\mathrm{E}[X|\mathcal{G}] \geq 0\}$. Then

$$\mathrm{E}[|\mathrm{E}[X|\mathcal{G}]|] = \mathrm{E}[\mathrm{E}[X|\mathcal{G}]\,\mathbf{1}_A] - \mathrm{E}[\mathrm{E}[X|\mathcal{G}]\,\mathbf{1}_{A^c}] = \mathrm{E}[X\mathbf{1}_A] - \mathrm{E}[X\mathbf{1}_{A^c}] \leq \mathrm{E}[|X|],$$

proving (e).

For (f), note that by the convexity, there are a, b such that $f(x) \geq ax + b$ and $f(x_0) = x_0$ for some x_0. Taking $x_0 = \mathrm{E}[X|\mathcal{G}]$,

$$f(\mathrm{E}[X|\mathcal{G}]) = a\,\mathrm{E}[X|\mathcal{G}] + b = \mathrm{E}[aX+b|\mathcal{G}] \leq \mathrm{E}[f(X)|\mathcal{G}]$$

by (c) and (d). ∎

Proposition 3.3

Let $(\Omega, \mathcal{F}, \mathrm{P})$ be a probability space, $X_n, X \in \mathcal{L}^1(\Omega)$ and $\mathcal{G} \subset \mathcal{F}$ be a σ-algebra. Then the followings are true:

(a) If $X_n \nearrow X$ and $X_n \geq 0$, then $\mathrm{E}[X_n|\mathcal{G}] \nearrow \mathrm{E}[X|\mathcal{G}]$ almost surely.

(b) If $X_n \to X$ almost surely and $|X_n| \leq Y \in \mathcal{L}^1(\Omega)$, then $\mathrm{E}[X_n|\mathcal{G}] \to \mathrm{E}[X|\mathcal{G}]$ almost surely.

(c) $\mathrm{E}[\liminf_{n\to\infty} X_n|\mathcal{G}] \leq \liminf_{n\to\infty} \mathrm{E}[X_n|\mathcal{G}]$ almost surely.

Proof. For (b), by assumption and LDCT we have $X_n \to X$ in $\mathcal{L}^1$. The proposition 3.2 (e) implies that $\mathrm{E}\,|\mathrm{E}[X_n - X|\mathcal{G}]| \leq \mathrm{E}[|X_n - X|] \to 0$. Hence (b) follows.

Now for (a), proposition 3.2 (d) implies that $\mathrm{E}[X_n|\mathcal{G}]$ is incresing. Since $X_n \leq |X| \in \mathcal{L}^1$, (b) shows that $\mathrm{E}[X_n|\mathcal{G}] \to \mathrm{E}[X|\mathcal{G}]$.

Again, by proposition 3.2 (d), $\mathrm{E}[\inf_{k\geq n} X_k|\mathcal{G}] \leq \mathrm{E}[X_m|\mathcal{G}]$ for every $m \geq n$. Thus

$$\mathrm{E}\left[\inf_{k\geq n} X_k|\mathcal{G}\right] \leq \inf_{m\geq n} \mathrm{E}[X_m|\mathcal{G}].$$

Since $\inf_{k\geq n} X_k \nearrow \liminf_{n\to\infty} X$, we may take $n \to \infty$ and the result from (a) gives

$$\mathrm{E}\left[\liminf_{n\to\infty} X_n|\mathcal{G}\right] \leq \liminf_{n\to\infty} \mathrm{E}\left[X_n|\mathcal{G}\right]$$

as desired. ∎

Proposition 3.4
If $\sigma(X) \subset \mathcal{G}$ and $\mathrm{E}\left[|Y|\right], \mathrm{E}\left[|XY|\right] < \infty$, then $\mathrm{E}\left[XY|\mathcal{G}\right] = X\,\mathrm{E}\left[Y|\mathcal{G}\right]$.

Proof. Suppose first that $X = \mathbf{1}_G$ where $G \in \mathcal{G}$. Then for any $A \in \mathcal{G}$,

$$\mathrm{E}\left[X\,\mathrm{E}\left[Y|\mathcal{G}\right]\mathbf{1}_A\right] = \mathrm{E}\left[\mathbf{1}_G\mathbf{1}_A\,\mathrm{E}\left[Y|\mathcal{G}\right]\right] = \mathrm{E}\left[\mathbf{1}_G Y\mathbf{1}_A\right] = \mathrm{E}\left[\mathrm{E}\left[\mathbf{1}_G Y|\mathcal{G}\right]\mathbf{1}_A\right] = \mathrm{E}\left[\mathrm{E}\left[XY|\mathcal{G}\right]\mathbf{1}_A\right]$$

since $G \cap A \in \mathcal{G}$. By the linearity this extends to the case where X is simple. For $X \geq 0$, there are simple functions $X_n \nearrow X$ almost surely and hence $X_n Y \to XY$ with $|X_n Y| \leq |XY| \in \mathcal{L}^1$. By proposition 3.3 (b), $\mathrm{E}\left[X_n Y|\mathcal{G}\right] \to \mathrm{E}\left[XY|\mathcal{G}\right]$ almost surely. Also, $|\mathrm{E}\left[X_n Y|\mathcal{G}\right]\mathbf{1}_A| \leq \mathrm{E}\left[|X_n Y|\,|\mathcal{G}\right]\mathbf{1}_A \leq \mathrm{E}\left[|XY|\right]\mathbf{1}_A \in \mathcal{L}^1$.

$$\mathrm{E}\left[X\,\mathrm{E}\left[Y|\mathcal{G}\right]\mathbf{1}_A\right] = \lim_{n\to\infty} \mathrm{E}\left[X_n\,\mathrm{E}\left[Y|\mathcal{G}\right]\mathbf{1}_A\right] = \lim_{n\to\infty} \mathrm{E}\left[\mathrm{E}\left[X_n Y|\mathcal{G}\right]\mathbf{1}_A\right] = \mathrm{E}\left[\mathrm{E}\left[XY|\mathcal{G}\right]\mathbf{1}_A\right]$$

where the first equality follows by $|X_n\,\mathrm{E}\left[Y|\mathcal{G}\right]\mathbf{1}_A| \leq |XY\mathbf{1}_A| \in \mathcal{L}^1$ and then applying proposition 3.2 (f) with $f(x) = |x|$, $X_n\,\mathrm{E}\left[Y|\mathcal{G}\right]\mathbf{1}_A \to X\,\mathrm{E}\left[Y|\mathcal{G}\right]\mathbf{1}_A$ and LDCT. The general case follows from the decomposition $X = X^+ - X^-$ and $X_n = X_n^+ - X_n^-$. ∎

Proposition 3.5 (Law of Iterated Expectation, Tower Property)
Let $(\Omega, \mathcal{F}, \mathrm{P})$ be a probability space and $\mathcal{H} \subset \mathcal{G} \subset \mathcal{F}$ are σ-algebras. Then

(a) $\mathrm{E}\left[\mathrm{E}\left[X|\mathcal{H}\right]|\mathcal{G}\right] = \mathrm{E}\left[X|\mathcal{H}\right]$.

(b) $\mathrm{E}\left[\mathrm{E}\left[X|\mathcal{G}\right]|\mathcal{H}\right] = \mathrm{E}\left[X|\mathcal{H}\right]$.

Proof. For (a), since $\mathrm{E}\left[X|\mathcal{H}\right]$ is $\mathcal{G}$-measurable, proposition 3.4 implies that $\mathrm{E}\left[\mathrm{E}\left[X|\mathcal{H}\right]|\mathcal{G}\right] = \mathrm{E}\left[X|\mathcal{H}\right]\mathrm{E}\left[1|\mathcal{G}\right] = \mathrm{E}\left[X|\mathcal{H}\right]$.

For (b), first not that both sides are $\mathcal{H}$-measurable. For $A \in \mathcal{H} \subset \mathcal{G}$,

$$\mathrm{E}\left[\mathrm{E}\left[\mathrm{E}\left[X|\mathcal{G}\right]|\mathcal{H}\right]\mathbf{1}_A\right] = \mathrm{E}\left[\mathrm{E}\left[X|\mathcal{G}\right]\mathbf{1}_A\right] = \mathrm{E}\left[X\mathbf{1}_A\right] = \mathrm{E}\left[\mathrm{E}\left[X|\mathcal{H}\right]\mathbf{1}_A\right].$$

The conclusion follows. ∎

Corollary 3.6
For random variables $X, Y \in \mathcal{L}^1$, we have

$$\mathrm{E}\left[\mathrm{E}\left[X|\mathcal{G}\right]Y\right] = \mathrm{E}\left[X\,\mathrm{E}\left[Y|\mathcal{G}\right]\right] = \mathrm{E}\left[\mathrm{E}\left[X|\mathcal{G}\right]\mathrm{E}\left[Y|\mathcal{G}\right]\right].$$

Proof. By the tower property,

$$\mathrm{E}\left[\mathrm{E}\left[X|\mathcal{G}\right]Y\right] = \mathrm{E}\left[\mathrm{E}\left[\mathrm{E}\left[X|\mathcal{G}\right]Y|\mathcal{G}\right]\right] = \mathrm{E}\left[\mathrm{E}\left[X|\mathcal{G}\right]\mathrm{E}\left[Y|\mathcal{G}\right]\right].$$

Similarly,

$$\mathrm{E}\left[X\,\mathrm{E}\left[Y|\mathcal{G}\right]\right] = \mathrm{E}\left[\mathrm{E}\left[X\,\mathrm{E}\left[Y|\mathcal{G}\right]|\mathcal{G}\right]\right] = \mathrm{E}\left[\mathrm{E}\left[X|\mathcal{G}\right]\mathrm{E}\left[Y|\mathcal{G}\right]\right].$$

■

Remark

If we view $\mathrm{E}\left[XY\right] = \langle X, Y\rangle$*, then corollary 3.6 is effectively saying that the conditional expectation operator is self-adjoint.*

Proposition 3.7

Let $(\Omega, \mathcal{F}, \mathrm{P})$ *be a probability space and* $\mathcal{G} \subset \mathcal{F}$ *be a* σ*-algebra. Consider the minimization problem*

$$\inf_{Z\in\mathcal{L}^2(\mathcal{G})} \mathrm{E}\left[(X-Z)^2\right]$$

where $X \in \mathcal{L}^2(\mathcal{F})$*. Then the minimum is attained in* $\mathcal{L}^2(\mathcal{G})$ *and the minimizer is given by* $Z = \mathrm{E}\left[X|\mathcal{G}\right]$*.*

Proof. First notice that $\mathcal{L}^2(\mathcal{G})$ is a closed convex subset of $\mathcal{L}^2(\mathcal{F})$. Hence the minimum is attained by a unique minimizer. Now rewrite the expression as

$$\mathrm{E}\left[(X-Z)^2\right] = \mathrm{E}\left[(X-\mathrm{E}\left[X|\mathcal{G}\right])^2\right] + \mathrm{E}\left[(Z-\mathrm{E}\left[X|\mathcal{G}\right])^2\right] - 2\,\mathrm{E}\left[(X-\mathrm{E}\left[X|\mathcal{G}\right])(Z-\mathrm{E}\left[X|\mathcal{G}\right])\right].$$

By the tower property, the third term becomes

$$\begin{aligned}\mathrm{E}\left[(X-\mathrm{E}\left[X|\mathcal{G}\right])(Z-\mathrm{E}\left[X|\mathcal{G}\right])\right] &= \mathrm{E}\left[\mathrm{E}\left[(X-\mathrm{E}\left[X|\mathcal{G}\right])(Z-\mathrm{E}\left[X|\mathcal{G}\right])|\mathcal{G}\right]\right]\\ &= \mathrm{E}\left[(\mathrm{E}\left[X|\mathcal{G}\right]-\mathrm{E}\left[X|\mathcal{G}\right])(Z-\mathrm{E}\left[X|\mathcal{G}\right])\right] = 0.\end{aligned}$$

To minimize the expression, it suffices to minimize the second term, which gives the solution $Z = \mathrm{E}\left[X|\mathcal{G}\right]$. ■

Remark

The proposition gives a characterization of conditional expectations. In fact, we can conversely view the conditional expectation as an projection operator $\mathrm{E}\left[\cdot|\mathcal{G}\right] : \mathcal{L}^2(\mathcal{F}) \to \mathcal{L}^2(\mathcal{G})$*. Since* $\mathcal{L}^2$ *is dense in* $\mathcal{L}^1$ *under* $\|\cdot\|_1$*, the projection operater extends to* $\mathrm{E}\left[\cdot|\mathcal{G}\right] : \mathcal{L}^1(\mathcal{F}) \to \mathcal{L}^1(\mathcal{G})$*. For* $X \in \mathcal{L}^1(\mathcal{F})$*, we may choose* $X_n \to X$ *in* $\mathcal{L}^1$ *where* $X_n \in \mathcal{L}^2$ *and define* $\mathrm{E}\left[X|\mathcal{G}\right] = \lim_{n\to\infty}\mathrm{E}\left[X_n|\mathcal{G}\right]$*.*

Definition 3.8

Let $X, Y \in \mathcal{L}^2$ *be random variables. For any* σ*-algebra* $\mathcal{G} \subset \mathcal{F}$*, the* ***conditional covariance*** *is defined as*

$$\mathrm{Cov}\left[X, Y|\mathcal{G}\right] = \mathrm{E}\left[(X-\mathrm{E}\left[X|\mathcal{G}\right])(Y-\mathrm{E}\left[Y|\mathcal{G}\right])|\mathcal{G}\right] = \mathrm{E}\left[XY|\mathcal{G}\right] - \mathrm{E}\left[X|\mathcal{G}\right]\mathrm{E}\left[Y|\mathcal{G}\right].$$

The ***conditional variance*** *is defined as* $\mathrm{Var}\left[X|\mathcal{G}\right] = \mathrm{Cov}\left[X, X|\mathcal{G}\right] = \mathrm{E}\left[X^2|\mathcal{G}\right] - \mathrm{E}\left[X|\mathcal{G}\right]^2$*.*

Lemma 3.9 (Law of Total Variance)
Let $(\Omega, \mathcal{F}, \mathrm{P})$ *be a probability space and* $\mathcal{G} \subset \mathcal{F}$ *be a* σ*-algebra. For any random variables* $X, Y \in \mathcal{L}^2$*,*

(a) $\mathrm{Var}[X] = \mathrm{E}[\mathrm{Var}[X|\mathcal{G}]] + \mathrm{Var}[\mathrm{E}[X|\mathcal{G}]]$.

(b) $\mathrm{Cov}[X, Y] = \mathrm{E}[\mathrm{Cov}[X, Y|\mathcal{G}]] + \mathrm{Cov}[\mathrm{E}[X|\mathcal{G}], \mathrm{E}[Y|\mathcal{G}]]$.

Proof. We prove only (b). (a) follows immediately by replacing Y with X. By the tower property,

$$\begin{aligned}\mathrm{Cov}[X, Y] &= \mathrm{E}[XY] - \mathrm{E}[X]\,\mathrm{E}[Y] = \mathrm{E}[\mathrm{E}[XY|\mathcal{G}]] - \mathrm{E}[\mathrm{E}[X|\mathcal{G}]]\,\mathrm{E}[\mathrm{E}[Y|\mathcal{G}]] \\ &= \mathrm{E}[\mathrm{Cov}[XY|\mathcal{G}]] + \mathrm{E}[(\mathrm{E}[X|\mathcal{G}])(\mathrm{E}[Y|\mathcal{G}])] - \mathrm{E}[\mathrm{E}[X|\mathcal{G}]]\,\mathrm{E}[\mathrm{E}[Y|\mathcal{G}]] \\ &= \mathrm{E}[\mathrm{Cov}[X, Y|\mathcal{G}]] + \mathrm{Cov}[\mathrm{E}[X|\mathcal{G}]]\,.\end{aligned}$$

■

3.2. Martingale

Definition 3.10
*Let $(\Omega, \mathcal{F}, \mathrm{P})$ be a probability space. A **stochastic process** is a collection of random variables $X_t : \Omega \to (S, \mathcal{S})$ where $t \in T$. T is a totally ordered index set.*

Remark
*T is often taken to be $\mathbb{Z}_+$ or $\mathbb{R}_+$, which represents the time. If T is countable, we say that X_t is a **discrete time** process and **continuous time** if T is some uncountable subset of $\mathbb{R}$.*

Definition 3.11
*A **filtration** $\{\mathcal{F}_t\}_{t \in T}$ is a collection of σ-algebras such that $\mathcal{F}_t \subset \mathcal{F}_s$ for every $t < s$, $t, s \in T$.*

Definition 3.12
*Let X_t be a stochastic process and $\mathcal{F}_t$ be a filtration. We say that X_t is **adapted** to $\mathcal{F}_t$ or $\mathcal{F}_t$-**adapted** if $\sigma(X_t) \subset \mathcal{F}_t$ for all t.*

Remark
In many cases, the filtration is not mentioned since we may consider the natural filtration generated by X_t through the definition $\mathcal{F}_t = \sigma(\{X_s | s \le t\})$.

Definition 3.13
*Let X_t be a $\mathbb{R}$-valued stochastic process. We say that X_t is an $\mathcal{F}_t$-**martingale** if*

(a) $\mathrm{E}[|X_t|] < \infty$.

(b) X_t *is* $\mathcal{F}_t$*-adapted.*

(c) $X_t = \mathrm{E}[X_s | \mathcal{F}_t]$ *for all* $s > t$.

*We say that X_t is a **supermartingale** if $X_t \ge \mathrm{E}[X_s | \mathcal{F}_t]$ and **submartingale** if $X_t \le \mathrm{E}[X_s | \mathcal{F}_t]$ for all $s > t$.*

Remark
X_t is a martingale if and only if X_t is both a submartingale and a supermartingale.

Proposition 3.14
Let X_n be a discrete time stochastic process.

(a) If $X_n \ge \mathrm{E}[X_{n+1} | \mathcal{F}_n]$ for all n, then X_n is a supermartingale.

(b) If $X_n \le \mathrm{E}[X_{n+1} | \mathcal{F}_n]$ for all n, then X_n is a submartingale.

Proof. The proof for (b) is similar to (a). We prove only the case (a). Let $m > n$.

$$\mathrm{E}[X_m | \mathcal{F}_n] = \mathrm{E}[\mathrm{E}[X_m | \mathcal{F}_{m-1}] | \mathcal{F}_n] \ge \mathrm{E}[X_{m-1} | \mathcal{F}_n] = \cdots \ge \mathrm{E}[X_n | \mathcal{F}_n] = X_n.$$

Hence X_n is a supermartingale. ∎

Example (Random Walk)
Let $\xi_i \in \mathcal{L}^1$ be independent and identically distributed with $\mathrm{E}[\xi_i] = 0$. Set $X_0 = 0$ and $X_n = X_{n-1} + \xi_n$ for $n \in \mathbb{N}$. Clearly X_n is adapted to the natural filtration $\mathcal{F}_n$ generated by X_n and $\mathrm{E}[|X_n|] \leq n\,\mathrm{E}[|\xi_i|] < \infty$ for given n. Also,

$$\mathrm{E}[X_{n+1}|\mathcal{F}_n] = \mathrm{E}[X_n + \xi_{n+1}|\mathcal{F}_n] = X_n.$$

Hence X_n is a martingale.

Example (Quadratic Martingale)
Let ξ_i be independent and identically distributed with $\mathrm{E}[\xi_i] = 0$ and $\mathrm{E}\left[\xi_i^2\right] = \sigma^2 < \infty$. Put $X_n = \sum_{i\leq n} \xi_i$. Then $M_n = X_n^2 - n\sigma^2$ is a martingale. It is clear that M_n is adapted to the natural filtration $\mathcal{F}_n$ generated by X_n and

$$\mathrm{E}[|M_n|] \leq \mathrm{E}\left[X_n^2\right] + n\sigma^2 = 2n\sigma^2 < \infty.$$

Also,

$$\begin{aligned}\mathrm{E}[M_{n+1}|\mathcal{F}_n] &= \mathrm{E}\left[X_{n+1}^2 - (n+1)\sigma^2|\mathcal{F}_n\right] = \mathrm{E}\left[X_n^2 + 2\xi_{n+1}X_n + \xi_{n+1}^2 - (n+1)\sigma^2|\mathcal{F}_n\right] \\ &= X_n^2 + \sigma^2 - (n+1)\sigma^2 = M_n.\end{aligned}$$

Hence M_n is a martingale.

Example (Exponential Martingale)
Let ξ_i be independent and identically distributed with $M(t) = \mathrm{E}[\exp(t\xi_i)] < \infty$ for given t. Put $X_i = \frac{\exp(t\xi_i)}{M(t)}$ and thus $\mathrm{E}[X_i] = 1$. Let $M_n = \prod_{i\leq n} X_i$. M_n is adapted to the natural filtration $\mathcal{F}_n$ generated by X_n and

$$\mathrm{E}[|M_n|] = \mathrm{E}\left[\prod_{i\leq n} X_i\right] = \prod_{i\leq n} \mathrm{E}[X_i] = 1.$$

Also,

$$\mathrm{E}[M_{n+1}|\mathcal{F}_n] = M_n\,\mathrm{E}[X_{n+1}|\mathcal{F}_n] = M_n\,\mathrm{E}[X_{n+1}] = M_n.$$

Hence M_n is a martingale.

Lemma 3.15
Let X_t be a $\mathcal{F}_t$-martingale and φ is a function such that $\mathrm{E}[|\varphi(X_t)|] < \infty$.

(a) If φ is convex, then $\varphi(X_t)$ is a submartingale.

(b) If φ is concave, then $\varphi(X_t)$ is a supermartingale

Proof. The proof for (b) is similar to (a). We only prove the case (a). Let $s > t$.

$$\mathrm{E}[\varphi(X_s)|\mathcal{F}_t] \geq \varphi(\mathrm{E}[X_s|\mathcal{F}_t]) = \varphi(X_t).$$

Hence $\varphi(X_t)$ is a submartingale. ■

Lemma 3.16

Let X_t be a stochastic process and φ is an increasing function such that $\mathrm{E}[|\varphi(X_t)|] < \infty$.

(a) If X_t is a $\mathcal{F}_t$-submartingale and φ is convex, then $\varphi(X_t)$ is a $\mathcal{F}_t$-submartingale.

(b) If X_t is a $\mathcal{F}_t$-supermartingale and φ is concave, then $\varphi(X_t)$ is a $\mathcal{F}_t$-supermartingale.

Proof. For (a), let $s > t$.

$$\mathrm{E}[\varphi(X_s)|\mathcal{F}_t] \geq \varphi(\mathrm{E}[X_s|\mathcal{F}_t]) \geq \varphi(X_t).$$

Hence $\varphi(X_t)$ is a submartingale.

For (b),

$$\mathrm{E}[\varphi(X_s)|\mathcal{F}_t] \leq \varphi(\mathrm{E}[X_s|\mathcal{F}_t]) \leq \varphi(X_t).$$

Hence $\varphi(X_t)$ is a supermartingale. ■

Definition 3.17

*Let X_n be a stochastic process and $\mathcal{F}_n$ be the filtration generated by X_n. A stochastic process H_n is said to be **predictable** if H_{n+1} is $\mathcal{F}_n$-adapted.*

Definition 3.18

*Let X_n, Y_n be two discrete-time stochastic processes. The **discrete-time stochastic integral** is defined as*

$$(X \cdot Y)_n = \sum_{i \leq n} X_i(Y_i - Y_{i-1}).$$

Theorem 3.19

If X_n is a $\mathcal{F}_n$-supermartingale and $H_n \geq 0$ is predictable and is bounded for each n. Then $(H \cdot X)_n$ is a supermartingale.

Proof. First, it is clear that $(H \cdot X)_n$ is $\mathcal{F}_n$-adapted since the discrete-time stochastic integral is a function of $(H_1, \ldots H_n, X_0, \ldots, X_n)$. Second, since H_n is bounded, say by c_i,

$$\mathrm{E}[|(H \cdot X)_n|] \leq \sum_{i \leq n} \mathrm{E}[|H_i|\,|X_i - X_{i-1}|] \leq \sum_{i \leq n} c_i \max_{1 \leq i \leq n} 2\,\mathrm{E}[|X_i|] < \infty$$

for given n since X_n is a supermartingale and satisfies that $X_n \in \mathcal{L}^1$. Finally,

$$\begin{aligned}
\mathrm{E}[(H \cdot X)_{n+1}|\mathcal{F}_n] &= (H \cdot X)_n + \mathrm{E}[H_{n+1}(X_{n+1} - X_n)|\mathcal{F}_n] \\
&= (H \cdot X)_n + H_{n+1}\,\mathrm{E}[X_{n+1}|\mathcal{F}_n] - H_{n+1}X_n \\
&\leq (H \cdot X)_n + H_{n+1}X_n - H_{n+1}X_n = (H \cdot X)_n.
\end{aligned}$$

Hence $(H \cdot X)_n$ is a supermartingale. ■

Remark

If we instead assume that X_n is a $\mathcal{F}_n$-submartingale, then $(H \cdot X)_n$ is a submartingale. Furthermore, if X_n is a $\mathcal{F}_n$-martingale, then for every predictable H_n, $(H \cdot X)_n$ is a martingale.

Definition 3.20
*A random variable $\tau \in T$ is a **stopping time** if $\{\tau \leq t\} \in \mathcal{F}_t$.*

Example (Hitting Time)
Let X_t be a stochastic process and $\mathcal{F}_t$ be the natural filtration generated by X_t. The hitting time of A is $\tau = \inf\{s \in T \mid X_s \in A\}$. Then τ is a stopping time.

Proposition 3.21
Suppose that τ_1 and τ_2 are stopping times with respect to $\mathcal{F}_t$. Then $\tau_1 \wedge \tau_2$ and $\tau_1 \vee \tau_2$ are stopping times.

Proof. Notice that

$$\{\tau_1 \wedge \tau_2 \leq t\} = \{\tau_1 \leq t\} \cup \{\tau_2 \leq t\} \in \mathcal{F}_t, \quad \text{and} \quad \{\tau_1 \vee \tau_2 \leq t\} = \{\tau_1 \leq t\} \cap \{\tau_2 \leq t\} \in \mathcal{F}_t.$$

Hence $\tau_1 \wedge \tau_2$ and $\tau_1 \vee \tau_2$ are stopping times. ∎

Remark
Since any constant n is also a stopping time, $n \wedge \tau$ is a stopping time provided that τ is. The conclusion from theorem 3.19 also applies to the strategy of the form $H'_n = H_{n \wedge \tau}$. In particular, taking the predictable process

$$H_n = \begin{cases} 1 & \tau \leq n \\ 0 & \tau > n \end{cases}$$

shows that $X_{n \wedge \tau}$ is a martingale provided that X_n is.

Definition 3.22
*Let X_t be a stochastic process with index set T and $a < b$ be constants. Given ω, the $[a, b]$-**crossing time** $N_a^b(\omega)$ is defined as the supremum over $n \in \mathbb{Z}_+$ such that there is $s_1 < t_1 < s_2 < t_2 < \cdots < s_n < t_n$, $s_i, t_i \in T$ with $X_{s_1}(\omega) \leq a$ and $X_{t_1}(\omega) \geq b$.*

Lemma 3.23 (Doob's Upcrossing Lemma)
Let X_n be a submartingale on $\mathbb{Z}_+$ and $N_a^b(n)$ be the crossing time up to $n \in \mathbb{Z}_+$. Then

$$\mathrm{E}\left[N_a^b(n)\right] \leq \frac{1}{b-a} \mathrm{E}\left[(X_n - a)^+\right]$$

for all $n \in \mathbb{Z}_+$.

Proof. Let $Y_n = a + (X_n - a)^+$. By lemma 3.16, Y_n is also a submartingale since $x \mapsto a + (x-a)^+$ is a convex increasing function. Notice that the crossing time up to time n for Y_n is also $N_a^b(n)$. Now define the stopping times $0 = \tau_0 \leq \sigma_1 < \tau_1 < \cdots$ by

$$\sigma_k = \inf\{n \geq \tau_{k-1} \mid Y_n = a\}, \quad \text{and} \quad \tau_k = \inf\{n \geq \sigma_k \mid Y_n \geq b\}.$$

Next, we consider the predictable process

$$H_n = \sum_{k\geq 1} \mathbf{1}\{\sigma_k < n \leq \tau_k\}.$$

By theorem 3.19, $((1-H)\cdot Y)_n$ is a submartingale and thus

$$\mathrm{E}[((1-H)\cdot Y)_n] \geq \mathrm{E}[((1-H)\cdot Y)_0] = 0.$$

Notice that $(H\cdot Y)_n \geq (b-a)N_a^b(n)$. We have

$$\mathrm{E}\left[(X_n - a)^+\right] = \mathrm{E}[(H\cdot Y)_n] + \mathrm{E}[((1-H)\cdot Y)_n] \geq (b-a)\,\mathrm{E}\left[N_a^b(n)\right].$$

Rearranging the inequality gives the desired result. ∎

Theorem 3.24 (Martingale Convergence Theorem I)
Let X_n be a $\mathcal{F}_n$-submartingale. Suppose that $\sup_n \mathrm{E}\left[X_n^+\right] < \infty$. Then there is $X \in \mathcal{L}^1$ such that $X_n \to X$ almost surely.

Proof. For $a, b \in \mathbb{Q}$, consider the event $A_{a,b} = \{\liminf_n X_n < a < b < \limsup_n X_n\}$. Put S be the union of all $A_{a,b}$ over $a, b \in \mathbb{Q}$. For given $a < b$, consider the crossing time N_a^b on $\mathbb{Z}_+$ and $N_a^b(n)$ which is up to time n. Clearly $N_a^b(n) \nearrow N_a^b$. By LMCT and Doob's upcrossing lemma,

$$\mathrm{E}\left[N_a^b\right] = \lim_{n\to\infty} \mathrm{E}\left[N_a^b(n)\right] \leq \limsup_{n\to\infty} \frac{1}{b-a}\,\mathrm{E}\left[(X_n - a)^+\right] \leq C \sup_n \mathrm{E}\left[X_n^+\right]$$

for some constant C. Hence $\mathrm{P}(N_a^b < \infty) = 0$. This implies that X_n crosses $[a, b]$ only finite times almost surely. Thus $\mathrm{P}(A_{a,b}) = 0$ and $\mathrm{P}(S) = 0$. We see that

$$\mathrm{P}(\liminf_n X_n < \limsup_n X_n) = 0.$$

Thus the limit of X_n exists, possibly in extended sense, almost surely. Denote the limit of X_n as X. It now remains to show that $X \in \mathcal{L}^1$.

$$\mathrm{E}\left[X^+\right] = \mathrm{E}\left[\lim_{n\to\infty} X_n^+\right] \leq \liminf_{n\to\infty} \mathrm{E}\left[X_n^+\right] \leq \sup_n \mathrm{E}\left[X_n^+\right] < \infty$$

by Fatou's lemma. Also, $X_n^- = X_n^+ - X_n$.

$$\mathrm{E}\left[X_n^-\right] = \mathrm{E}\left[X_n^+\right] - \mathrm{E}[X_n] \leq \mathrm{E}\left[X_n^+\right] - \mathrm{E}[X_0] < \infty.$$

Hence $X \in \mathcal{L}^1$. The proof is complete. ∎

Remark
The theorem also holds for supermartingale. If X_n is a $\mathcal{F}_n$-supermartingale and $\sup_n \mathrm{E}\left[X_n^-\right] < \infty$. Then there is $X \in \mathcal{L}^1$ such that $X_n \to X$ almost surely.

Corollary 3.25
If $X_n \geq 0$ is a supermartingale, then there is $X \in \mathcal{L}^1$ such that $X_n \to X$ almost surely and $\mathrm{E}[X] \leq \mathrm{E}[X_0]$.

Proof. Since $\sup_n \mathrm{E}\left[X_n^-\right] = 0$, the martingale convergence theorem implies that there is $X \in \mathcal{L}^1$ such that $X_n \to X$ almost surely. By Fatou's lemma,

$$\mathrm{E}[X] = \mathrm{E}\left[\liminf_{n\to\infty} X_n\right] \leq \liminf_{n\to\infty} \mathrm{E}[X_n] \leq \liminf_{n\to\infty} \mathrm{E}[X_0] = \mathrm{E}[X_0].$$

The proof is complete. ∎

Remark
The hard bound on X_n is neccessary. Consider ξ_i with $\mathrm{P}(\xi_i = \pm 1) = 1/2$. Let $X_n = \sum_{i\leq n} \xi_i$ and $X_0 = 1$. Consider the stopping time $\tau = \inf\{n \mid X_n = 0\}$. Then $\mathrm{P}(\tau < \infty) = 1$. Since $X_{n\wedge\tau}$ is a martingale, $\mathrm{E}[X_{n\wedge\tau}] = X_0 = 1$. Also, the martingale convergence theorem shows that there is X where $X_{n\wedge\tau} \to X$ as $n \to \infty$. Since $\tau < \infty$ almost surely, we have $X = 0$ almost surely and $\mathrm{E}[X] = 0$.

Example (Pólya's Urn)
Let $X_0, Y_0 \in \mathbb{N}$. Define

$$(X_n, Y_n) = \begin{cases} (X_{n-1}+1, Y_{n-1}), & \textit{prob.} = \frac{X_{n-1}}{X_{n-1}+Y_{n-1}}, \\ (X_{n-1}, Y_{n-1}+1), & \textit{prob.} = \frac{Y_{n-1}}{X_{n-1}+Y_{n-1}}. \end{cases}$$

Let $Z_n = X_n/(X_n+Y_n)$. Notice that

$$\begin{aligned} \mathrm{E}[Z_{n+1}|\mathcal{F}_n] &= Z_n \cdot \frac{X_n+1}{X_n+1+Y_n} + (1-Z_n)\cdot\frac{X_n}{X_n+Y_n+1} = \frac{X_n}{X_n+Y_n}\cdot\frac{1}{X_n+1+Y_n} + \frac{X_n}{X_n+Y_n+1} \\ &= \frac{X_n}{X_n+Y_n} = Z_n. \end{aligned}$$

Hence Z_n is a martingale. Since Z_n is clearly bounded, the martingale convergence theorem implies that there is some $Z \in \mathcal{L}^1$ such that $Z_n \to Z$ almost surely. In fact, one can show that $Z \sim Beta(X_0, Y_0)$.

Theorem 3.26 (Optional Stopping Theorem)
Let τ be a stopping time and X_n be a martingale. Then $\mathrm{E}[|X_\tau|] < \infty$ and $\mathrm{E}[X_\tau] = \mathrm{E}[X_0]$ if one of the followings holds:

(a) There is some N such that $\tau \leq N$ almost surely.

(b) There is some M such that $|X_{\tau\wedge n}| \leq M$ and $\mathrm{P}(\tau < \infty) = 1$.

(c) $\mathrm{E}[\tau] < \infty$ and $|X_{n+1} - X_n| \leq M$ almost surely for some M.

Proof. For (a), we have that

$$|X_{n\wedge\tau} - X_0| \leq \sum_{k=1}^{n\wedge\tau} |X_k - X_{k-1}| \leq \sum_{k=1}^{N} |X_k - X_{k-1}| \in \mathcal{L}^1$$

for every n. Hence by the LDCT, since $\tau < \infty$ almost surely and $X_{n\wedge\tau} \to X_\tau$,

$$\mathrm{E}\left[X_\tau\right] = \mathrm{E}\left[\lim_{n\to\infty} X_{n\wedge\tau}\right] = \lim_{n\to\infty} \mathrm{E}\left[X_{n\wedge\tau}\right] = \mathrm{E}\left[X_0\right]$$

by the fact that $X_{n\wedge\tau}$ is a martingale by theorem 3.19.

For (b), by the bounded convergence theorem and that $\tau < \infty$ almost surely,

$$\mathrm{E}\left[X_\tau\right] = \mathrm{E}\left[\lim_{n\to\infty} X_{n\wedge\tau}\right] = \lim_{n\to\infty} \mathrm{E}\left[X_{n\wedge\tau}\right] = \mathrm{E}\left[X_0\right].$$

For (c), notice that

$$|X_{n\wedge\tau} - X_0| \leq \sum_{k=1}^{n\wedge\tau} |X_k - X_{k-1}| \leq M\tau \in \mathcal{L}^1$$

since $\tau \in \mathcal{L}^1$. Thus $|X_{n\wedge\tau}| \leq M\tau \in \mathcal{L}^1$. LDCT implies that

$$\mathrm{E}\left[X_\tau\right] = \mathrm{E}\left[\lim_{n\to\infty} X_{n\wedge\tau}\right] = \lim_{n\to\infty} \mathrm{E}\left[X_{n\wedge\tau}\right] = \mathrm{E}\left[X_0\right].$$

■

Corollary 3.27 (Wald's Identity)
Let X_i be independent and identically distributed with $\mathrm{E}\left[X_i\right] = \mu < \infty$. $|X_i| \leq K$ almost surely. Let $S_n = \sum_{i\leq n} X_i$, $S_0 = 0$ and τ be a stopping time with $\mathrm{E}\left[\tau\right] < \infty$. Then $\mathrm{E}\left[S_\tau\right] = \mu\,\mathrm{E}\left[\tau\right]$.

Proof. Consider $M_n = S_n - n\mu$. Notice that

$$\mathrm{E}\left[M_{n+1}|\mathcal{F}_n\right] = \mathrm{E}\left[X_{n+1} - \mu|\mathcal{F}_n\right] + S_n - n\mu = M_n.$$

Thus M_n is a martingale. Also,

$$|M_{n+1} - M_n| = |X_{n+1} - \mu| \leq K + |\mu|$$

almost surely. By the optional stopping theorem (c),

$$\mathrm{E}\left[S_\tau - \tau\mu\right] = \mathrm{E}\left[M_\tau\right] = \mathrm{E}\left[M_0\right] = 0.$$

Hence $\mathrm{E}\left[S_\tau\right] = \mu\,\mathrm{E}\left[\tau\right]$. ■

Example
Consider a random walk $X_n = \sum_{i\leq n} \xi_i$ where $\mathrm{P}(\xi_i = \pm 1) = 1/2$ are independent and $X_0 = x \in$

$\mathbb{Z}$. *Define the hitting time* $\tau = \inf\{k \mid X_k = a, b\}$ *where* $a < x < b$, $a, b \in \mathbb{Z}$. *We first show that*

$$\mathrm{P}(X_\tau = a) = \frac{b-x}{b-a}.$$

Since X_n *is a martingale,* $X_{n\wedge\tau}$ *is again a martingale by theorem 3.19. Thus*

$$\begin{aligned} x &= \mathrm{E}\left[X_{n\wedge\tau}\right] = \mathrm{E}\left[X_\tau \mathbf{1}\left\{n \geq \tau\right\}\right] + \mathrm{E}\left[X_n \mathbf{1}\left\{n < \tau\right\}\right] \\ &= a\,\mathrm{P}(X_\tau = a, \tau \leq n) + b\,\mathrm{P}(X_\tau = b, \tau \leq n) + \mathrm{E}\left[X_n \mathbf{1}\left\{n < \tau\right\}\right]. \end{aligned}$$

Notice that $|X_n|\,\mathbf{1}\left\{n < \tau\right\} \leq |a| \vee |b|$ *and* $|X_n|\,\mathbf{1}\left\{n < \tau\right\} \to 0$ *since* $\tau < \infty$ *almost surely. Letting* $n \to \infty$ *gives*

$$x = a\,\mathrm{P}(X_\tau = a) + b\,\mathrm{P}(X_\tau = b) = a\,\mathrm{P}(X_\tau = a) + b(1 - \mathrm{P}(X_\tau = a)), \quad \Rightarrow \quad \mathrm{P}(X_\tau = a) = \frac{b-x}{b-a}.$$

Next, we show that

$$\mathrm{E}\left[\tau\right] = (b-x)(x-a).$$

Consider $M_n = X_n^2 - n$. *Then clearly the integrability and measurability holds. Also,*

$$\mathrm{E}\left[M_{n+1}|\mathcal{F}_n\right] = \mathrm{E}\left[X_n^2 + 2X_n\xi_{n+1} + \xi_{n+1}^2 - (n+1)|\mathcal{F}_n\right] = M_n.$$

Hence M_n *is a martingale. Again, by theorem 3.19,* $M_{n\wedge\tau}$ *is still a martingale.*

$$\begin{aligned} x^2 &= \mathrm{E}\left[M_{n\wedge\tau}\right] = \mathrm{E}\left[X_\tau^2 \mathbf{1}\left\{\tau \leq n\right\}\right] + \mathrm{E}\left[X_n^2 \mathbf{1}\left\{n < \tau\right\}\right] - \mathrm{E}\left[n \wedge \tau\right] \\ &= a^2\,\mathrm{P}(X_\tau = a, \tau \leq n) + b^2\,\mathrm{P}(X_\tau = b, \tau \leq n) + \mathrm{E}\left[X_n^2 \mathbf{1}\left\{n < \tau\right\}\right] - \mathrm{E}\left[n \wedge \tau\right]. \end{aligned}$$

Notice that $\tau < \infty$ *almost surely and thus* $n \wedge \tau \nearrow \tau$. *Also,* $X_n^2 \mathbf{1}\left\{n < \tau\right\} < (a^2 \vee b^2)$ *and* $X_n^2 \mathbf{1}\left\{n < \tau\right\} \to 0$ *as* $n \to \infty$. *By the LMCT and LDCT, taking* $n \to \infty$,

$$x^2 = a^2\,\mathrm{P}(X_\tau = a) + b^2\,\mathrm{P}(X_\tau = b) - \mathrm{E}\left[\tau\right] = a^2\frac{b-x}{b-a} + b^2\frac{x-a}{b-a} - \mathrm{E}\left[\tau\right].$$

Hence $\mathrm{E}\left[\tau\right] = (b-x)(x-a)$.

Theorem 3.28 (Doob's Decomposition)
Let X_n *be an integrable and* $\mathcal{F}_n$*-adapted process. Then there is an almost surely unique decomposition* $X_n = M_n + A_n$ *where* M_n *is an* $\mathcal{F}_n$*-martingale and* A_n *is* $\mathcal{F}_n$*-predictable with* $A_0 = 0$. *In particular, if* X_n *is a submartingale, then* A_n *is non-decreasing almost surely.*

Proof. Put $A_n = \sum_{k\leq n} \mathrm{E}\left[X_k|\mathcal{F}_{k-1}\right] - X_{k-1}$ with $A_0 = 0$ and $M_n = X_n - A_n$. Notice that A_n is $\mathcal{F}_{n-1}$ measurable and hence $\mathcal{F}_n$ predictable. Also, M_n is integrable and $\mathcal{F}_n$-adapted. We verify the martingale condition:

$$\begin{aligned} \mathrm{E}\left[X_n|\mathcal{F}_{n-1}\right] &= \mathrm{E}\left[M_n|\mathcal{F}_{n-1}\right] + A_n = \mathrm{E}\left[M_n|\mathcal{F}_{n-1}\right] + A_{n-1} + \mathrm{E}\left[X_n|\mathcal{F}_{n-1}\right] - X_{n-1} \\ &= \mathrm{E}\left[M_n|\mathcal{F}_{n-1}\right] - M_{n-1} + \mathrm{E}\left[X_n|\mathcal{F}_{n-1}\right]. \end{aligned}$$

Rearranging the equation shows that $M_{n-1} = \mathrm{E}\left[M_n|\mathcal{F}_{n-1}\right]$. Thus M_n is indeed a martingale.

To see the uniqueness, suppose that $X_n = M'_n + A'_n$ is another decomposition. Then

$$\mathrm{E}\left[X_n|\mathcal{F}_{n-1}\right] = M_{n-1} + A_n = X_{n-1} + A_n - A_{n-1} = X_{n-1} + A'_n - A'_{n-1}.$$

Since $A_0 = A'_0 = 0$ and $A_n - A_{n-1} = A'_n - A'_{n-1}$ for all n, it follows that A_n are unique and also the M_n.

Now suppose that X_n is a submartingale,

$$A_{n-1} + M_{n-1} = X_{n-1} \leq \mathrm{E}\left[X_n|\mathcal{F}_{n-1}\right] = M_{n-1} + A_n.$$

Hence $A_{n-1} \leq A_n$ almost surely for all n. ∎

Example

Let X_n be independent with $\mathrm{E}\left[X_i\right] = 0$ and $\sigma_i^2 = \mathrm{E}\left[X_i^2\right] < \infty$. Let $S_n = \sum_{i\leq n} X_i$ be a martingale. Then S_n^2 is a submartingale. Consider the Doob's decomposition $S_n^2 = M_n + A_n$. We claim that $M_n = S_n^2 - \sum_{i\leq n}\sigma_i^2$ and $A_n = \sum_{i\leq n}\sigma_i^2$ is the Doob's decomposition for S_n^2. First, the integrability and adaptedness of M_n are clear.

$$\mathrm{E}\left[M_{n+1}|\mathcal{F}_n\right] = S_n^2 + 2S_n\,\mathrm{E}\left[X_{n+1}|\mathcal{F}_n\right] + \mathrm{E}\left[X_{n+1}^2|\mathcal{F}_n\right] - \sum_{i\leq n+1}\sigma_i^2 = S_n^2 - \sum_{i\leq n}\sigma_i^2 = M_n.$$

Thus M_n is a martingale. Also, A_n is increasing almost surely and predictable. We conclude that M_n and A_n are the desired decomposition.

Theorem 3.29

X_n is a martingale with $|X_{n+1} - X_n| \leq M$. Let

$$C = \left\{\lim_{n\to\infty} X_n \text{ exists and is finite}\right\}, \quad D = \left\{\liminf_{n\to\infty} X_n = -\infty, \text{ and } \limsup_{n\to\infty} X_n = \infty\right\}.$$

Then $\mathrm{P}(C \cup D) = 1$.

Proof. Without loss of generality, we assume that $X_0 = 0$. For fixed $k > 0$, define the stopping time $\tau = \inf\{n \mid X_n \leq k\}$. Since the increments of X_n are bounded, we have $X_{n\wedge\tau} \geq -k - M$. Since $X_{n\wedge\tau}$ is a martingale, by the martingale convergence theorem $X_{n\wedge\tau} \to X$ for some $X \in \mathcal{L}^1$ almost surely. On $\{\tau = \infty\}$, $\lim_{n\to\infty} X_n = \lim_{n\to\infty} X_{n\wedge\tau} = X$ is finite. Now taking $k \to \infty$, then on the event $\{\liminf_{n\to\infty} X_n > -\infty\}$, $\lim_{n\to\infty} X_n$ exists and is finite almost surely. The same argument for $-X_n$ shows that on $\{\limsup_{n\to\infty} < \infty\}$, $\lim_{n\to\infty} X_n$ exists and is finite. Combining the results,

$$D^c = \left\{\liminf_{n\to\infty} X_n > -\infty\right\} \cup \left\{\limsup_{n\to\infty} < \infty\right\} \subset C$$

with some exception of measure zero. Hence $\mathrm{P}(C \cup D) = 1$. ∎

Theorem 3.30 (Borel-Cantelli III)
Let $\mathcal{F}_n$ be a filtration with $\mathcal{F}_0 = \{\varnothing, \Omega\}$ and $A_n \in \mathcal{F}_n$. Then

$$\mathrm{P}\left\{A_n \text{ i.o.}\right\} = \mathrm{P}\left\{\sum_n \mathrm{P}(A_n|\mathcal{F}_{n-1}) = \infty\right\}.$$

Proof. Let $X_0 = 0$ and $X_n = \sum_{k\le n} \mathbf{1}_{A_k} - \mathrm{P}(A_k|\mathcal{F}_{k-1})$. Notice that X_n is bounded and $\mathcal{F}_n$-adapted. Also,

$$\mathrm{E}\left[X_{n+1}|\mathcal{F}_n\right] = X_n + \mathrm{E}\left[\mathbf{1}_{A_{n+1}}|\mathcal{F}_n\right] - \mathrm{P}(A_{n+1}|\mathcal{F}_n) = X_n.$$

Thus X_n is a martingale with bounded increments. By theorem 3.29,

$$C = \left\{\lim_{n\to\infty} X_n \text{ exists and is finite}\right\}, \quad \text{and} \quad D = \left\{\liminf_{n\to\infty} X_n = -\infty, \text{ and } \limsup_{n\to\infty} X_n = \infty\right\}$$

satisfy that $\mathrm{P}(C \cup D) = 1$. Notice that $\left\{\sum_n \mathbf{1}_{A_n} = \infty\right\} = \{A_n \text{ i.o.}\}$. On C, $\sum_n \mathbf{1}_{A_n} = \infty$ if and only if $\sum_n \mathrm{P}(A_n|\mathcal{F}_{n-1}) = \infty$. On D, we have $\sum_n \mathbf{1}_{A_n} = \infty$ and $\sum_n \mathrm{P}(A_n|\mathcal{F}_{n-1}) = \infty$. Since $\mathrm{P}(C \cup D) = 1$, we conclude that $\{A_n \text{ i.o.}\} = \{\sum_n \mathrm{P}(A_n|\mathcal{F}_{n-1}) = \infty\}$ up to some measure zero sets and the result follows. ∎

Lemma 3.31
Let X_n be a submartingale and τ be a stopping time with $\tau \le T$. Then

$$\mathrm{E}\left[X_0\right] \le \mathrm{E}\left[X_\tau\right] \le \mathrm{E}\left[X_T\right].$$

Proof. Note that $X_{n\wedge\tau}$ is again a submartingale. Then

$$\mathrm{E}\left[X_0\right] = \mathrm{E}\left[X_{0\wedge\tau}\right] \le \mathrm{E}\left[\mathrm{E}\left[X_{T\wedge\tau}|\mathcal{F}_0\right]\right] = \mathrm{E}\left[X_\tau\right].$$

Also, for fixed $0 \le k \le T$,

$$\mathrm{E}\left[X_\tau \mathbf{1}\left\{\tau = k\right\}\right] = \mathrm{E}\left[X_k \mathbf{1}\left\{\tau = k\right\}\right] \le \mathrm{E}\left[\mathrm{E}\left[X_T|\mathcal{F}_k\right]\mathbf{1}\left\{\tau = k\right\}\right] = \mathrm{E}\left[X_T \mathbf{1}\left\{\tau = k\right\}\right].$$

Summing over $0 \le k \le T$ gives $\mathrm{E}\left[X_\tau\right] \le \mathrm{E}\left[X_T\right]$. ∎

Theorem 3.32 (Doob's Submatingale Inequality)
Let X_n be a submartingale and $\lambda > 0$. Define $X_n^* = \max_{0\le k\le n} X_k^+$. Then

$$\mathrm{P}\left\{X_n^* \ge \lambda\right\} \le \frac{1}{\lambda}\mathrm{E}\left[X_n \mathbf{1}\left\{X_n^* \ge \lambda\right\}\right] \le \frac{1}{\lambda}\mathrm{E}\left[X_n^+\right].$$

Proof. Let $A = \left\{X_n^* \ge \lambda\right\}$ and $\tau = \inf\left\{k \mid X_k \ge \lambda\right\} \wedge n$. On A, $X_\tau \ge \lambda$ and hence

$$\mathrm{P}\left\{X_n^* \ge \lambda\right\} = \mathrm{E}\left[\mathbf{1}_A\right] \le \frac{1}{\lambda}\mathrm{E}\left[X_\tau \mathbf{1}_A\right].$$

Notice that by lemma 3.31, $\mathrm{E}\left[X_\tau\right] \le \mathrm{E}\left[X_n\right]$. Also, $X_\tau \mathbf{1}_{A^c} = X_n \mathbf{1}\tau_{A^c}$. Thus $\mathrm{E}\left[X_\tau \mathbf{1}_A\right] \le$

$\mathrm{E}[X_n \mathbf{1}_A]$. The first inequality holds. The second inequality is trivial since $X_n \mathbf{1}_A \leq X_n^+ \mathbf{1}_A \leq X_n^+$. The proof is complete. ■

Corollary 3.33
Let M_n be a martingale and $\mathrm{E}[|M_n|^p] < \infty$ for some $p \geq 1$. For any $\lambda > 0$,

$$\mathrm{P}\left\{\max_{0\leq k\leq n} |M_k| \geq \lambda\right\} \leq \frac{1}{\lambda^p} \mathrm{E}\left[|M_n|^p\right].$$

Proof. For $p \geq 1$, $x \mapsto |x|^p$ is convex and thus $|M_n|^p$ is a submartingale. By the Doob's submartingale inequality,

$$\mathrm{P}\left\{\max_{0\leq k\leq n} |M_k| \geq \lambda\right\} = \mathrm{P}\left\{\max_{0\leq k\leq n} |M_k|^p \geq \lambda^p\right\} \leq \frac{1}{\lambda^p} \mathrm{E}\left[|M_k|^p\right].$$

■

Theorem 3.34 ($\mathcal{L}^p$ Maximal Inequality)
Let X_n be a submartingale and $X_n^* = \max_{0\leq k\leq n} X_k^+$. For $1 < p < \infty$, we have

$$\mathrm{E}\left[\left|X_n^*\right|^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathrm{E}\left[X_n^+\right]^p.$$

Furthermore, if X_n is a martingale, then

$$\mathrm{E}\left[\left(\max_{0\leq k\leq n} |X_k|\right)^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathrm{E}\left[|X_n|^p\right].$$

Proof. Suppose first that $X_n^* \in \mathcal{L}^p$. By lemma 2.12, Doob's submartingale inequality, Fubini theorem and Hölder's inequality,

$$\begin{aligned}
\mathrm{E}\left[(X_n^*)^p\right] &= \int_0^\infty p\lambda^{p-1} \mathrm{P}(X_n^* \geq \lambda) d\lambda \leq \int_0^\infty p\lambda^{p-1} \frac{1}{\lambda} \mathrm{E}\left[X_n \mathbf{1}\left\{X_n^* \geq \lambda\right\}\right] d\lambda \\
&\leq \int_0^\infty p\lambda^{p-2} \mathrm{E}\left[X_n^+ \mathbf{1}\left\{X_n^* \geq \lambda\right\}\right] d\lambda = p\,\mathrm{E}\left[X_n^+ \int_0^{X_n^*} \lambda^{p-2} d\lambda\right] \\
&= \frac{p}{p-1} \mathrm{E}\left[X_n^+ (X_n^*)^{p-1}\right] \leq \frac{p}{p-1} \mathrm{E}\left[\left|X_n^+\right|^p\right]^{1/p} \mathrm{E}\left[\left|X_n^*\right|^p\right]^{(p-1)/p}.
\end{aligned}$$

Rearranging the inequality gives

$$\mathrm{E}\left[\left|X_n^*\right|^p\right] \leq \left(\frac{p}{p-1}\right)^p \mathrm{E}\left[(X_n^+)^p\right].$$

For general X_n, consider $Y_n \in \mathcal{L}^p$ such that $Y_n \nearrow X_n$. Since $Y_n^* \in \mathcal{L}^p$ as well, the result extends to X_n by the monotone convergence theorem. For the second part, apply the previous result

to the submartingale $|X_n|$. Then

$$\mathrm{E}\left[\left(\max_{0\le k\le n} X_k\right)^p\right] \le \left(\frac{p}{p-1}\right)^p \mathrm{E}\left[|X_n|^p\right].$$

This finishes the proof. ∎

Example
Let S_n be a symmetric simple random walk on $\mathbb{Z}$ with $S_0 = 1$. Let $\tau = \inf\{k \mid S_k = 0\}$ be the hitting time. Put $X_n = S_{n\wedge\tau}$. Then

$$\mathrm{P}\left\{\sup_{k\ge 0} X_k \ge M\right\} = \frac{1}{M}.$$

To see this, note that by the martingale convergence theorem, X_k converges to some $X \in \mathcal{L}^1$ almost surely since X_k is bounded. By the optional stopping theorem, for the hitting time $N = \inf\{k \mid X_k = M \text{ or } X_k = 0\}$,

$$1 = \mathrm{E}[S_0] = \mathrm{E}[X_0] = \mathrm{E}[X_N] = M\,\mathrm{P}(X_N = M) = M\,\mathrm{P}\left\{\sup_{k\ge 0} X_k \ge M\right\}.$$

Rearranging the equality shows the desired equation. Now, by the LMCT,

$$\lim_{n\to\infty} \mathrm{E}\left[\max_{k\le n}|X_k|\right] \ge \lim_{n\to\infty} \mathrm{E}\left[\max_{k\le n} X_k\right] = \mathrm{E}\left[\sup_{k\ge 0} X_k\right] = \sum_{M\ge 1} \mathrm{P}\left\{\sup_{k\ge 0} X_k \ge M\right\} = \sum_{M\ge 1}\frac{1}{M} = \infty.$$

However, $\mathrm{E}[X_n] = \mathrm{E}[S_{n\wedge\tau}] = \mathrm{E}[S_0] = 1$. We see that the $\mathcal{L}^p$ inequality fails for $p = 1$.

Theorem 3.35 (Martingale Convergence Theorem II)
Let X_n be a martingale and $\sup_n \mathrm{E}[|X_n|^p] < \infty$ for some $p > 1$. Then $X_n \to X$ almost surely and in $\mathcal{L}^p$.

Proof. First, note that $|X_n|^p$ is a submartingale. The almost sure convergence follows immediately from the martingale convergence theorem I. Next, since $|X_n - X|^p \le 2\sup_n |X_n|^p$ and by LMCT and the $\mathcal{L}^p$ martingale inequality,

$$\mathrm{E}\left[\sup_n |X_n|^p\right] = \lim_{n\to\infty} \mathrm{E}\left[\max_{0\le k\le n}|X_k|^p\right] \le \lim_{n\to\infty}\left(\frac{p}{p-1}\right)^p \mathrm{E}\left[|X_n|^p\right] \le \left(\frac{p}{p-1}\right)^p \sup_n \mathrm{E}\left[|X_n|^p\right] < \infty.$$

We see that $2\sup_n |X_n|^p$ is integrable and hence by LDCT, $X_n \to X$ in $\mathcal{L}^p$. ∎

3.3. Uniform Integrability

Definition 3.36

A collection of random variable $\{X_i\}_{i\in I}$ is uniformly integrable if

$$\lim_{M\to\infty}\sup_{i\in I}\mathrm{E}\left[|X_i|\,\mathbf{1}\left\{|X_i|>M\right\}\right]=0.$$

Remark

The uniform integrability implies that $\sup_i \mathrm{E}\left[|X_i|\right]<\infty$. *To see this, note that we can find M such that*

$$\sup_i \mathrm{E}\left[|X_i|\,\mathbf{1}\left\{|X_i|>M\right\}\right]<\epsilon.$$

Then

$$\sup_i \mathrm{E}\left[|X_i|\right]=\sup_i \mathrm{E}\left[|X_i|\,\mathbf{1}\left\{|X_i|>M\right\}\right]+\mathrm{E}\left[|X_i|\,\mathbf{1}\left\{|X_i|\le M\right\}\right]\le\epsilon+M<\infty.$$

Proposition 3.37

If $\{X_i\}$ and $\{Y_i\}$ are two collections of random variables, then $\{X_i+Y_i\}$ is also uniformly integrable.

Proof. Note that

$$\begin{aligned}|X_i+Y_i|\,\mathbf{1}\left\{|X_i+Y_i|>M\right\}&\le 2\max\left\{|X_i|,|Y_i|\right\}\mathbf{1}\left\{2\max\left\{|X_i|,|Y_i|\right\}>M\right\}\\&\le 2(|X_i|\,\mathbf{1}\left\{|X_i|>M/2\right\}+|Y_i|\,\mathbf{1}\left\{|Y_i|>M/2\right\}).\end{aligned}$$

Then

$$\begin{aligned}&\lim_{M\to\infty}\sup_i\mathrm{E}\left[|X_i+Y_i|\,\mathbf{1}\left\{|X_i+Y_i|>M\right\}\right]\\&\le\lim_{M\to\infty}2\left(\sup_i\mathrm{E}\left[|X_i|\,\mathbf{1}\left\{|X_i|>M/2\right\}\right]+\sup_i\mathrm{E}\left[|Y_i|\,\mathbf{1}\left\{|Y_i|>M/2\right\}\right]\right)=0.\end{aligned}$$

■

Proposition 3.38

If $\sup_i \mathrm{E}\left[|X_i|^p\right]<\infty$ *for $p>1$, then $\{X_i\}$ is uniformly integrable.*

Proof. Let $C=\sup_i \mathrm{E}\left[|X_i|^p\right]^{1/p}<\infty$. Notice that by the Hölder's inequality,

$$\mathrm{E}\left[|X_i|\,\mathbf{1}\left\{|X_i|>M\right\}\right]\le\mathrm{E}\left[|X_i|^p\right]^{1/p}\mathrm{E}\left[\mathbf{1}\left\{|X_i|>M\right\}\right]^{1/q}\le C\,\mathrm{P}\left\{|X_i|>M\right\}^{1/q}.$$

Suppose that for every M we can find a subsequence such that $\mathrm{P}\left\{\left|X_{i_k}\right|>M\right\}\ge\delta>0$. Then $C\ge\mathrm{E}\left[\left|X_{i_k}\right|^p\right]\ge\mathrm{E}\left[\left|X_{i_k}\right|^p\mathbf{1}\left\{\left|X_{i_k}\right|>M\right\}\right]\ge M^p\delta\to\infty$ as $M\to\infty$, which is a contradiction. Hence $\sup_i \mathrm{P}\left\{|X_i|>M\right\}\to 0$ as $M\to\infty$ and

$$\sup_i\mathrm{E}\left[|X_i|\,\mathbf{1}\left\{|X_i|>M\right\}\right]\le C\,\mathrm{P}\left\{|X_i|>M\right\}^{1/q}\to 0$$

as $M \to \infty$. We conclude that $\{X_i\}$ is uniformly integrable. ∎

Example
If we only have $\sup_i \mathrm{E}[|X_i|] < \infty$*, then the result does not hold in general. For instance, consider* X_n *such that* $\mathrm{P}(X_n = 0) = 1 - 1/n$ *and* $\mathrm{P}(X_n = n) = 1/n$*. Then* $\mathrm{E}[X_n] = 1 < \infty$ *but*

$$\lim_{M\to\infty} \sup_n \mathrm{E}[|X_n| \mathbf{1}\{|X_n| > M\}] = 1 \neq 0.$$

Theorem 3.39
$\{X_i\}$ *is uniformly integrable if and only if* $\sup_i \mathrm{E}[|X_i|] < \infty$ *and for all* $\epsilon > 0$*, there is* $\delta > 0$ *such that* $\mathrm{P}(A) \leq \delta$ *implies that* $\mathrm{E}[|X_i| \mathbf{1}_A] < \epsilon$*.*

Proof. Assume that $\{X_i\}$ is uniformly integrable. Then $\sup_i \mathrm{E}[|X_i|] < \infty$. For every $\epsilon > 0$, we can find M such that $\mathrm{E}[|X_i| \mathbf{1}\{|X_i| > M\}] < \epsilon/2$. Pick $\delta = \epsilon/(2M)$. We have

$$\begin{aligned}\mathrm{E}[|X_i| \mathbf{1}_A] &= \mathrm{E}[|X_i| \mathbf{1}\{|X_i| > M\} \mathbf{1}_A] + \mathrm{E}[|X_i| \mathbf{1}\{|X_i| \leq M\} \mathbf{1}_A] \\ &\leq \mathrm{E}[|X_i| \mathbf{1}\{|X_i| > M\}] + M\,\mathrm{P}(A) \leq \frac{\epsilon}{2} + M\frac{\epsilon}{2M} = \epsilon.\end{aligned}$$

For the converse, notice that by the Markov inequality,

$$\mathrm{P}\{|X_i| > M\} \leq \frac{1}{M}\mathrm{E}[|X_i|] \leq \frac{1}{M}\sup_i \mathrm{E}[|X_i|].$$

For any $\epsilon > 0$, there is some $\delta > 0$ such that if $\mathrm{P}(A) \leq \delta$, $\mathrm{E}[|X_i| \mathbf{1}_A] < \epsilon$. Put $A_M = \{|X_i| > M\}$. For any $M > M_0 = \delta^{-1} \sup_i \mathrm{E}[|X_i|]$, $\mathrm{P}(A_M) \leq \delta$ and hence $\mathrm{E}\left[|X_i| \mathbf{1}_{A_M}\right] < \epsilon$. Since the choice of M_0 is uniform in i, $\{X_i\}$ is uniformly integrable. ∎

Lemma 3.40
$X_n \to 0$ *in* $\mathcal{L}^1$ *if and only if* $X_n \xrightarrow{p} 0$ *and* $\{X_n\}$ *is uniformly integrable.*

Proof. Assume that $X_n \to 0$ in $\mathcal{L}^1$. Then $X_n \xrightarrow{p} 0$. Note that $\{X_n\}$ is convergent in $\mathcal{L}^1$ and hence must be $\mathcal{L}^1$ bounded. We have $\sup_n \mathrm{E}[|X_n|] < \infty$. For every $\epsilon > 0$, we can find N such that for $n > N$, $\mathrm{E}[|X_n|] < \epsilon$. Pick M such that $\mathrm{E}[|X_n| \mathbf{1}\{|X_n| \geq M\}] < \epsilon$ for all $n \leq N$. Hence X_n is uniformly integrable.

Conversely, for fixed M and ϵ, write

$$\mathrm{E}[|X_n|] = \mathrm{E}[|X_n| \mathbf{1}\{|X_n| \geq M\}] + \mathrm{E}[|X_n| \mathbf{1}\{\epsilon \leq |X_n| < M\}] + \mathrm{E}[|X_n| \mathbf{1}\{|X_n| < \epsilon\}].$$

The first term converges to 0 by the uniform integrability. For the second term,

$$\mathrm{E}[|X_n| \mathbf{1}\{\epsilon \leq |X_n| < M\}] \leq M\,\mathrm{P}(|X_n| \geq \epsilon) \to 0$$

by the convergence in probability. The last term is bounded by ϵ. Hence $\mathrm{E}[|X_n|] \to \epsilon$ as $n \to \infty$. Since ϵ can be arbitrarily small, $X_n \to 0$ in $\mathcal{L}^1$. ∎

Theorem 3.41
Let $X_n \xrightarrow{p} X$ and $\mathrm{E}[|X_n|] < \infty$. Then the followings are equivalent:

(a) $\{X_n\}$ *is uniformly integrable.*

(b) $X_n \to X$ *in* $\mathcal{L}^1$.

(c) $\mathrm{E}[|X_n|] \to \mathrm{E}[|X|]$.

Proof. We first prove that (a) implies (b). By the unifom integrability, $\sup_n \mathrm{E}[|X_n|] < \infty$. By the Fatou's lemma (corollary 2.9),

$$\mathrm{E}[|X|] \le \liminf_{n\to\infty} \mathrm{E}[|X_n|] \le \sup_n \mathrm{E}[|X_n|] < \infty.$$

Hence X itself is uniformly integrable. By proposition 3.37, $\{X_n - X\}$ is uniformly integrable. Let $Y_n = |X_n - X| \xrightarrow{p} 0$. It now suffices to prove that $\mathrm{E}[Y_n] \to 0$. Notice that for $M > 0$,

$$\begin{aligned}\mathrm{E}[|Y_n|] &= \mathrm{E}[|Y_n|\,\mathbf{1}\{|Y_n| \le M\}] + \mathrm{E}[|Y_n|\,\mathbf{1}\{|Y_n| > M\}] \\ &\le \mathrm{E}[|Y_n|\,\mathbf{1}\{|Y_n| \le M\}] + \sup_n \mathrm{E}[|Y_n|\,\mathbf{1}\{|Y_n| > M\}]\end{aligned}$$

Taking $n \to \infty$, the first term converges to 0 by corollary 2.9 since $|Y_n|\,\mathbf{1}\{|Y_n| \le M\} \le M \in \mathcal{L}^1$. We see that

$$\limsup_{n\to\infty} \mathrm{E}[|Y_n|] \le \sup_n \mathrm{E}[|Y_n|\,\mathbf{1}\{|Y_n| > M\}].$$

Taking $M \to \infty$, the uniform integrability implies that the right hand side converges to 0. We conclude that $\mathrm{E}[|Y_n|] \to$ and $X_n \to X$ in $\mathcal{L}^1$.

Now assume (b) holds.

$$|\mathrm{E}[|X_n|] - \mathrm{E}[|X|]| \le \mathrm{E}[|X_n - X|] \to 0$$

by the assumption.

Suppose that (c) holds. Note that for $M > 0$,

$$\mathrm{E}[|X_n|\,\mathbf{1}\{|X_n| > M\}] = \mathrm{E}[|X_n|] - \mathrm{E}[|X_n|\,\mathbf{1}\{|X_n| \le M\}] \to \mathrm{E}[|X|] - \mathrm{E}[|X|\,\mathbf{1}\{|X| \le M\}]$$

as $n \to \infty$ by corollary 2.9 with $|X_n|\,\mathbf{1}\{|X_n| \le M\} \to |X|\,\mathbf{1}\{|X| \le M\}$ and that $|X_n|\,\mathbf{1}\{|X_n| \le M\} \le M \in \mathcal{L}^1$. Notice that the right hand side convegres to 0 as $M \to \infty$ by LMCT. Hence, for $\epsilon > 0$, we can find M_0 so that if $M > M_0$, there is some $n_0(M_0)$ such that $\mathrm{E}[|X_n|\,\mathbf{1}\{|X_n| > M\}] < \epsilon$ for all $n > n_0$. Note that $\{X_n\}_{n \le n_0}$ is uniformly integrable. We conclude that $\{X_n\}$ is uniformly integrable. ∎

Theorem 3.42
If $X \in \mathcal{L}^1$, then $\{\mathrm{E}[X|\mathcal{G}] \mid \mathcal{G} \subset \mathcal{F}$ is a σ-algebra$\}$ is uniformly integrable.

Proof. To prove the theorem, we use the characterization theorem 3.39. Note that for $\mathcal{G} \subset \mathcal{F}$,

$$\mathrm{E}\left[\left|\mathrm{E}\left[X|\mathcal{G}\right]\right|\right] \leq \mathrm{E}\left[\mathrm{E}\left[\left|X\right||\mathcal{G}\right]\right] = \mathrm{E}\left[\left|X\right|\right] < \infty.$$

Hence $\sup_{\mathcal{G}\subset\mathcal{F}} \mathrm{E}\left[\left|\mathrm{E}\left[X|\mathcal{G}\right]\right|\right] < \infty$. Also, for $M > 0$,

$$\begin{aligned}\mathrm{E}\left[\left|\mathrm{E}\left[X|\mathcal{G}\right]\right|\mathbf{1}\left\{\left|\mathrm{E}\left[X|\mathcal{G}\right]\right| > M\right\}\right] &\leq \mathrm{E}\left[\mathrm{E}\left[\left|X\right||\mathcal{G}\right]\mathbf{1}\left\{\mathrm{E}\left[\left|X\right||\mathcal{G}\right] > M\right\}\right] \\ &= \mathrm{E}\left[\left|X\right|\mathbf{1}\left\{\mathrm{E}\left[\left|X\right||\mathcal{G}\right] > M\right\}\right].\end{aligned}$$

Since $\{X\}$ is uniformly integrable, for any $\epsilon > 0$, there is $\delta > 0$ such that $\mathrm{P}(A) \leq \delta$ implies that $\mathrm{E}\left[\left|X\right|\mathbf{1}_A\right] < \epsilon$. For M sufficiently large such that

$$\mathrm{P}\left\{\mathrm{E}\left[\left|X\right||\mathcal{G}\right] > M\right\} \leq \frac{1}{M}\mathrm{E}\left[\mathrm{E}\left[\left|X\right||\mathcal{G}\right]\right] = \frac{1}{M}\mathrm{E}\left[\left|X\right|\right] \leq \delta,$$

we have

$$\mathrm{E}\left[\left|\mathrm{E}\left[X|\mathcal{G}\right]\right|\mathbf{1}\left\{\left|\mathrm{E}\left[X|\mathcal{G}\right]\right| > M\right\}\right] \leq \mathrm{E}\left[\left|X\right|\mathbf{1}\left\{\mathrm{E}\left[\left|X\right||\mathcal{G}\right] > M\right\}\right] < \epsilon.$$

Since the chocie of M does not depend on $\mathcal{G}$, we see that $\{\mathrm{E}\left[X|\mathcal{G}\right] \mid \mathcal{G} \subset \mathcal{F} \text{ is a } \sigma\text{-algebra}\}$ is indeed uniformly integrable. ∎

Theorem 3.43

Let X_n be a submartingale. Then the followings are equivalent.

(a) X_n is uniformly integrable.

(b) X_n converges almost surely and in $\mathcal{L}^1$.

(c) X_n converges in $\mathcal{L}^1$.

If, furthermore, X_n is a martingale, then (a), (b) and (c) are also equivalent to that there is an $X \in \mathcal{L}^1$ such that $X_n = \mathrm{E}\left[X|\mathcal{F}_n\right]$.

Proof. Assume (a). By the uniform integrability we have $\sup_n \mathrm{E}\left[\left|X_n\right|\right] < \infty$. By the martingale convergence theorem, X_n converges almost surely. By theorem 3.41, X_n converges in $\mathcal{L}^1$. Hence (b) holds.

(b) to (c) is trivial.

Assume (c). Since X_n converges in $\mathcal{L}^1$, it also converges in probability. Also, $\mathrm{E}\left[\left|X_n\right|\right] < \infty$ by the submartingale property. By theorem 3.41, X_n is uniform integrable. (a) holds.

Now suppose that X_n is a martingale. If there is an X such that $X_n = \mathrm{E}\left[X|\mathcal{F}_n\right]$. Then by theorem 3.42, $\{X_n\}$ is uniformly integrable. Conversely, assume (c). Write $X_n \to X$ in $\mathcal{L}^1$. Let $A \in \mathcal{F}_n$. By the martingale property, $\mathrm{E}\left[X_m|\mathcal{F}_n\right] = X_n$ for all $m > n$ and $\mathrm{E}\left[X_m\mathbf{1}_A\right] = \mathrm{E}\left[X_n\mathbf{1}_A\right]$. Also, since

$$\mathrm{E}\left[\left|X_m\mathbf{1}_A - X\mathbf{1}_A\right|\right] \leq \mathrm{E}\left[\left|X_m - X\right|\right] \to 0,$$

$\mathrm{E}\left[X_m\mathbf{1}_A\right] \to \mathrm{E}\left[X\mathbf{1}_A\right]$. We see that $\mathrm{E}\left[X_n\mathbf{1}_A\right] = \mathrm{E}\left[X\mathbf{1}_A\right]$ and $X_n = \mathrm{E}\left[X|\mathcal{F}_n\right]$. ∎

Theorem 3.44 (Levy's Upward Theorem)
Let $X \in \mathcal{L}^1$ and $\mathcal{F}_n \nearrow \mathcal{F}$. Then $\mathrm{E}[X|\mathcal{F}_n] \to \mathrm{E}[X|\mathcal{F}]$ almost surely and in $\mathcal{L}^1$.

Proof. Let $Y_n = \mathrm{E}[X|\mathcal{F}_n]$. By theorem 3.43, Y_n converges almost surely and in $\mathcal{L}^1$, say to Y, which we can pick to be $\mathcal{F}$-measurable. It now suffices to verify that $Y = \mathrm{E}[X|\mathcal{F}]$. To see this, let $\mathcal{P}$ be the union of $\mathcal{F}_n$. Since $\mathcal{F}_n$ form a filtration, $\mathcal{P}$ is a π-system. Consider

$$\mathcal{L} = \{A \in \mathcal{F} \mid \mathrm{E}[Y\mathbf{1}_A] = \mathrm{E}[X\mathbf{1}_A]\}.$$

We claim that $\mathcal{L}$ is a λ-system. First,

$$\mathrm{E}[Y\mathbf{1}_\Omega] = \mathrm{E}[Y] = \mathrm{E}\left[\lim_{n\to\infty} \mathrm{E}[X|\mathcal{F}_n]\right] = \lim_{n\to\infty} \mathrm{E}[\mathrm{E}[X|\mathcal{F}_n]] = \mathrm{E}[X] = \mathrm{E}[X\mathbf{1}_\Omega].$$

Hence $\Omega \in \mathcal{L}$. Next, if $A, B \in \mathcal{L}$ and $A \subset B$, then

$$\mathrm{E}[Y\mathbf{1}_{B-A}] = \mathrm{E}[Y\mathbf{1}_B] - \mathrm{E}[Y\mathbf{1}_A] = \mathrm{E}[X\mathbf{1}_B] - \mathrm{E}[X\mathbf{1}_A] = \mathrm{E}[X\mathbf{1}_{B-A}].$$

Hence $B - A \in \mathcal{L}$. Finally, if $A_n \in \mathcal{L}$ and $A_n \nearrow A$, then by the LMCT,

$$\begin{aligned}\mathrm{E}[Y\mathbf{1}_A] &= \mathrm{E}\left[Y^+\mathbf{1}_A\right] - \mathrm{E}[Y^-\mathbf{1}_A] = \lim_{n\to\infty} \mathrm{E}\left[Y^+\mathbf{1}_{A_n}\right] - \lim_{n\to\infty} \mathrm{E}\left[Y^-\mathbf{1}_{A_n}\right] \\ &= \lim_{n\to\infty} \mathrm{E}\left[X^+\mathbf{1}_{A_n}\right] - \lim_{n\to\infty} \mathrm{E}\left[X^-\mathbf{1}_{A_n}\right] = \mathrm{E}\left[X^+\mathbf{1}_A\right] - \mathrm{E}[X^-\mathbf{1}_A] = \mathrm{E}[X\mathbf{1}_A]\end{aligned}$$

We conclude that $A \in \mathcal{L}$ and hence $\mathcal{L}$ is a λ-system. For any $A \in \mathcal{P}$, $A \in \mathcal{F}_n$ for some n.

$$\mathrm{E}[Y\mathbf{1}_A] = \mathrm{E}\left[\lim_{n\to\infty} Y_n\mathbf{1}_A\right] = \mathrm{E}\left[\lim_{n\to\infty} \mathrm{E}[X\mathbf{1}_A|\mathcal{F}_n]\right] = \lim_{n\to\infty} \mathrm{E}[\mathrm{E}[X\mathbf{1}_A|\mathcal{F}_n]] = \mathrm{E}[X\mathbf{1}_A].$$

Thus $A \in \mathcal{L}$ and $\mathcal{P} \subset \mathcal{L}$. By the π-λ theorem, for all $A \in \mathcal{F}$, $\mathrm{E}[X\mathbf{1}_A] = \mathrm{E}[Y\mathbf{1}_A]$. This shows that $Y = \mathrm{E}[X|\mathcal{F}]$. ∎

Corollary 3.45 (Levy's Zero-One Law)
If $\mathcal{F}_n \nearrow \mathcal{F}$ and $A \in \mathcal{F}$, then $\mathrm{P}(A|\mathcal{F}_n) \to \mathbf{1}_A$ almost surely.

Proof. By Levy's upward theorem, $\mathrm{P}(A|\mathcal{F}_n) = \mathrm{E}[\mathbf{1}_A|\mathcal{F}_n] \to \mathrm{E}[\mathbf{1}_A|\mathcal{F}] = \mathbf{1}_A$ almost surely. ∎

Corollary 3.46
Suppose that $X_n \to X$ almost surely with $X_n \in \mathcal{L}^1$. If $|X_n| \le Y$ for some $Y \in \mathcal{L}^1$ and $\mathcal{F}_n \nearrow \mathcal{F}$, $\mathrm{E}[X_n|\mathcal{F}_n] \to \mathrm{E}[X|\mathcal{F}]$ almost surely.

Proof. Notice that

$$|\mathrm{E}[X_n|\mathcal{F}_n] - \mathrm{E}[X|\mathcal{F}]| \le |\mathrm{E}[X_n|\mathcal{F}_n] - \mathrm{E}[X|\mathcal{F}_n]| + |\mathrm{E}[X|\mathcal{F}_n] - \mathrm{E}[X|\mathcal{F}]|.$$

It now sufices to prove that the two terms on the right hand side converges almost surely to 0. For the first term, let $Z_N = \sup_{m,n\ge N} |X_m - X_n| \le 2Y$. Note that $Z_N \to 0$ almost surely. For

fixed N,

$$\limsup_{n\to\infty} \mathrm{E}\left[|X_n - X|\,|\mathcal{F}_n\right] \le \lim_{n\to\infty} \mathrm{E}\left[Z_N|\mathcal{F}_n\right] = \mathrm{E}\left[Z_N|\mathcal{F}\right]$$

almost surely by Levy's upward theorem. Taking $N \to \infty$ shows that

$$\left|\mathrm{E}\left[X_n|\mathcal{F}_n\right] - \mathrm{E}\left[X|\mathcal{F}_n\right]\right| \le \mathrm{E}\left[|X_n - X|\,|\mathcal{F}_n\right] \to 0.$$

almost surely since $\mathrm{E}\left[Z_N|\mathcal{F}\right] \to 0$ by LDCT. For the second term, directly applying Levy's upward theorem gives

$$\left|\mathrm{E}\left[X|\mathcal{F}_n\right] - \mathrm{E}\left[X|\mathcal{F}\right]\right| \to 0$$

almost surely. Hence the conclusion follows. ∎

Theorem 3.47 (Optional Stopping Theorem II)
Let X_n be a uniformly integrable martingale and $\sigma \le \tau$ almost surely are stopping times. Then $\mathrm{E}\left[X_\tau\right] < \infty$ and

$$X_\sigma = \mathrm{E}\left[X_\tau|\mathcal{F}_\sigma\right].$$

In particular, τ can be taken as ∞.

Proof. Since X_n is a uniformly integrable martingale, there is $X_\infty \in \mathcal{L}^1$ such that $X_n = \mathrm{E}\left[X_\infty|\mathcal{F}_n\right]$ by theorem 3.43. We may take X_∞ to be $\mathcal{F}_\infty$-measurable. We first show that

$$X_\tau = \mathrm{E}\left[X_\infty|\mathcal{F}_\tau\right].$$

Define $Y_n = \mathrm{E}\left[X_\infty^+|\mathcal{F}_n\right]$ and $Z_n = \mathrm{E}\left[X_\infty^-|\mathcal{F}_n\right]$ and $Y_\infty = X_\infty^+$, $Z_\infty = X_\infty^-$. Clearly, $X_n = Y_n - Z_n$. For $A \in \mathcal{F}_\tau$,

$$\mathrm{E}\left[Y_\infty \mathbf{1}_{A\cap\{\tau=\infty\}}\right] = \mathrm{E}\left[Y_\tau \mathbf{1}_{A\cap\{\tau=\infty\}}\right]$$

since $Y_n = \mathrm{E}\left[Y_\infty|\mathcal{F}_n\right] \to Y_\infty$ almost surely by Levy's upward theorem. Now, since $A\cap\{\tau = k\} \in \mathcal{F}_\tau$,

$$\begin{aligned}\mathrm{E}\left[Y_\infty \mathbf{1}_{A\cap\{\tau<\infty\}}\right] &= \sum_{k\ge 0} \mathrm{E}\left[Y_\infty \mathbf{1}_{A\cap\{\tau=k\}}\right] = \sum_{k\ge 0} \mathrm{E}\left[Y_k \mathbf{1}_{A\cap\{\tau=k\}}\right] \\ &= \sum_{k\ge 0} \mathrm{E}\left[Y_\tau \mathbf{1}_{A\cap\{\tau=k\}}\right] = \mathrm{E}\left[Y_\tau \mathbf{1}_{A\cap\{\tau<\infty\}}\right].\end{aligned}$$

Thus, $\mathrm{E}\left[Y_\tau \mathbf{1}_A\right] = \mathrm{E}\left[Y_\infty \mathbf{1}_A\right]$ and $\mathrm{E}\left[Y_\infty|\mathcal{F}_\tau\right] = Y_\tau$ almost surely. Similarly, $\mathrm{E}\left[Z_\infty|\mathcal{F}_\tau\right] = Z_\tau$ and hence $\mathrm{E}\left[X_\infty|\mathcal{F}_\tau\right] = X_\tau$. Now for $\sigma \le \tau$,

$$\mathrm{E}\left[X_\tau|\mathcal{F}_\sigma\right] = \mathrm{E}\left[\mathrm{E}\left[X_\infty|\mathcal{F}_\tau\right]|\mathcal{F}_\sigma\right] = \mathrm{E}\left[X_\infty|\mathcal{F}_\sigma\right] = X_\sigma.$$

The proof is complete. ∎

Lemma 3.48
For $n \ge 1$, let X_n and Y_n be random variables such that

(a) $X_n \xrightarrow{p} c$,

(b) $\{Y_n\}, \{X_n Y_n\}$ *are uniformly integrable,*

(c) $\mathrm{E}[Y_n] \to 1$.

Then $\mathrm{E}[X_n Y_n] \to c$.

Proof. It is equivalent to show that $\mathrm{E}[(X_n - c)Y_n] \to 0$. Since $X_n Y_n$ and cY_n are unifomly integrable, $(X_n - c)Y_n$ is also uniformly integrable. Now by lemma 3.40, it suffices to show that $(X_n - c)Y_n \xrightarrow{p} 0$. Indeed, let $\epsilon > 0$. If $|X_n - c|\,|Y_n| > \epsilon$, either $|X_n - c| > \epsilon/M$ or $|Y_n| > M$. Hence

$$\mathrm{P}(|X_n - c|\,|Y_n| > \epsilon) \le \mathrm{P}(|X_n - c| > \epsilon/M) + \mathrm{P}(|Y_n| > M) \to 0$$

as $M \to \infty$ since $X_n \xrightarrow{p} c$ and Y_n is unifomly integrable. ∎

Theorem 3.49 (Martingale Central Limit Theorem)
Let $X_{n,k}$, $1 \le k \le m_n$ *be an array of martingale differences, i.e.,* $\mathrm{E}\left[X_{n,k+1}|\mathcal{F}_{n,k}\right] = 0$. *Let* $S_n = \sum_{k \le m_n} X_{n,k}$. *Suppose that*

$$\mathrm{E}\left[\max_{k \le m_n} \left|X_{n,k}\right|\right] \to 0 \quad \textit{and} \quad \sum_{k=1}^{m_n} X_{n,k}^2 \xrightarrow{p} \sigma^2.$$

Then $S_n \xrightarrow{d} N(0, \sigma^2)$.

Proof. Without loss of generality, it suffices to prove the case where $\sigma = 1$. Consider

$$Y_{n,1} = X_{n,1} \quad \text{and} \quad Y_{n,k} = X_{n,k} \mathbf{1}\left\{\sum_{j \le k-1} X_{n,j}^2 \le 2\right\}$$

for $k = 2, \ldots, m_n$. Now

$$\mathrm{E}\left[Y_{n,k}|\mathcal{F}_{n,k-1}\right] = \mathrm{E}\left[X_{n,k}|\mathcal{F}_{n,k-1}\right] \mathbf{1}\left\{\sum_{j \le k-1} X_{n,j}^2 \le 2\right\} = 0.$$

$Y_{n,k}$ is also an array of martingale differences. Put

$$\tau_n = \inf\left\{k \;\middle|\; \sum_{j \le k} X_{n,j}^2 > 2\right\} \wedge m_n.$$

We have

$$\mathrm{P}(X_{n,k} \ne Y_{n,k} \text{ for some } k \le m_n) = \mathrm{P}(\tau_n \le m_n - 1) \le \mathrm{P}\left(\sum_{j \le m_n} X_{n,j}^2 > 2\right) \to 0$$

by the assumption that $\sum_{k=1}^{m_n} X_{n,k}^2 \xrightarrow{p} 1$. Let $T_n = \sum_{k \le m_n} Y_{n,k}$. We have seen that $|S_n - T_n| \xrightarrow{p} 0$. It now suffices to show that $T_n \xrightarrow{d} N(0,1)$.

Observe that

$$\log(1+ix) = \int_0^x \frac{i}{1+it} dt = \int_0^x t + i - \frac{t^2}{t-i} dt$$

since

$$\frac{i}{1+it} = \frac{1}{t-i} = \frac{t+i}{t^2+1} = t + i + (t+i)\left(\frac{1}{1+t^2} - 1\right) = t + i - \frac{(t+i)t^2}{1+t^2} = t + i - \frac{t^2}{t-i}.$$

Then

$$\left|\log(1+ix) - (ix + x^2/2)\right| \le \int_0^x \left|\frac{t^2}{t-i}\right| dt = \int_0^x \frac{t^2}{\sqrt{1+t^2}} dt \le \int_0^x t^2 dt = \frac{|x|^3}{3}$$

and

$$e^{ix} = (1+ix)e^{-\frac{x^2}{2} + r(x)}$$

with $|r(x)| \le |x|^3$.

Now,

$$\mathrm{E}\left[\exp(itT_n)\right] = \mathrm{E}\left[\left(\prod_{k=1}^{m_n}(1+itY_{n,k})\right)\exp\left(-\frac{t^2}{2}\sum_{k=1}^{m_n} Y_{n,k}^2 + \sum_{k=1}^{m_n} r(tY_{n,k})\right)\right].$$

For any fixed t,

$$\left|\sum_{k=1}^{m_n} r(tY_{n,k})\right| \le \sum_{k=1}^{m_n} |t|^3 \left|Y_{n,k}\right|^3 \le |t|^3 \sum_{k=1}^{m_n} \left|X_{n,k}\right|^3 \le |t|^3 \max_{1\le k\le m_n} \left|X_{n,k}\right| \sum_{k=1}^{m_n} \left|X_{n,k}\right|^2 \xrightarrow{p} 0$$

by the assumption (The condition $\mathrm{E}\left[\max_k \left|X_{n,k}\right|\right] \to 0$ is the $\mathcal{L}^1$ convergence and this implies $\max_{1\le k\le m_n} \left|X_{n,k}\right| \xrightarrow{p} 0$). Since $\mathrm{P}(X_{n,k} \ne Y_{n,k} \text{ for some } k \le m_n) \to 0$ as $n \to \infty$, we obtain that

$$\exp\left(-\frac{t^2}{2}\sum_{k=1}^{m_n} Y_{n,k}^2 + \sum_{k=1}^{m_n} r(tY_{n,k})\right) \xrightarrow{p} \exp\left(-\frac{t^2}{2}\right).$$

Also,

$$\begin{aligned}\mathrm{E}\left[\prod_{k=1}^{m_n}(1+itY_{n,k})\right] &= \mathrm{E}\left[\prod_{k=1}^{m_n-1}(1+itY_{n,k})\,\mathrm{E}\left[(1+itY_{n,m_n})|\mathcal{F}_{n,m_n-1}\right]\right] \\ &= \mathrm{E}\left[\prod_{k=1}^{m_n-1}(1+itY_{n,k})\right] = \cdots = 1.\end{aligned}$$

Notice that

$$\left|\left(\prod_{k=1}^{m_n}(1+itY_{n,k})\right)\exp\left(-\frac{t^2}{2}\sum_{k=1}^{m_n}Y_{n,k}^2+\sum_{k=1}^{m_n}r(tY_{n,k})\right)\right|=|\exp(itT_n)|=1.$$

It remains to show that $\prod_{k=1}^{m_n}(1+itY_{n,k})$ is unifom integrable. Note that $|1+ix|^2=1+x^2\le e^{x^2}$. We have

$$\begin{aligned}\prod_{k=1}^{m_n}\left|1+itY_{n,k}\right|&=\left|1+itX_{n,\tau_n}\right|\prod_{k=1}^{\tau_n-1}\left|1+itX_{n,k}\right|\le\exp\left(\frac{t^2}{2}\sum_{k=1}^{\tau_n-1}X_{n,k}^2\right)\left|1+itX_{n,\tau_n}\right|\\&\le e^{t^2}(1+|t|\max_{k\le m_n}\left|X_{n,k}\right|).\end{aligned}$$

Since $\max_{k\le m_n}\left|X_{n,k}\right|\to 0$ in $\mathcal{L}^1$, we see that $\prod_{k=1}^{m_n}\left|1+itY_{n,k}\right|$ is indeed unifomly integrable. It follows by lemma 3.48 that $\mathrm{E}\left[\exp(itT_n)\right]\to e^{-t^2/2}$. The proof is complete. ■

3.4. Backward Martingale

Definition 3.50

*Consider the filtration $\cdots \subset \mathcal{F}_{-1} \subset \mathcal{F}_0$. A process X_n is called an $\mathcal{F}_n$-**backward martingale** if it is $\mathcal{F}_n$-adapted and $\mathrm{E}[X_{n+1}|\mathcal{F}_n] = X_n$ for all $n < 0$.*

Remark

Alternatively, we can define $Y_n = X_{-n}$ and $\mathcal{G}_n = \mathcal{F}_{-n}$. Then the underlying process X_n is a $\mathcal{F}_n$-backward martingale if and only if Y_n is $\mathcal{G}_n$-adapted and $\mathrm{E}[Y_n|\mathcal{G}_{n+1}] = Y_{n+1}$.

Theorem 3.51

Let X_n be a backward martingale. Then $X_n \to X_{-\infty}$ as $n \to -\infty$ for some $X_{-\infty} \in \mathcal{L}^1$ almost surely and in $\mathcal{L}^1$. Furthermore, $X_{-\infty} = \mathrm{E}[X_0| \cap_{n\le 0} \mathcal{F}_n]$.

Proof. Fix $n < 0$. Let $N_a^b[n, 0]$ be the number of upcrossing of $X_n, \ldots, X_0$. By the upcrossing lemma, we have $(b-a)\,\mathrm{E}\left[N_a^b[n,0]\right] \le \mathrm{E}[(X_0 - a)^+] < \infty$. Since $N_a^b[n,0] \nearrow N_a^b[-\infty, 0]$ as $n \to -\infty$, by the LMCT we have $(b-a)\,\mathrm{E}\left[N_a^b[-\infty,0]\right] \le \mathrm{E}[(X_0-a)^+] < \infty$. This holds for all $a, b \in \mathbb{Q}$. Take $A_{a,b} = \{\liminf_{n\to-\infty} X_n < a < b < \limsup_{n\to-\infty} X_n\}$ and S be the union of $A_{a,b}$ over all $a, b \in \mathbb{Q}$. Since $N_a^b[-\infty, 0]$ is almost surely finite, we see that $\mathrm{P}(A_{a,b}) = 0$ and hence $\mathrm{P}(S) = 0$. $X_n \to X_{-\infty}$ as $n \to -\infty$. $X_n = \mathrm{E}[X_0|\mathcal{F}_n]$ by the martingale property. By theorem 3.43, the convergence of $X_n \to X_{-\infty}$ is almost surely and in $\mathcal{L}^1$.

Lastly, we show that $X_{-\infty} = \mathrm{E}[X_0| \cap_{n\le 0} \mathcal{F}_n]$. Let $\mathcal{F}_{-\infty} = \cap_{n\le 0}\mathcal{F}_n$. First, it is clear that $X_{-\infty}$ is $\mathcal{F}_{-\infty}$-measurable. For $A \in \cap_{n\le 0}\mathcal{F}_n$,

$$\mathrm{E}[X_{-\infty}\mathbf{1}_A] = \mathrm{E}\left[\lim_{n\to-\infty} X_n\mathbf{1}_A\right] = \lim_{n\to-\infty} \mathrm{E}[X_n\mathbf{1}_A] = \lim_{n\to\infty} \mathrm{E}\left[\mathrm{E}[X_0|\mathcal{F}_n]\,\mathbf{1}_A\right] = \mathrm{E}[X_0\mathbf{1}_A]\,.$$

Hence $X_{-\infty} = \mathrm{E}[X_0|\mathcal{F}_{-\infty}]$. ∎

Corollary 3.52

Let $\mathcal{G}_n \searrow \mathcal{G}$ and $X \in \mathcal{L}^1$. Then $\mathrm{E}[X|\mathcal{G}_n] \to \mathrm{E}[X|\mathcal{G}]$ almost surely and in $\mathcal{L}^1$.

Proof. Take $\mathcal{F}_n = \mathcal{G}_{-n}$ and $Y_n = \mathrm{E}[X|\mathcal{F}_{-n}]$. Then Y_n is a backward martingale with respect to $\mathcal{F}_n$. It follows by theorem 3.51 that Y_n converges almost surely to $\mathrm{E}[Y_0| \cap_{n\le 0} \mathcal{F}_{-n}] = \mathrm{E}[\mathrm{E}[X|\mathcal{G}_0]\,|\mathcal{G}] = \mathrm{E}[X|\mathcal{G}]$ both almost surely and in $\mathcal{L}^1$. ∎

Definition 3.53

*Let $f : \mathbb{R}^k \to \mathbb{R}$ and $X_1, \ldots X_n$ be samples. The **U-statistics** is defined by*

$$A_n(f) = \frac{1}{\left|I_{k,n}\right|} \sum_{\sigma\in I_{k,n}} f(X_{\sigma_1}, \ldots, X_{\sigma_k})$$

where $I_{k,n}$ collects all distinct combinations of k indices among $\{1, \ldots, n\}$ and $\left|I_{k,n}\right| = n\cdots(n-k+1)$ is the number of the combinations.

Remark

If f is symmetric, i.e., $f(X_{\sigma_1}, \ldots, X_{\sigma_k}) = f(X_{\sigma'_1}, \ldots, X_{\sigma'_k})$ for all permutations σ, σ' to the set $\{i_1, \ldots, i_k\}$, then

$$A_n(f) = \frac{1}{\binom{n}{k}} \sum_{\sigma \in J_{k,n}} f(X_{\sigma_1}, \ldots, X_{\sigma_k})$$

where $J_{k,n}$ is the subset of $I_{k,n}$ consisting of all the increasing sequence.

Definition 3.54

*Consider the product space $\Omega = \mathbb{R}^{\mathbb{N}}$. $X_n(\omega) = \omega_n$ for all $\omega \in \Omega$ are independent and identically distributed. Consider all events such that $\pi^{-1}A := \{\omega \mid \pi\omega \in A\} = A$ for any π permuting the first n-coordinate. Denote the collection of all such events by $\mathcal{E}_n$. The **exchangeable σ-algebra** is defined by $\mathcal{E} = \cap_n \mathcal{E}_n$.*

Remark

$A_n(f)$ is $\mathcal{E}_n$-measurable and thus

$$A_n(f) = \mathrm{E}\left[A_n(f)|\mathcal{E}_n\right] = \frac{1}{|I_{k,n}|} \sum_{\sigma \in I_{k,n}} \mathrm{E}\left[f(X_{\sigma_1}, \ldots, X_{\sigma_k})|\mathcal{E}_n\right] = \mathrm{E}\left[f(X_1, \ldots, X_k)|\mathcal{E}_n\right].$$

Lemma 3.55

Let $X \in \mathcal{L}^2$. If $\mathrm{E}[X|\mathcal{G}]$ is $\mathcal{F}$-measurable and X is independent of $\mathcal{F}$, then $\mathrm{E}[X|\mathcal{G}] = \mathrm{E}[X]$.

Proof. Let $Y = \mathrm{E}[X|\mathcal{G}]$. By the independence, $\mathrm{E}[XY] = \mathrm{E}[X]\,\mathrm{E}[Y] = \mathrm{E}[Y]^2$ due to $\mathrm{E}[Y] = \mathrm{E}[\mathrm{E}[X|\mathcal{G}]] = \mathrm{E}[X]$. Then

$$\begin{aligned}\mathrm{Var}[Y] &= \mathrm{E}\left[Y^2\right] - \mathrm{E}[Y]^2 = \mathrm{E}\left[Y^2 - XY\right] + \mathrm{E}[XY] - \mathrm{E}[Y]^2 \\ &= \mathrm{E}[\mathrm{E}[\mathrm{E}[X|\mathcal{G}]\,(X - \mathrm{E}[X|\mathcal{G}])|\mathcal{G}]] = 0.\end{aligned}$$

We see that $\mathrm{E}[X|\mathcal{G}]$ is a constant and hence $\mathrm{E}[X|\mathcal{G}] = \mathrm{E}[\mathrm{E}[X|\mathcal{G}]] = \mathrm{E}[X]$. ∎

Lemma 3.56

If f is bounded and measurable, then $A_n(f) \to \mathrm{E}[f(X_1, \ldots, X_k)]$ almost surely.

Proof. Note that $\mathcal{E}_n \searrow \mathcal{E}$. Hence by corollary 3.52,

$$A_n(f) = \mathrm{E}[f(X_1, \ldots, X_k)|\mathcal{E}_n] \to \mathrm{E}[f(X_1, \ldots, X_k)|\mathcal{E}]$$

almost surely. It now suffices to show that $\mathrm{E}[f(X_1, \ldots, X_k)|\mathcal{E}] = \mathrm{E}[f(X_1, \ldots, X_k)]$. Notice that there are $k(n-1)\cdots(n-1-(k-1)+1) = k(n-1)\cdots(n-k+1)$ terms involving X_1 in $A_n(f)$. Hence

$$\frac{1}{|I_{k,n}|} \sum_{\sigma \in I_{k,n}, 1 \in \sigma} f(X_{\sigma_1}, \ldots, X_{\sigma_k}) \le \frac{k(n-1)\cdots(n-k+1)}{n\cdots(n-k+1)} \sup f \to 0$$

as $n \to \infty$. Hence $\mathrm{E}\left[f(X_1, \ldots, X_k)|\mathcal{E}\right]$ is $\sigma(X_2, \ldots)$-measurable. Repeating the procedure, we see that $\mathrm{E}\left[f(X_1, \ldots, X_k)|\mathcal{E}\right]$ is $\sigma(X_{k+1}, \ldots)$-measurable. By lemma 3.55, since $f(X_1, \ldots, X_k)$ is independent of $\sigma(X_{k+1}, \ldots)$,

$$\mathrm{E}\left[f(X_1, \ldots, X_k)|\mathcal{E}\right] = \mathrm{E}\left[f(X_1, \ldots, X_k)\right].$$

The proof is complete. ∎

Theorem 3.57 (Hewitt-Savage Zero-One Law)
If X_n is independent and identically distributed, then $\mathcal{E}$ is trivial.

Proof. By lemma 3.56, we see that

$$\mathrm{E}\left[f(X_1, \ldots, X_k)|\mathcal{E}\right] = \mathrm{E}\left[f(X_1, \ldots, X_k)\right]$$

for all bounded f. This implies that $\mathcal{E}$ is independent of $\mathcal{G}_k = \sigma(X_1, \ldots, X_k)$. Let $\mathcal{G} = \cup_k \mathcal{G}_k$. Clearly $\mathcal{G}$ is a π-system. By theorem 1.52, $\sigma(\mathcal{G})$ is independent of $\mathcal{E}$. In other words, $\mathcal{E}$ and $\sigma(X_1, \ldots)$ are independent. Also notice that $\mathcal{E} \subset \sigma(\mathcal{G})$. Then for all $A \in \mathcal{E}$, $\mathrm{P}(A) = \mathrm{P}(A \cap A) = \mathrm{P}(A)^2$ and thus $\mathrm{P}(A) \in \{0, 1\}$. ∎

Definition 3.58
*A sequence of random variables X_n is **exchangeable** if the joint distribution is invariant under finite permutation.*

Example
Consider X_n being drawn from $Ber(1/2)$ or $Ber(1/3)$ with probability $1/2$. Then X_n is exchangeable but not independent and identically distributed.

Proposition 3.59
Suppose that $X_n \in \mathcal{L}^2$ are exchangeable. Then $\mathrm{Cov}\left[X_1, X_2\right] \geq 0$.

Proof. By the exchangeability, $\mu = \mathrm{E}\left[X_i\right]$, $\sigma^2 = \mathrm{Var}\left[X_i\right]$ and $\rho = \mathrm{Cov}\left[X_i, X_j\right]$ for $i \neq j$. For any n,

$$0 \leq \mathrm{Var}\left[X_1, \ldots, X_n\right] = n\sigma^2 + n(n-1)\rho.$$

Hence $\rho \geq \sigma^2/(n-1)$. This holds for all n and thus $\rho \geq 0$. ∎

Theorem 3.60 (de Finetti)
If X_n is exchangeable, then conditional on $\mathcal{E}$, X_n are independent and identically distributed. Equivalently, for bounded $f_1, \ldots, f_k$,

$$\mathrm{E}\left[f_1(X_1) \cdots f_k(X_k)|\mathcal{E}\right] = \prod_{i=1}^{k} \mathrm{E}\left[f_i(X_1)|\mathcal{E}\right].$$

Proof. When $k = 1$, the result is trivial. Suppose that the reuslt holds for k. Let

$$f(X_1, \ldots, X_{k+1}) = \prod_{i=1}^{k+1} f_i(X_i)$$

where f_i are bounded. Set $g(X_1, \ldots, X_k) = \prod_{i=1}^{k} f_i(X_i)$. By lemma 3.56, we again see that

$$A_n(f) = \mathrm{E}\left[g(X_1, \ldots, X_k) f_{k+1}(X_{k+1}) | \mathcal{E}_n\right] \to \mathrm{E}\left[g(X_1, \ldots, X_k) f_{k+1}(X_{k+1}) | \mathcal{E}\right]$$

almost surely. Also,

$$\begin{aligned}
A_n(g)A_n(f_{k+1}) &= \left(\frac{1}{n \cdots (n-k+1)} \sum_{\sigma \in I_{k,n}} g(X_{\sigma_1}, \ldots, X_{\sigma_k})\right)\left(\frac{1}{n}\sum_{i=1}^{n} f_{k+1}(X_i)\right) \\
&= \frac{1}{n}\frac{1}{n \cdots (n-k+1)} \sum_{\sigma \in I_{k+1,n}} g(X_{\sigma_1}, \ldots, X_{\sigma_k}) f_{k+1}(X_{\sigma_{k+1}}) \\
&\quad + \frac{1}{n}\frac{1}{n \cdots (n-k+1)} \sum_{\sigma \in I_{k,n}} \sum_{j=1}^{k} g(X_{\sigma_1}, \ldots, X_{\sigma_k}) f_{k+1}(X_{\sigma_j}) \\
&= \frac{n-k}{n} A_n(f) + \frac{1}{n}\sum_{j=1}^{k} A_n(g(x_1, \ldots, x_k) f_{k+1}(x_j)).
\end{aligned}$$

Hence

$$A_n(f) = \frac{n}{n-k} A_n(g)A_n(f_{k+1}) - \frac{1}{n-k}\sum_{j=1}^{k} A_n(g(x_1, \ldots, x_k) f_{k+1}(x_j)).$$

Then,

$$A_n(f) = \mathrm{E}\left[A_n(f) | \mathcal{E}_n\right] \to \mathrm{E}\left[g(X_1, \ldots X_k) | \mathcal{E}\right] \mathrm{E}\left[f_{k+1}(X_{k+1}) | \mathcal{E}\right].$$

It follows that

$$\mathrm{E}\left[g(X_1, \ldots, X_k) f_{k+1}(X_{k+1}) | \mathcal{E}\right] = \mathrm{E}\left[g(X_1, \ldots X_k) | \mathcal{E}\right] \mathrm{E}\left[f_{k+1}(X_{k+1}) | \mathcal{E}\right] = \prod_{i=1}^{k+1} \mathrm{E}\left[f_i(X_1) | \mathcal{E}\right]$$

By induction, the result holds for any k and the proof is complete. ∎

Remark

X_n is exchangeable if and only if it is independent and identically distributed conditional on $\mathcal{E}$. Also, if $\mathcal{E}$ is trivial, then X_n are independent and identically distributed.

Corollary 3.61

If X_n is a sequence of exchangeable random variables taking value in $\{0, 1\}$, then there is a random variable ξ taking value in $[0, 1]$ such that

$$\mathrm{P}(X_1 = \cdots = X_k = 1, X_{k+1} = \cdots = X_n = 0) = \mathrm{E}\left[\xi^k (1-\xi)^{n-k}\right].$$

Proof. Define

$$\xi = \lim_{n\to\infty} \frac{1}{n} \sum_{i\le n} X_i.$$

Note that ξ is $\mathcal{E}$-measurable. By de Finetti's theorem,

$$\mathrm{P}(X_1 = \cdots = X_k = 1, X_{k+1} = \cdots = X_n = 0|\mathcal{E}) = \xi^k (1-\xi)^{n-k}.$$

Hence

$$\mathrm{P}(X_1 = \cdots = X_k = 1, X_{k+1} = \cdots = X_n = 0) = \mathrm{E}\left[\xi^k (1-\xi)^{n-k}\right].$$

■

4. Discrete-Time Stochastic Processes

4.1. Markov Property

Definition 4.1
*Let $(S, \mathcal{S})$ be a measurable space. A **transition kernel** (or **Markov kernel**) is a function $p : (S, \mathcal{S}) \to [0, 1]$ such that*

(a) For all $x \in S$, $B \mapsto p(x, B)$ is a probability measure.

(b) For all $B \in \mathcal{S}$, $x \mapsto p(x, B)$ is $\mathcal{S}$-measurable.

Remark
If S is countable, then it is equivalent to define the Markov kernel as $p : S^2 \to [0, 1]$ with $\sum_{y \in S} p(x, y) = 1$.

Definition 4.2
*A process X_n on $(S, \mathcal{S})$ is a **Markov chain** with Markov kernels p_n with respect to a filtration $\mathcal{F}_n$ if*

$$\mathrm{P}(X_{n+1} \in B | \mathcal{F}_n) = p_n(X_n, B)$$

for all $B \in \mathcal{S}$.

Definition 4.3
*Let X_n be a Markov chain on S with kernel p. The **transition operator** P is given by*

$$(Pf)(x) = \int p(x, dy) f(y).$$

The right multiplication is defined by

$$(P'\mu)(B) = (\mu P)(B) = \int \mu(dx) p(x, B).$$

Remark
One can think the right multiplication as the evolution of the probability measure over time. In the case where S is countable, the operator has a matrix represntation with rows summing up to 1. Another observation is that when S is countable, we can associate P with a matrix (possibly with infinitely many entries) with row summing up to 1.

Remark
One can regard the measures as dual of the functions. Thus we also use μf to denote the expectation of f under μ.

Definition 4.4
*A Markov chain X_n is said to be **time-homogeneous** if the transition kernels are invariant, i.e., $p_n = p$.*

Remark

In the rest of the note, unless explicitly pointed out, we will assume that the Markov chains are time-homogeneous.

Remark

Every time-inhomogeneous Markov chain can be integrated into a time-homogeneous one by considering (X_n, n) *on* $(S, \mathcal{S}) \times (T, \mathcal{T})$.

Definition 4.5

A space $(S, \mathcal{S})$ *is said to be* ***Borel*** *if it is isomorphic in terms of measurability to* $(\mathbb{R}, \mathcal{B})$. *In other words, there is a bijection* $\phi : (S, \mathcal{S}) \to (\mathbb{R}, \mathcal{B})$ *such that* ϕ *and* ϕ^{-1} *are measurable.*

Remark

Since there is also a measurable bijection between $\mathbb{R}$ *and* $[0, 1]$. *It is equivalent that* S *is Borel and that there is a measurable bijection between* S *and* $[0, 1]$.

Theorem 4.6

Let X_n *be a Markov process taking value in a Borel space* $(S, \mathcal{S})$ *with the Markov kernel* p. *Let* μ *be the initial distribution of* X_0. *Then the finite-dimensional distribution defined by*

$$\mathrm{P}_\mu(X_j \in B_j | 0 \le j \le n) = \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_n} p(x_{n-1}, dx_n)$$

extends uniquely to a distribution on $S^{\mathbb{Z}_+}$.

Proof. Note that the space is Borel. Thus it suffices to show the result when $(S, \mathcal{S}) = (\mathbb{R}, \mathcal{B})$. For every n, let $A = (\prod_{j=1}^n B_j) \times \mathbb{R} \subset S^{n+1}$. Then

$$\begin{aligned}
\mathrm{P}_\mu(X_j \in A_j, X_{n+1} \in \mathbb{R} | 0 \le j \le n) &= \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_n} p(x_{n-1}, dx_n) \int_{\mathbb{R}} p(x_n, dx_{n+1}) \\
&= \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_n} p(x_{n-1}, dx_n) \\
&= \mathrm{P}_\mu(X_j \in A_j | 0 \le j \le n).
\end{aligned}$$

Hence the probability measures are consistent. By the Kolmogorov extension theorem, the finite-dimensional distributions extends uniquely to a distribution defined on $S^{\mathbb{Z}_+}$. ∎

Remark

The notation P_μ *denotes the case where the initial distribution is* μ. *When the initial distribution is a fixed constant* $X_0 = x$, *we also write* $\mathrm{P}_{\delta_x} = \mathrm{P}_x$. *Notice also that*

$$\mathrm{P}_\mu(B) = \int_S \mu(dx)\, \mathrm{P}_x(B).$$

We also define the expectation E_μ *in the way*

$$\mathrm{E}_\mu [f(\{X_n\})] = \int f d\, \mathrm{P}_\mu$$

where $f : S^{\mathbb{Z}_+} \to \mathbb{R}$ *is measurable.*

Proposition 4.7
Let X_n *be a process on a Borel space* S*. Then the followings are equivalent:*

(a) X_n *is a Markov chain with kernels* p *on* S*.*

(b) There is a measurable function $f : S \times [0,1] \to S$ *and random variables* $\xi_n \overset{iid}{\sim} U[0,1]$ *independent of* X_0 *such that* $X_{n+1} = f(X_n, \xi_{n+1})$ *almost surely.*

Proof. Suppose (b) holds. Then

$$\mathrm{P}(X_{n+1} \in B|\mathcal{F}_n) = \mathrm{P}(f(X_n, \xi_{n+1} \in B)|\mathcal{F}_n) = p(X_n, B)$$

where $p(x, B) = \mathrm{P}(f(x, \xi) \in B)$ is the kernel. Thus X_n is a Markov chain.

Conversely, suppose that X_n is a Markov chain with kernel p. Since S is Borel, it suffices to prove the case where $S = [0,1]$. Define

$$f(x, \xi) = \sup \{y \in [0,1] \mid p(x, [0, y]) < \xi\}.$$

Observe that f is measurable in $[0,1]^2$. Now take $\xi \sim U[0,1]$.

$$\mathrm{P}(f(x, \xi) \le z) = \mathrm{P}(\xi \le p(x, [0, z])) = p(x, [0, z]).$$

Hence $f(x, \xi)$ has the probability measure $p(x, \cdot)$. Then by the Markov property,

$$\mathrm{P}(X_{n+1} \in B|\mathcal{F}_n) = p(X_n, B) = \mathrm{P}(f(X_n, \xi_{n+1}) \in B|X_n)$$

for all B. This implies that $X_{n+1} = f(X_n, \xi_{n+1})$ in distribution. Now iterating by $\tilde{X}_n = f(\tilde{X}_{n-1}, \xi_n)$ from X_0 to get $\{\tilde{X}_n\}$. We may find the replication of $\{\tilde{X}_n\}$ on the space of $\{X_n\}$ such that they equals almost surely. The proof is complete. ∎

Proposition 4.8
Let X_n *be a Markov chain with kernel* p *in* $(S, \mathcal{S})$ *with respect to the natural filtration* $\mathcal{F}_n = \sigma(X_0, \ldots, X_n)$ *and* μ *be a distribution on* S*. Then* X_n *is a Markov chain under* P_μ*. That is, for all* $B \in \mathcal{S}$*,*

$$\mathrm{P}_\mu(X_{n+1} \in B|\mathcal{F}_n) = p(X_n, B).$$

Proof. To prove the result, we need to show that

$$\mathrm{E}\left[\mathbf{1}\left\{X_{n+1} \in B\right\} \mathbf{1}_A\right] = \mathrm{E}\left[p(X_n, B)\mathbf{1}_A\right]$$

for all $A \in \mathcal{F}_n$. If $A = \{X_0 \in B_0, \ldots, X_n \in B_n\}$, then

$$\begin{aligned}\mathrm{E}\left[\mathbf{1}\left\{X_{n+1} \in B\right\}\mathbf{1}_A\right] &= \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_n} p(x_{n-1}, dx_n) \int_S p(x_n, dx_{n+1}) \mathbf{1}_B(x_{n+1}) \\ &= \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_n} p(x_{n-1}, dx_n) p(x_n, B)\end{aligned}$$

We claim that for every bounded measurable $f : S \to \mathbb{R}$, we have

$$\mathrm{E}\left[\mathbf{1}_A f(X_n)\right] = \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_n} p(x_{n-1}, dx_n) f(x_n).$$

Suppose first that $f = \mathbf{1}_B$ where $B \in \mathcal{S}$. Then

$$\begin{aligned}\mathrm{E}\left[\mathbf{1}_A \mathbf{1}_B(X_n)\right] &= \mathrm{E}\left[\mathbf{1}\left\{X_0 \in B_0, \ldots, X_{n-1} \in B_{n-1}, X_n \in B_n \cap B\right\}\right] \\ &= \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_n \cap B} p(x_{n-1}, dx_n) \\ &= \int_{B_0} \mu(dx_0) \int_{B_1} p(x_0, dx_1) \cdots \int_{B_n} p(x_{n-1}, dx_n) \mathbf{1}_B(x_n).\end{aligned}$$

By linearity, this extends to the simple functions. For a general bounded measurable f. Write $f = f^+ - f^-$ and approximate each part by simple function. By the monotone convergence theorem, the claim follows. As a result,

$$\mathrm{E}\left[\mathbf{1}\left\{X_{n+1} \in B\right\}\mathbf{1}_A\right] = \mathrm{E}\left[p(X_n, B)\mathbf{1}_A\right]$$

for every A of the form $\{X_0 \in B_0, \ldots, X_n \in B_n\}$.

Next, observe that all $A \in \mathcal{F}_n$ of the form $\{X_0 \in B_0, \ldots, X_n \in B_n\}$ forms a π-system and all $A \in \mathcal{F}_n$ such that $\mathrm{E}\left[\mathbf{1}\left\{X_{n+1} \in B\right\}\mathbf{1}_A\right] = \mathrm{E}\left[p(X_n, B)\mathbf{1}_A\right]$ forms a λ-system. It follows by the Dynkin's π-λ theorem that $\mathrm{E}\left[\mathbf{1}\left\{X_{n+1} \in B\right\}\mathbf{1}_A\right] = \mathrm{E}\left[p(X_n, B)\mathbf{1}_A\right]$ holds for all $A \in \mathcal{F}_n$ since the events of the form $\{X_0 \in B_0, \ldots, X_n \in B_n\}$ generates $\mathcal{F}_n$. The proof is complete. ∎

Remark

As a result, we also have

$$\mathrm{E}\left[f(X_{n+1})|\mathcal{F}_n\right] = \int_S p(X_n, dy) f(y)$$

for all bounded measurable $f : S \to \mathbb{R}$. To see this, note that when f is a characteristic function, the statement collapse to the proposition. By linearity, the statement holds for all simple f. For all bounded measurable f, write $f = f^+ - f^-$ and approximate each term using simple functions. We see that the statement follows by the monotone convergence theorem.

Remark

This also implies that

$$\mathrm{E}_\mu\left[\prod_{i=0}^{m} f_i(X_i)\right] = \int \mu(dx_0) f_0(x_0) \int p(x_0, dx_1) f_1(x_1) \cdots \int p(x_{m-1}, dx_m) f(x_m).$$

To see this, note that

$$\mathrm{E}\left[f_m(X_m)|\mathcal{F}_{m-1}\right] = \int p(x_{m-1}, dx_m) f(x_m)$$

and

$$\mathrm{E}_\mu\left[\prod_{i=0}^{m} f_i(X_i)\right] = \int \mu(dx_0)\, \mathrm{E}\left[f_0\, \mathrm{E}\left[\cdots|\mathcal{F}_1\right]|\mathcal{F}_0\right].$$

The result follows by induction.

Definition 4.9

*Let $\Omega = S^{\mathbb{Z}_+}$ and $\omega = (\omega_1, \ldots) \in \Omega$. The **shift operator** is given by $\theta : \theta \mapsto (\omega_2, \ldots)$.*

Theorem 4.10

Let X_n be a Markov chain on S, $\Omega = S^{\mathbb{Z}_+}$ and $Y : \Omega \to \mathbb{R}$ be bounded and measurable. Then

$$\mathrm{E}_\mu\left[Y \circ \theta^n|\mathcal{F}_n\right] = \mathrm{E}_{X_n}\left[Y\right].$$

Proof. We suppose first that $Y = \prod_{0\le i\le m} f_i(X_i)$. Let $A = \prod_{i=0}^{n} A_i$ where $A_i \subset S$ is measurable. Then

$$\begin{aligned}\mathrm{E}_\mu\left[\prod_{0\le i\le m} f_i(X_{m+i})\mathbf{1}_A\right] &= \int_{A_0} \mu(dx_0) \int_{A_1} p(x_0, dx_1) \cdots \int_{A_n} p(x_{n-1}, dx_n) f_0(x_n) \\ &\quad \cdot \int p(x_n, dx_{n+1}) f_1(x_{n+1}) \cdots \int p(x_{n+m-1}, dx_{n+m}) f_m(x_{n+m}) \\ &= \mathrm{E}_\mu\left[\mathrm{E}_{X_n}\left[\prod_{0\le i\le m} f_i(X_i)\right]\mathbf{1}_A\right]\end{aligned}$$

Observe that the sets A satisfying the equation above forms a λ-system and that the sets of the form $A = \prod_{i=0}^{n} A_i$ forms a π-system. It follows that the above equation holds for all $A \in \mathcal{F}_n$ by Dynkin's π-λ theorem.

Next, observe that all the bounded measurable Y can be approximated by the linear combination of the functions of the form $\left\{\prod_{0\le i\le m} f_i(x_i)\right\}_m$. The equation thus holds for general Y. The proof is complete. ∎

Corollary 4.11

Let X_n be a Markov chain and $\mathcal{F}_n$ be the filtration generated by X_n. If $Z \in \mathcal{L}^1$ and is $\sigma(X_n, \ldots)$-measurable, then

$$\mathrm{E}\left[Z|\mathcal{F}_n\right] = \mathrm{E}\left[Z|X_n\right].$$

Proof. By the assumption, we can write $Z = Y \circ \theta^n$. Then theorem 4.10 implies that $\mathrm{E}\left[Z|\mathcal{F}_n\right] = \mathrm{E}\left[Y \circ \theta^n|\mathcal{F}_n\right] = \mathrm{E}_{X_n}\left[Y\right] = \mathrm{E}\left[Z|X_n\right]$. ∎

Corollary 4.12 (Chapman-Kolmogorov Equation)
Let X_n be a Markov chain. Then

$$\mathrm{P}_\mu(X_{m+n} \in B) = \mathrm{E}_\mu \left[\mathrm{P}_{X_m}(X_n \in B)\right].$$

Proof. Note that

$$\mathrm{P}_x(X_{m+n} \in B) = \mathrm{E}_x \left[\mathrm{P}_x(X_{m+n} \in B|\mathcal{F}_m)\right] = \mathrm{E}_x \left[\mathrm{P}_{X_m}(X_n \in B)\right].$$

Then

$$\mathrm{P}_\mu(X_{m+n} \in B) = \int \mathrm{P}_x(X_{m+n} \in B)\mu(dx) = \int \mathrm{E}_x \left[\mathrm{P}_{X_m}(X_n \in B)\right] \mu(dx) = \mathrm{E}_\mu \left[\mathrm{P}_{X_m}(X_n \in B)\right].$$

■

Proposition 4.13
Let $\mathcal{F}_n$ be a filtration and τ be a stopping time. Then

$$\mathcal{F}_\tau = \{A \mid A \cap \{\tau = n\} \in \mathcal{F}_n\}$$

is a σ-algebra.

Proof. It is clear that $\varnothing \in \mathcal{F}_\tau$ and for $A \in \mathcal{F}_\tau$, $A^c \cap \{\tau = n\} = (A \cup \{\tau = n\}^c)^c \in \mathcal{F}_n$. For $A_i \nearrow A$ with $A_i \in \mathcal{F}_n$,

$$A \cap \{\tau = n\} = \cup_i (A_i \cap \{\tau = n\}) \in \mathcal{F}_n.$$

We conclude that $\mathcal{F}_\tau$ is a σ-algebra. ■

Theorem 4.14 (Strong Markov Property)
Let X be a Markov process and τ be a stopping time. Then for all $Y_n : S^{\mathbb{Z}_+} \to \mathbb{R}$ with $|Y_n| \leq K$,

$$\mathrm{E}_\mu \left[Y_\tau \circ \theta^\tau | \mathcal{F}_\tau\right] = \mathrm{E}_{X_\tau} \left[Y_\tau\right]$$

on $\{\tau < \infty\}$.

Proof. Note that on $\{\tau < \infty\}$,

$$Y_\tau \circ \theta^\tau = \sum_{n=0}^{\infty} Y_n \circ \theta^n \mathbf{1} \left\{\tau = n\right\}.$$

For $A \in \mathcal{F}_\tau$, by Fubini's theorem,

$$\mathrm{E}_\mu \left[Y_\tau \circ \theta^\tau \mathbf{1}_A\right] = \sum_{n=0}^{\infty} \mathrm{E} \left[Y_n \circ \theta^n \mathbf{1} \left\{\tau = n\right\} \mathbf{1}_A\right] = \sum_{n=0}^{\infty} \mathrm{E} \left[\mathrm{E}_{X_n} \left[Y_n\right] \mathbf{1} \left\{\tau = n\right\} \mathbf{1}_A\right] = \mathrm{E} \left[\mathrm{E}_{X_\tau} \left[Y_\tau\right] \mathbf{1}_A\right].$$

Hence $\mathrm{E}_\mu \left[Y_\tau \circ \theta^\tau | \mathcal{F}_\tau\right] = \mathrm{E}_{X_\tau} \left[Y_\tau\right]$ on $\{\tau < \infty\}$. ■

4.2. Markov Chain on Countable Spaces

Definition 4.15
*Suppose that S is countable. Let $x, y \in S$ and P be the transition operator of a Markov chain on S. We say that y is **accessible** from x if there exists $n \in \mathbb{Z}_+$ such that $P^n(x,y) > 0$. We denote by $x \rightsquigarrow y$. The probability of visiting in finite time $\mathrm{P}_x(\tau_y < \infty)$ is denoted by ρ_{xy}.*

Definition 4.16
*Suppose that S is countable. Let $x, y \in S$ with a Markov chain on S. We say that x **communicates** with y if $x \rightsquigarrow y$ and $y \rightsquigarrow x$. We denote by $x \leftrightsquigarrow y$.*

Proposition 4.17
Let S be a countable space. Then $\leftrightsquigarrow$ is a equivalence relation on S.

Proof. $x \leftrightsquigarrow x$ and $x \leftrightsquigarrow y$ implying $y \leftrightsquigarrow x$ are trivial. If $x \leftrightsquigarrow y$ and $y \leftrightsquigarrow z$, then

$$P^{m+n}(x,z) = P^m(x,y)P^n(y,z) > 0$$

for some $m, n \in \mathbb{Z}_+$ by the Chapman-Kolmogorov equation. We conclude that $\leftrightsquigarrow$ is indeed an equivalence relation. ∎

Definition 4.18
*Let S be a countable space. The **communicating class** of x is the equivalence class defined by $\leftrightsquigarrow$, denoted by $C[x]$.*

Definition 4.19
*Let S be a countable space. $x \in S$ is **absorbing** if $p(x,x) = 1$.*

Theorem 4.20
Let S be a countable space and $\tau_y^{(0)} = 0$. Consider the k-th hitting time defined recursively by

$$\tau_y^{(k)} = \inf\left\{ n > \tau_y^{(k-1)} \;\middle|\; X_n = y \right\}.$$

Then

$$\mathrm{P}_x(\tau_y^{(k)} < \infty) = \rho_{xy}\rho_{yy}^{k-1}.$$

Proof. We prove the statement by induction on k. If $k = 1$, the result is trivial. Suppose that the statement holds for $k-1$. Let

$$Y = \mathbf{1}\left\{\text{there is } n > 0 \text{ such that } X_n = y\right\}, \text{ and } \tau = \tau_y^{(k-1)}.$$

By the strong Markov property and the induction hypothesis,

$$\begin{aligned}
\mathrm{P}_x(\tau_y^{(k)} < \infty) &= \mathrm{E}_x\left[\mathbf{1}\left\{\tau_y^{(k)} < \infty\right\}\right] = \mathrm{E}_x\left[Y \circ \theta^\tau \mathbf{1}\left\{\tau < \infty\right\}\right] = \mathrm{E}_x\left[\mathrm{E}_{X_\tau}[Y]\,\mathbf{1}\left\{\tau < \infty\right\}\right] \\
&= \mathrm{E}_x\left[\mathrm{E}_y[Y]\,\mathbf{1}\left\{\tau < \infty\right\}\right] = \mathrm{E}_x\left[\mathbf{1}\left\{\tau < \infty\right\}\right]\mathrm{E}_y\left[\mathbf{1}\left\{\tau_y^{(1)} < \infty\right\}\right] \\
&= \mathrm{P}_x(\tau_y^{(k-1)} < \infty)\,\mathrm{P}_y\left[\tau_y^{(1)} < \infty\right] = \rho_{xy}\rho_{yy}^{k-1}.
\end{aligned}$$

The proof is complete. ■

Definition 4.21

*Let S be a countable space and $x \in S$. We say that $x \in S$ is **recurrent** if $\rho_{xx} = 1$ and **transient** if $\rho_{xx} < 1$.*

Proposition 4.22

Let S be a countable space, $x \in S$ and τ_x be the hitting time of x. Then the followings are true:

(a) If x is recurrent, then $\mathrm{P}_x(X_n = x \text{ i.o.}) = 1$.

(b) If x is transient, then

$$\sum_{n\geq 0} \mathbf{1}\{X_n = x\} \sim 1 + Geom(1 - \rho_{xx}).$$

Proof. For (a), by theorem 4.20,

$$\mathrm{P}_x(X_n = x \text{ i.o.}) = \lim_{k\to\infty} \mathrm{P}_x(\tau_x^{(k)} < \infty) = \rho_{xx}^k = 1.$$

For (b), using theorem 4.20 again,

$$\begin{aligned}\mathrm{P}_x\left(\sum_{n\geq 0} \mathbf{1}\{X_n = x\} = k+1\right) &= \mathrm{P}_x\left(\sum_{n\geq 1} \mathbf{1}\{X_n = x\} = k\right) = \mathrm{P}_x(\tau_x^{(k)} < \infty, \tau_x^{(k+1)} = \infty) \\ &= \rho_{xx}^k(1 - \rho_{xx} < \infty)\end{aligned}$$

where the last equality utilizes the strong Markov property. Hence

$$\sum_{n\geq 0} \mathbf{1}\{X_n = x\} \sim 1 + \mathrm{Geom}(1 - \rho_{xx}).$$

■

Remark

The expected number of hitting is also available. Let $x, y \in S$ and the number of hitting y be

$$N(y) = \sum_{n=1}^{\infty} \mathbf{1}\{X_n = y\}.$$

Then using the formula from theorem 4.20,

$$\begin{aligned}\mathrm{E}_x[N(y)] = \mathrm{E}_x\left[\sum_{k=1}^{N(y)} 1\right] = \mathrm{E}_x\left[\sum_{k=1}^{\infty} \mathbf{1}\{N(y) \geq k\}\right] &= \sum_{k=1}^{\infty} \mathrm{P}_x(N(y) \geq k) = \sum_{k=1}^{\infty} \mathrm{P}_x(\tau_y^{(k)} < \infty) \\ = \sum_{k=1}^{\infty} \rho_{xy}\rho_{yy}^{k-1} &= \frac{\rho_{xy}}{1 - \rho_{yy}}.\end{aligned}$$

As a direct consequence, we have the following observations:

(a) If $\rho_{xy} > 0$ and y is recurrent, then $\mathrm{E}_x[N(y)] = \infty$.

(b) *If* $\rho_{xy} > 0$ *and* y *is transient, then* $\mathrm{E}_x[N(y)] < \infty$.

(c) *If* $\rho_{xy} = 0$, *then* $\mathrm{E}_x[N(y)] = 0$.

Taking $x = y$, *we see that* y *is recurrent if and only if* $\mathrm{E}_y[N(y)] = \sum_{k=1}^{\infty} P^k(y,y) = \infty$ *where* P *is the transition operator.*

Example

Consider a random walk S_n *on* $\mathbb{Z}$ *with increments* ξ_i *such that* $\mathrm{P}(\xi_i = 1) = p$ *and* $\mathrm{P}(\xi_i = -1) = 1 - p$. *Let* $S_0 = x$. *Now,*

$$\begin{aligned}\mathrm{E}_x[N(x)] &= \sum_{k=1}^{\infty} P^k(x,x) = \sum_{k=1}^{\infty} P^{2k}(x,x) = \sum_{k=1}^{\infty} \binom{2k}{k} p^k (1-p)^k \\ &= \sum_{k=1}^{\infty} \frac{\sqrt{4\pi k}\left(\frac{2k}{e}\right)^{2k}(1+O(1/k))}{4\pi k\left(\frac{k}{e}\right)^{2k}(1+O(1/k))} p^k(1-p)^k = \sum_{k=1}^{\infty} \frac{1+O(1/k)}{\sqrt{4\pi k}} 4^k p^k (1-p)^k\end{aligned}$$

by the Stirling formula. If $p = 1/2$, *then the above sum diverges. We see that* x *is recurrent. If* $p \neq 1/2$, *then the above sum converges and* x *is transient.*

Example

Consider $\xi_i = (\pm 1, \ldots, \pm 1) \in \mathbb{Z}^d$ *with equal probabilities on* $(\xi_i)_j = \pm 1$ *for* $j = 1, \ldots d$ *and each coordinate is independent. Then* $S_n = \sum_{i \leq n} \xi_i$ *is a symmetric random walk on* $\mathbb{Z}^d$. *Let* $S_n = 0$. *Using the Stirling formula again,*

$$\sum_{k=1}^{\infty} P^{2k}(0,0) = \sum_{k=1}^{\infty} \left[\binom{2k}{k} 4^{-k}\right]^d = \sum_{k=1}^{\infty} \left[\frac{1+O(1/k)}{\sqrt{4\pi k}}\right]^d.$$

For $d = 1, 2$, *the sum diverges and for* $d \geq 3$ *the sum converges. Hence* 0 *is transient for* $d \geq 3$ *and is recurrent for* $d = 1, 2$.

Theorem 4.23

Let S *be a countable space. If* $x \in S$ *is recurrent and* $x \rightsquigarrow y$, *then* y *is recurrent and* $x \leftrightsquigarrow y$ *with* $\rho_{xy} = \rho_{yx} = 1$.

Proof. Observe that we can choose a path from x to y, $x_0 = x, \ldots, x_k = y$ with $x_i \neq x$ for all $i = 1, \ldots k$ that happens with positive probability. If $\rho_{yx} < 1$, then

$$\mathrm{P}_x(\tau_x = \infty) \geq p(x_0, x_1) \cdots p(x_{k-1}, x_k)(1 - \rho_{yx}) > 0.$$

Then $\rho_{xx} < 1$, which is a contradiction. Hence $\rho_{yx} = 1$ and $x \leftrightsquigarrow y$. Also, there are $m_{xy}, m, m_{yx} \in \mathbb{Z}_+$ such that $P^{m_{xy}}(x,y), P^m(x,x), P^{m_{yx}}(y,x) > 0$. By the Chapman-Kolmogorov equation,

$$P^{m_{xy}+m+m_{yx}}(y,y) \geq P^{m_{xy}}(x,y) P^m(x,x) P^{m_{yx}}(y,x)$$

Summing over all possible m,

$$\sum_{m=0}^{\infty} P^m(y,y) \geq \sum_{m=0}^{\infty} P^{m_{xy}}(x,y) P^m(x,x) P^{m_{yx}}(y,x) = P^{m_{xy}} P^{m_{yx}}(y,x) \sum_{m=0}^{\infty} P^m(x,x) = \infty$$

since x is recurrent. Hence y is recurrent as well. Switching the roles of x and y gives $\rho_{xy} = 1$. ■

Definition 4.24
*A non-empty subset $C \subset S$ is **closed** if $\rho_{xy} = 0$ for all $x \in C$ and $y \in C^c$.*

Definition 4.25
*A non-empty subset $C \subset S$ is **irreducible** if for all $x, y \in C$, $\rho_{xy} > 0$ and $\rho_{yx} > 0$.*

Remark
If C is irreducible and $x \in C$ is recurrent, then all $y \in C$ are recurrent by theorem 4.23.

Proposition 4.26
Let S be a countable space. If $x \in S$ is recurrent, then $C[x]$ is closed.

Proof. Let $y \in C[x]$ and $z \notin C[x]$. If $\rho_{yz} > 0$, then $y \rightsquigarrow z$. Since y is recurrent, this implies that z is recurrent and $\rho_{yz} = \rho_{zy} = 1$ by theorem 4.23. Then $z \in C[x]$, a contradiction. Hence $\rho_{yz} = 0$ and $C[x]$ is closed. ■

Proposition 4.27
Let S be a countable space. If $C \subset S$ is closed and finite, then C has a recurrent state.

Proof. Suppose that all states in C are transient. Then for $x, y \in C$,

$$\mathrm{E}_x\left[N(y)\right] = \frac{\rho_{xy}}{1 - \rho_{yy}} < \infty.$$

This implies that

$$\infty > \sum_{y \in C} \mathrm{E}_x\left[N(y)\right] = \sum_{y \in C} \sum_{k=1}^{\infty} P^k(x,y) = \sum_{k=1}^{\infty} \sum_{y \in C} P^k(x,y) = \sum_{k=1}^{\infty} 1 = \infty,$$

which is absurd. Hence there is at least one recurrent state. ■

Theorem 4.28 (Canonical Decomposition)
Let S be a countable space. Then $S = T \cup C_1 \cup \cdots \cup C_n$ where T is the set of transient states and $C_1, \ldots, C_n$ are closed and irreducible recurrent states.

Proof. Let $C = S - T$. If $C = \varnothing$, then we are done. Suppose that C is non-empty. By proposition 4.27, C is closed. We need to prove that if $C[x], C[y] \subset C$ and $C[x] \cap C[y] \neq \varnothing$, then $C[x] = C[y]$. To see this, note that if $z \in C[x] \cap C[y]$, then for any $x \in C[x]$ and $y \in C[y]$, $x \leftrightsquigarrow z \leftrightsquigarrow y$. This implies that $x \in C[y]$ and $y \in C[x]$. Hence $C[x] = C[y]$. The proof is complete. ■

Example (Branching Process)
Let $Z_{t,k}$ be a independent and identically distributed on $\mathbb{Z}_+$ with $\mathrm{P}(Z_{t,k} = i) = p_i$. $X_{t+1} = \sum_{k=1}^{X_t} Z_{t,k}$ with $X_0 = 1$. Suppose $\mu = \mathrm{E}\left[Z_{t,k}\right] < \infty$ and $\pi = \mathrm{P}(X_t \to 0)$ is the probability of dying out. Now $\mathrm{P}_k(X_t \to 0) = \pi^k$ since k branches are independent. Then

$$\pi = \mathrm{P}_1(X_t \to 0) = \sum_{k=0}^{\infty} \mathrm{P}_1(X_t \to 0, X_1 = k) = \sum_{k=0}^{\infty} p_k \pi^k = \mathrm{E}\left[\pi^Z\right]$$

where $\mathrm{P}(Z = k) = p_k$. Now set $g(x) = \mathrm{E}\left[x^Z\right] = \sum_{k=0}^{\infty} p_k x^k$. Observe that g is increasing and convex on $[0, 1]$. π is a solution to the fixed point of g. We claim that π is the smallest fixed point of g. Notice that $\mathrm{E}\left[x^{X_t}\right] = g^t(x)$ where g^t is understood as acting g t-times. Since 0 is absorbing, $g^t(0) = \mathrm{P}_1(X_t = 0)$ is increasing in t and $\mathrm{P}_1(X_t = 0$ for some $t) = \pi$. If $x \in [0, 1]$ is a fixed point,

$$\pi = \lim_{t\to\infty} g^t(0) \leq \lim_{t\to\infty} g^t(x) = x.$$

This proves that π is the smallest fixed point.

4.3. Stationarity on Countable Spaces

Definition 4.29
*Let P be a transition operator. A non-trivial measure μ is **invariant** or **stationary** if $P'\mu = \mu$. If μ is a distribution, then we say that μ is a **stationary distribution**.*

Remark
Equivalently, the Markov process X_t associated with P has invariant finite-dimensional distribution under shifts, $(X_{t_1}, \dots, X_{t_n}) \overset{d}{=} (X_{t_1+s}, \dots, X_{t_n+s})$ for all $s \geq 0$.

Definition 4.30
*A Markov matrix P is said to be **bistochastic** if P' is also a stochastic matrix.*

Remark
Constant measures are always stationary for a bistochastic matrix since if $\mu(x) = c$ for all $x \in S$, $P'\mu = c = \mu$.

Definition 4.31
*A measure μ is said to be **reversible** for a Markov kernel p if $\mu(x)p(x,y) = \mu(y)p(y,x)$ for all $x, y \in S$.*

Remark
If μ is reversible for p, then μ is also stationary since

$$(P'\mu)(y) = \sum_{x \in S} \mu(x)p(x,y) = \sum_{x \in S} \mu(y)p(y,x) = \mu(y).$$

Example
Let G be a locally finite graph, i.e., for each vertex x, $\deg(x) < \infty$. A symmetric random walk on G is given by the kernel

$$p(x,y) = \frac{1}{\deg(x)}$$

if y is connected to x and 0 otherwise. Notice that $\mu(x) = \deg(x)$ is stationary.

Theorem 4.32
Let S be a countable space. If π is a stationary distribution, then $y \in \operatorname{supp}(\pi)$ is recurrent.

Proof. Note that

$$\mathrm{E}_\pi\left[N(y)\right] = \sum_{k=1}^{\infty} \mathrm{P}_\pi(X_n = y) = \sum_{k=1}^{\infty} \pi(y) = \infty.$$

On the other hand,

$$\mathrm{E}_\pi\left[N(y)\right] = \sum_{x \in S} \pi(x)\, \mathrm{E}_x\left[N(y)\right] = \sum_{x \in S} \pi(x) \frac{\rho_{xy}}{1 - \rho_{yy}} \leq \frac{1}{1 - \rho_{yy}}.$$

Hence $\rho_{yy} = 1$ and y is recurrent. ∎

Theorem 4.33
Let S be countable. Suppose that x is recurrent, then the measure

$$\mu_x(y) = \mathrm{E}_x\left[\sum_{k=0}^{\tau_x - 1} \mathbf{1}\left\{X_k = y\right\}\right] = \sum_{k=1}^{\infty} \mathrm{P}_x(X_k = y, \tau_x > k)$$

where $\tau_x = \inf\{k > 0 \mid X_k = x\}$, is stationary.

Proof. Note that if x is absorbing, then the theorem is trivial. Assume that x is not absorbing. Notice that $\operatorname{supp}(\mu_x) = C[x]$. For all $y \in C[x]$, $\mu_x(y) < \infty$ since τ_x is finite almost surely. We now verify that $P'\mu_x = \mu_x$. Note that

$$\sum_{y \in S} \mu_x(y) p(y, z) = \sum_{n=0}^{\infty} \sum_{y \in S} \mathrm{P}_x(X_n = y, \tau_x > n) p(y, z).$$

If $z \neq x$,

$$\mathrm{P}_x(X_n = y, \tau_x > n) p(y, z) = \mathrm{P}_x(X_n = y, X_{n+1} = z, \tau_x > n + 1)$$

and

$$\sum_{y \in S} \mu_x(y) p(y, z) = \sum_{n=0}^{\infty} \mathrm{P}_x(X_{n+1} = z, \tau_x > n + 1) = \mu_x(z)$$

If $z = x$,

$$\sum_{y \in S} \mathrm{P}_x(X_n = y, \tau_x > n) p(y, z) = \mathrm{P}_x(\tau_x = n + 1).$$

Hence

$$\sum_{y \in S} \mu_x(y) p(y, x) = \sum_{n=0}^{\infty} \mathrm{P}_x(\tau_x = n + 1) = 1$$

and it is clear that $\mu_x(x) = 1$. Hence μ_x is stationary. ∎

Remark
Notice that

$$\sum_{y \in S} \mu_x(y) = \mathrm{E}_x\left[\tau_x\right].$$

If $\mathrm{E}_x[\tau_x] < \infty$, then we can normalize μ_x to be a stationary distribution.

Definition 4.34
*Let x be recurrent. If $\mathrm{E}_x[\tau_x] < \infty$, then x is **positive recurrent**; if $\mathrm{E}_x[\tau_x] = \infty$, then x is **null recurrent**.*

Theorem 4.35
If p is irreducible and recurrent, then for every stationary measure μ, $\mu = c\mu_x$ for some $c > 0$.

Proof. From theorem 4.33, the stationary measure exists. Suppose that μ is a stationary measure. For $y \in S$, we show that $\mu(y) = \mu(x)\mu_x(y)$. Note that if $y = x$, then the result is

trivial. If $y \neq x$, then

$$\begin{aligned}
\mu(y) &= \sum_{z \in S} \mu(z)p(z,y) = \mu(x)p(x,y) + \sum_{z \neq x} \mu(z)p(z,y) \\
&= \mu(x)p(x,y) + \mu(x) \sum_{z \neq x} p(x,z)p(z,y) + \sum_{z,w \neq x} \mu(w)p(w,z)p(z,y) \\
&\vdots \\
&= \mu(x) \sum_{m=1}^{n} \mathrm{P}_x(X_k \neq x \text{ for } 1 \leq k \leq m, X_m = y) + \sum_{a \neq x} \mu(a)\, \mathrm{P}_a(X_k \neq x \text{ for } 1 \leq k \leq n, X_n = y) \\
&\geq \mu(x) \sum_{m=1}^{n} \mathrm{P}_x(X_k \neq x \text{ for } 1 \leq k \leq m, X_m = y).
\end{aligned}$$

Take $n \to \infty$,

$$\mu(y) \geq \mu(x) \sum_{m=1}^{\infty} \mathrm{P}_x(X_k \neq x \text{ for } 1 \leq k \leq m, X_m = y) = \mu(x)\mu_x(y)$$

by definition. Notice that

$$\mu(x) \geq \mu(x) \sum_{m=1}^{\infty} \mathrm{P}_x(X_k \neq x \text{ for } 1 \leq k \leq m, X_m = y) = \mu(x)\mu_x(x) = \mu(x).$$

This implies that the inequality are in fact equality and thus $\mu(y) = \mu(x)\mu_x(y)$ for all y such that $x \rightsquigarrow y$. By the irreducibility, all y satisfies that $x \rightsquigarrow y$. The conclusion follows. ∎

Theorem 4.36

If p is irreducible and admits a stationary distribution π, then

$$\pi(x) = \frac{1}{\mathrm{E}_x[\tau_x]}.$$

Proof. By theorem 4.35, $\pi(y) = c_x\mu_x(y)$ for some constant c_x. To normalize the stationary measure μ_x,

$$c_x = \frac{1}{\mathrm{E}_x[\tau_x]}.$$

Hence

$$\pi(x) = c_x\mu_x(x) = c_x = \frac{1}{\mathrm{E}_x[\tau_x]}.$$

∎

Theorem 4.37

Let S be a countable space. If p is irreducible, then the followings are true.

(a) There is a stationary distribution μ.

(b) There is a positive recurrent state.

(c) All states are positive recurrent.

Proof. Assume (a). Since p is irreducible, $\text{supp}(()\mu) = S$. From theorem 4.36, $\mathrm{E}_x[\tau_x] < \infty$ for all $x \in S$. This proves (c).

Assume (b). Suppose that x is positive recurrent. Then by theorem 4.33, there is a stationary measure μ_x and in particular we can normalize it to a stationary distribution. This proves (a).

(c) to (b) is trivial. The proof is complete. ∎

Remark

If S is finite and p is irreducible, then there is a unique stationary distribution. To see this, notice that S is closed and by proposition 4.27, there is a recurrent state x. We now show that x is positive recurrent. By the assumption, there is $n \in \mathbb{N}$ and $\epsilon > 0$ such that $p^n(y,x) > \epsilon$ for all y. Thus

$$\mathrm{E}_x[\tau_x] = \sum_{k=1}^{\infty} \mathrm{P}_x(\tau_x \geq k) \leq n\,\mathrm{P}_x(\tau_x \geq 1) + n\,\mathrm{P}_x(\tau_x \geq n+1) + \cdots \leq n(1 + (1-\epsilon) + \cdots) = \frac{n}{\epsilon} < \infty.$$

Hence x is positive recurrent. It follows that every state is positive recurrent and there is a unique stationary distribution.

Example

Consider random walk on $\mathbb{Z}$ with $S_n = \sum_{i \leq n} \xi_i$ and $\mathrm{P}(\xi_i = 1) = p$ and $\mathrm{P}(\xi_i = -1) = 1 - p$. If $p \neq 1/2$, then there are multiple stationary measures that are not linearly dependent. $\mu(x) = c$ for $c > 0$ is one of the class of the staionary measure. Also, $\mu(x) = \left(\frac{p}{1-p}\right)^x$ is another class of stationary measures. To see this, obserbe that the aboves are solutions to the following equation.

$$\mu(x) = p\mu(x-1) + (1-p)\mu(x+1).$$

If $p = 1/2$, then every state is null recurrent. To see this, notice that $\mu(x) = (\mu(x-1) + \mu(x))/2$ has unique solution $\mu(x) = c$. There is no stationary distribution and hence no state is positive recurrent.

Theorem 4.38

Suppose that y is recurrent. Let $N_n(y)$ be the number of visits to y before n. Then

$$\frac{1}{n} N_n(y) \to \frac{\mathbf{1}\{\tau_y < \infty\}}{\mathrm{E}_y[\tau_y]}.$$

almost surely.

Proof. Notice that $N_n(y) = \sum_{i \leq n} \mathbf{1}\{X_i = y\}$. Suppose that we start at y. Let $\tau_y^{(k)}$ be the k-th return time of y with $\tau_y^{(0)} = 0$. Now $\tau_y^{(k)} - \tau_y^{(k-1)}$ are independent and identically distributed. The strong law of large number gives

$$\frac{1}{k}\tau_y^{(k)} \to \mathrm{E}_y\left[\tau_y^{(1)}\right] = \mathrm{E}_y[\tau_y]$$

almost surely. Observe that

$$\tau_y^{N_n(y)} \le n < \tau_y^{N_n(y)+1}.$$

Then

$$\frac{\tau_y^{N_n(y)}}{N_n(y)} \le \frac{n}{N_n(y)} < \frac{\tau_y^{N_n(y)+1}}{N_n(y)+1}\frac{N_n(y)+1}{N_n(y)}.$$

Since y is recurrent, $N_n(y) \to \infty$ almost surely when $n \to \infty$. Then

$$\frac{\tau_y^{N_n(y)}}{N_n(y)} \to \mathrm{E}_y\left[\tau_y\right], \quad \frac{\tau_y^{N_n(y)+1}}{N_n(y)+1}\frac{N_n(y)+1}{N_n(y)} \to \mathrm{E}_y\left[\tau_y\right]$$

almost surely. This implies that

$$\frac{n}{N_n(y)} \to \mathrm{E}_y\left[\tau_y\right]$$

almost surely.

Now for general case where we start at $x \neq y$, on $\{\tau_y < \infty\}$, the strong Markov property applies and $\tau_y^{(k)} - \tau_y^{(k-1)}$ are independent and identically distributed. The strong law of large number again gives

$$\frac{1}{k}\tau_y^{(k)} = \frac{\tau_y^{(1)}}{k} + \frac{\sum_{k=2}^{\infty} \tau_y^{(k)} - \tau_y^{(k-1)}}{k} \to \mathrm{E}_y\left[\tau_y\right]$$

almost surely. Similarly, we see that $\frac{n}{N_n(y)} \to \mathrm{E}_y\left[\tau_y\right]$ on $\{\tau_y < \infty\}$. Notice that on $\{\tau_y = \infty\}$, $N_n(y) = 0$. Combining two cases finishes the proof. ∎

Remark

Since $N_n(y)/n \le 1$, LDCT implies that

$$\frac{1}{n}\mathrm{E}_x\left[N_n(y)\right] \to \frac{\mathrm{P}_x(\tau_y < \infty)}{\mathrm{E}_y\left[\tau_y\right]}.$$

Then if the Markov chain is irreducible with stationary measure π,

$$\frac{1}{n}\mathrm{E}_x\left[N_n(y)\right] = \frac{1}{n}\sum_{k=1}^{n} P^k(x,y) \to \pi(y)$$

as $n \to \infty$. That is, the sequence $P^k(x,y)$ always converges in Cesaro's sense to the stationary distribution. This also holds to transient y, since $\sum_{k=1}^{\infty} P^k(x,y) = \mathrm{E}_x\left[N(y)\right] < \infty$ in that case and $\pi(y) = 0$.

Definition 4.39

*Let x be recurrent. Let $I_x = \{n \ge 1 \mid P^n(x,x) > 0\}$. The **period** is defined by $d_x = \gcd(I_x)$.*

Remark

For $a, b \in I_x$, $a + b \in I_x$. To see this, note that if $P^a(x,x), P^b(x,x) > 0$, then $P^{a+b}(x,x) \ge P^a(x,x)P^b(x,x) > 0$.

Lemma 4.40
If $x \leftrightsquigarrow y$ and x, y are recurrent, then $d_x = d_y$.

Proof. By assumption, there are $m, n \in \mathbb{N}$ such that $P^m(x,y), P^n(y,x) > 0$. Thus $P^{m+n}(x,x) \geq P^m(x,y)P^n(y,x) > 0$. Then $m+n \in I_x$ and $d_x | m+n$. If $k \in I_y$, then $P^k(y,y) > 0$ and $P^{m+k+n}(x,x) \geq P^m(x,y)P^k(y,y)P^n(y,x) > 0$. Hence $d_x | m+k+n$. It follows that for all $k \in I_y$, $d_x | k$. Then $d_x | d_y$. By a symmetric argument, $d_y | d_x$ and thus $d_x = d_y$. ∎

Definition 4.41
*Let P be irreducible and recurrent. If $d_x = 1$, we say that P is **aperiodic**.*

Remark
If $a, a+1 \in I_x$, then P is aperiodic. If $P(x,x) > 0$ for some x, then P is also aperiodic.

Proposition 4.42
If P is irreducible and recurrent with period $d \geq 2$, then there is a partition of space C_i, $i = 1, \ldots, d$ such that $P(x,y) > 0$ if and only if $x \in C_i$ and $y \in C_{i+1}$ with $C_{d+1} = C_1$.

Proof. Note that we can start by x and collect all y such that $x \rightsquigarrow y$ into C_2. Set $C_3 = \{z \mid y \rightsquigarrow z, y \in C_2\}$. Repeat this process until $C_{s+1} = C_1$ contains x. Since the period is s, we know that $s = d$. It then follows by the construction that $P(x,y) > 0$ if and only if $x \in C_i$ and $y \in C_{i+1}$ with $C_{d+1} = C_1$. ∎

Lemma 4.43
Let P be irreducible, recurrent and aperiodic. Then there exists n_0 such that $P^n(x,x) > 0$ for all $n \geq n_0$.

Proof. Observe that if $k, k+1 \in I_x$, then $2k, 2k+1, 2k+2 \in I_x$, $3k, 3k+1, 3k+2, 3k+3, 3k+4 \in I_x$. So on and so forth, at least we would find that for all $n \geq k^2$, $n \in I_x$. The statement follows if we can find two consecutive integers in I_x. By assumption, $\gcd(I_x) = 1$ and we can find $a_1, \ldots, a_m \in I_x$ such that $t_1 a_1 + \cdots + t_m a_m = 1$ for some $t_i \in \mathbb{Z}$ for $i \in 1, \ldots m$. If $t_i \geq 0$ for all i, we can replace t_1 and t_2 with $t_1' = t_1 + s a_2$ and $t_2' = t_2 - s a_1$. For sufficiently large s, t_2' would be negative. Hence we may suppose that there are positive and negative integers in t_i. Then $t_1^+ a_1 + \cdots + t_m^+ a_m = t_1^- a_1 + \cdots + t_m^- a_m + 1$. $t_1^+ a_1 + \cdots + t_m^+ a_m$ and $t_1^- a_1 + \cdots + t_m^- a_m$ are consecutive integers in I_x. The proof is complete. ∎

Definition 4.44
*Let $(S, \mathcal{S})$ be a measurable space. The space $\mathcal{M}(S)$ collecting all locally finite measure on S equipped with the **total variation norm** $\|\mu\|_{TV} = \sup_{A \in \mathcal{S}} |\mu(A)|$ is a Banach space. The space $\mathcal{P}(S)$ collects the probability measures among $\mathcal{M}(S)$.*

Theorem 4.45
Let S be a countable space. If P is irreducible and aperiodic with stationary distribution π, then for all $\mu \in \mathcal{P}(S)$,

$$\|\mu P^n - \pi\|_{TV} \to 0$$

as $n \to \infty$.

Proof. We rely on the coupling technique to proved the theorem. Let X_n and Y_n be the Markov chains with transition operator P, starting from μ and π, respectively. Consider $Z_n = (X_n, Y_n)$ with $Z_0 \sim \mu \otimes \pi$. Z_n evolutes according to the transition operator

$$Q((x,y),(x',y')) = \begin{cases} P(x,x')P(y,y') & x \neq y \\ P(x,x') & x = y, \quad x' = y' \\ 0 & \text{otherwise} \end{cases}.$$

Observe that Z_n has marginals same as X_n and Y_n. Let $\tau = \inf\{n \mid X_n = Y_n\}$ be the first time X_n and Y_n agrees.

$$\begin{aligned} \|\mu P^n - \pi\|_{TV} &= \sum_{y \in S} \left| P_\mu(X_n = y) - P_\pi(Y_n = y) \right| \leq \sum_{y \in S} \left| P_\mu(X_n = y, Y_n \neq y) - P_\pi(Y_n = y, X_n \neq y) \right| \\ &\leq 2P(X_n \neq Y_n) = 2P(\tau > n). \end{aligned}$$

Consider the transition operator on S^2, $\tilde{Q}((x,y),(x',y')) = P(x,x')P(y,y')$. Since $\tilde{Q}$ is irreducible and $\pi \otimes \pi$ is a stationary distribution. It follows from theorem 4.37 that $\tilde{Q}$ is recurrent. Hence $\tau < \infty$ almost surely. Taking $n \to \infty$ shows $\|\mu P^n - \pi\|_{TV} \to 0$. ∎

Theorem 4.46 (Markov Chain Strong Law of Large Number)
Suppose that P is irreducible with stationary distribution π and $f : S \to \mathbb{R}$ with $\pi|f| < \infty$. Then

$$\frac{1}{n} \sum_{k=1}^{n} f(X_k) \to \pi f$$

almost surely.

Proof. Without loss of generality, we may assume that $f \geq 0$. Fix $x \in S$ and let τ_m be the time of m-th visits of x. Thus

$$\frac{1}{n} \sum_{k=1}^{n} f(X_k) = \frac{1}{n} \sum_{k=1}^{\tau_1} f(X_k) + \frac{1}{n} \sum_{k=1}^{N_n - 1} \sum_{k=\tau_s+1}^{\tau_{s+1}} f(X_k) + \frac{1}{n} \sum_{k=\tau_{N_n}+1}^{n} f(X_k)$$

where N_n is the number of visits to x up to n. For the first term, since $\tau_1 < \infty$ almost surely,

$$\frac{1}{n} \sum_{k=1}^{\tau_1} f(X_k) \to 0$$

almost surely as $n \to \infty$. For the third term,

$$\frac{1}{n} \sum_{k=\tau_{N_n}+1}^{n} f(X_k) \leq \frac{1}{n} \sum_{k=\tau_{N_n}+1}^{\tau_{N_n+1}} f(X_k) \to 0$$

almost surely as $n \to \infty$ since P is recurrent and $\tau_{N_n+1} - \tau_{N_n} < \infty$ almost surely. For the

second term,

$$\frac{1}{n}\sum_{k=1}^{N_n-1}\sum_{k=\tau_s+1}^{\tau_{s+1}} f(X_k) = \frac{N_n-1}{n}\cdot\frac{1}{N_n-1}\sum_{k=1}^{N_n-1}\sum_{k=\tau_s+1}^{\tau_{s+1}} f(X_k) \to \pi(x)\,\mathrm{E}_x\left[\sum_{k=1}^{\tau_2} f(X_k)\right]$$

$$= \pi(x)\,\mathrm{E}_x\left[\sum_{y\in S} f(y)\#(\text{visits to y up to } \tau_2)\right]$$

$$= \pi(x)\sum_{y\in S}\mu_x(y)f(y) = \pi(x)\sum_{y\in S}\pi(x)\frac{\pi(y)}{\pi(x)}f(y) = \pi f.$$

The convergence is due to $(N_n-1)/n \to \mathrm{E}_x\left[\tau_x\right]^{-1} = \pi(x)$ almost surely by theorem 4.38, and $\sum_{k=\tau_s+1}^{\tau_{s+1}} f(X_k)$ are independent and identically distributed by the Markov property, which implies that

$$\frac{1}{N_n-1}\sum_{k=1}^{N_n-1}\sum_{k=\tau_s+1}^{\tau_{s+1}} f(X_k) \to \mathrm{E}_x\left[\sum_{k=1}^{\tau_2} f(X_k)\right]$$

almost surely by the strong law of large number. ∎

Lemma 4.47

Let S be a countable space and P is irreducible and admits a stationary distribution. Consider the adjoint of P, P^*, on $\mathcal{L}^2(\pi)$. Then the followings are true.

(a) $P^*(x,y) = \pi(y)P(y,x)/\pi(x)$.

(b) P^* *is an irreducible transition operator with staionary distribution* π.

Proof. By definition, $\langle Pf,\, g\rangle_\pi = \langle f,\, P^*g\rangle_\pi$ for all $f, g \in \mathcal{L}^2(\pi)$. Hence

$$\sum_{x\in S}\sum_{y\in S} P(x,y)f(y)g(x)\pi(x) = \sum_{x\in S}\sum_{y\in S} P^*(x,y)g(y)f(x)\pi(x).$$

Plugging in $P^*(x,y) = \pi(y)P(y,x)/\pi(x)$ solves the above equation. (a) follows by the uniqueness of adjoint. Now

$$\sum_{y\in S} P^*(x,y) = \sum_{y\in S}\pi(y)P(y,x)/\pi(x) = 1.$$

Since P is irreducible, P^* is also irreducible. Finally,

$$\sum_{x\in S}\pi(x)P^*(x,y) = \sum_{x\in S}\pi(x)\pi(y)P(y,x)/\pi(x) = \pi(y).$$

This shows (b). ∎

Theorem 4.48

Let S be a countable space and P is irreducible and admits a stationary distribution π. Then for all $f \in \mathcal{L}^2(\pi)$ with $\pi f = 0$, the solution to the ***Poisson equation***

$$(I-P)g = f$$

exists. Furthermore, if g_1, g_2 are both solutions, then there is some constant c such that $g_1 = g_2 + c$.

Proof. By lemma 4.47, P^* is again irreducible with stationary idstirbution π. We claim that the only solution in $\mathcal{L}^2(\pi)$ to the equation $(I - P^*)h = 0$ is constant. To see this, suppose that $h \in \mathcal{L}^2(\pi)$ is a solution. Then $P^*h = h$. By the Jensen's inequality,

$$((P^*h)(x))^2 = \left(\sum_{y\in S} P^*(x,y)h(y)\right)^2 \leq \sum_{y\in S} P^*(x,y)h^2(y) = (P(h^2))(x).$$

This implies that $(P^*h)^2 \leq P^*(h^2)$. Also, since $P^*h = h$, $(P^*h)^2 = h^2$ and hence $h^2 \leq P^*(h^2)$. But $\pi(h^2) \leq \pi P^*(h^2) = \pi(h^2)$. This implies that $P^*(h^2) = h^2 = (P^*h)^2$ π-almost surely and hence everywhere since P^* is irreducible.

Now suppose that $h(x) \neq h(y)$. Let X_n be the Markov chain governed by P^* and $M_n = h(X_n)$. If $\mathcal{F}_n$ is the filtration generated by X_n,

$$\mathrm{E}\left[M_{n+1}|\mathcal{F}_n\right] = \mathrm{E}\left[h(X_{n+1})|X_n\right] = (P^*h)(X_n) = h(X_n) = M_n.$$

M_n is a martingale. By the martingale convergence theorem in $\mathcal{L}^2$, $M_n \to M$ almost surely for some random variable M. By the assumption, P^* is irreducible and positive recurrent. This implies that X_n will visit x and y infinitely many time and M_n cannot converges to M, a contradiction.

From above, we see that $\ker(I - P^*)$ is spanned by $\mathbf{1}$. Also, since P^* is bounded, it is also closed. By the clsoed range theorem, $R(I - P) = (\ker(I - P)^*)^\perp = \ker(I - P^*)^\perp$. Since $\ker(I - P^*)$ is spanned by $\mathbf{1}$, $R(I - P)$ consists of all $f \in \mathcal{L}^2(\pi)$ such that $\langle f, \mathbf{1}\rangle_\pi = \pi f = 0$. This gives the existence of the solution to the Poisson equation.

Finally, suppose that g_1, g_2 are two solutions to the equation. Using the same argument as above, we see that $\ker(I - P)$ is spanned by 1 as well. Since $(I - P)(g_1 - g_2) = f - f = 0$, $g_1 - g_2 \in \ker(I - P)$ and $g_1 = g_2 + c$ for some constant c. ∎

Theorem 4.49 (Markov Chain Central Limit Theorem)
Let X_n be an irreducible Markov chain governed by P with stationary distribution π and $f \in \mathcal{L}^2(\pi)$ such that $\pi f = 0$. Then

$$\frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} f(X_k) \xrightarrow{d} N(0, \sigma_f^2)$$

with $\sigma_f^2 = \mathrm{Var}_\pi(f(X_0)) + 2\sum_{k=1}^{\infty} \mathrm{Cov}_\pi(f(X_0), f(X_k))$.

Proof. By assumption, there is $g \in \mathcal{L}^2(\pi)$ such that $(I-P)g = f$. Replacing f by g,

$$\begin{aligned}\frac{1}{\sqrt{n}}\sum_{k=0}^{n-1} f(X_k) &= \frac{1}{\sqrt{n}}\sum_{k=0}^{n-1}((I-P)g)(X_k)\\ &= \frac{1}{\sqrt{n}}\sum_{k=0}^{n-1}(g(X_k)-g(X_{k+1})) - \frac{1}{\sqrt{n}}\sum_{k=0}^{n-1}(Pg)(X_k)-g(X_{k+1})\\ &= \frac{1}{\sqrt{n}}(g(X_0)-g(X_n)) + \sum_{k=0}^{n-1}\frac{g(X_{k+1})-(Pg)(X_k)}{\sqrt{n}}.\end{aligned}$$

Let

$$Z_{n,k} = \frac{g(X_k)-(Pg)(X_{k-1})}{\sqrt{n}}.$$

Then if $\mathcal{F}_n$ is the filtration generated by X_n,

$$\mathrm{E}\left[Z_{n,k+1}|\mathcal{F}_k\right] = \frac{1}{\sqrt{n}}\mathrm{E}\left[g(X_{k+1})-(Pg)(X_k)|X_k\right] = \frac{1}{\sqrt{n}}((Pg)(X_k)-(Pg)(X_k)) = 0.$$

We see that $Z_{n,k}$ are martingale differences. Now

$$\mathrm{E}\left[\max_{1\le k\le n}\left|Z_{n,k}\right|\right] = \frac{1}{\sqrt{n}}\mathrm{E}\left[\max_{1\le k\le n}|g(X_k)-(Pg)(X_{k-1})|\right] \to 0$$

since all states are positive recurrent and $g \in \mathcal{L}^2(\pi)$. Also,

$$\sum_{k=1}^{n} Z_{n,k}^2 = \frac{1}{n}\sum_{k=1}^{n}(g(X_k)-(Pg)(X_{k-1}))^2 \to \sigma^2 \coloneqq \mathrm{E}_\pi\left[(g(X_1)-Pg(X_0))^2\right]$$

almost surely by the Markov chain strong law of large number. Hence by the martingale central limit theorem and that $\frac{1}{\sqrt{n}}(g(X_0)-g(X_n)) \xrightarrow{p} 0$,

$$\frac{1}{\sqrt{n}}\sum_{k=0}^{n-1} f(X_k) \xrightarrow{d} N(0,\sigma^2)$$

Note that $g = \sum_{k=0}^{\infty} P^k f$. The variance is, by the tower property,

$$\begin{aligned}\sigma^2 &= \mathrm{E}_\pi\left[g(X_1)^2 + (Pg(X_0))^2 - 2g(X_1)Pg(X_0)\right]\\ &= \mathrm{E}_\pi\left[g(X_0)^2\right] - \mathrm{E}_\pi\left[(Pg(X_0))^2\right]\\ &= \mathrm{E}_\pi\left[f(X_0)(g(X_0)+Pg(X_0))\right] = \mathrm{E}_\pi\left[f(X_0)(2g(X_0)-f(X_0))\right]\\ &= 2\sum_{k=0}^{\infty}\mathrm{E}_\pi\left[(P^k f(X_0))f(X_0)\right] - \mathrm{Var}_\pi\left[f(X_0)\right] = 2\sum_{k=1}^{\infty}\mathrm{Cov}\left[f(X_0),f(X_k)\right] + \mathrm{Var}\left[f(X_0)\right].\end{aligned}$$

■

4.4. Ergodicity

Definition 4.50
*A sequence of random varaibles X_n is **stationary** if*

$$(X_0, \ldots) \overset{d}{=} (X_k, \ldots)$$

for all $k \geq 0$.

Remark
Equivalently, it means that on the product space, $X = (X_0, \ldots)$ is invariant under shifts, i.e. $\theta X \overset{d}{=} X$.

Proposition 4.51
If X_n is stationary and $f : \mathbb{R}^{\mathbb{Z}_+} \to \mathbb{R}$ is measurable, then $Y_n = f(X_n, X_{n+1}, \ldots)$ is stationary.

Proof. It suffices to show that for all $n \geq 0$, $(Y_0, \ldots, Y_n) \overset{d}{=} (Y_k, \ldots, Y_{k+n})$. Indeed, for all measurable $B \subset \mathbb{R}^{n+1}$,

$$\mathrm{P}((Y_0, \ldots, Y_n) \in B) = \mathrm{P}((X_0, \ldots) \in A) = \mathrm{P}((X_k, \ldots) \in A) = \mathrm{P}((Y_k, \ldots, Y_{k+n}) \in B)$$

for some measurable $A \subset \mathbb{R}^{\mathbb{Z}_+}$. ∎

Definition 4.52
*Let $(\Omega, \mathcal{F}, \mathrm{P})$ be a probability space. A measurable mapping $T : \Omega \to \Omega$ is called **measure-preserving** if $\mathrm{P}(T^{-1}(A)) = \mathrm{P}(A)$ for all $A \in \mathcal{F}$.*

Remark
By the approximation argument, this also implies that $\mathrm{E}[X] = \mathrm{E}[XT]$ for all measurable $X \geq 0$ or $\mathrm{E}[|X|] < \infty$.

Proposition 4.53
Let Ω be a probability space, X be a random variable on Ω and $T : \Omega \to \Omega$ be measurable. Then if T is measure-preserving, (XT^n) is stationary. Conversely, if $\mathcal{F} = \sigma(X)$ and XT^n is stationary, then T is measure-preserving.

Proof. If T is measure-preserving, then $\mathrm{P}(XT^n \in B) = \mathrm{P}(T^n\omega \in X^{-1}(B)) = \mathrm{P}(\omega \in X^{-1}(B)) = \mathrm{P}(X \in B)$. Hence XT^n is stationary. Conversely, for every $A \in \mathcal{F}$, there is B such that $A = X^{-1}(B)$. Since (XT^n) is stationary, $\mathrm{P}(T^{-1}(A)) = \mathrm{P}(T^{-1}X^{-1}(B)) = \mathrm{P}(XT \in B) = \mathrm{P}(X \in B) = \mathrm{P}(A)$. Hence T is measure-preserving. ∎

Proposition 4.54
Let X_n be stationary in S. Consider $\Omega = S^{\mathbb{Z}_+}$. Then the shift operator on Ω is measure-preserving.

Proof. For $\omega = (\omega_0, \ldots)$, we can rewrite X_n on Ω with $X_n(\omega) = \omega_n$ and $X_n = \omega_n = X_0\theta^n$ by definition. Since X_n is stationary, θ is measure-preserving on Ω. ∎

Definition 4.55
*Suppose that $T : \Omega \to \Omega$ is measure-preserving. A set $A \in \mathcal{F}$ is **invariant** if $T^{-1}A = A$ almost surely, i.e., $\mathrm{P}((T^{-1}A)\triangle A) = 0$. The collection of invariant sets $\mathcal{I}$ is called the **invariant σ-algebra**.*

Remark
$\mathcal{I}$ is indeed a σ-algebra. First it is clear that $\varnothing \in \mathcal{F}$. If $A \in \mathcal{I}$, then

$$\begin{aligned}\mathrm{P}((T^{-1}A^c)\triangle A^c) &= \mathrm{P}((T^{-1}A^c - A^c) \cup (A^c - T^{-1}A^c)) \\ &= \mathrm{P}((A - T^{-1}A) \cup (T^{-1}A - A)) = \mathrm{P}((T^{-1}A)\triangle A) = 0.\end{aligned}$$

Hence $A^c \in \mathcal{I}$. Finally, if $A = \cup_n A_n$ with $A_n \in \mathcal{F}$, $T^{-1}A = \cup_n(T^{-1}A_n) = \cup_n A_n = A$ almost surely. Hence $\mathcal{I}$ is a σ-algebra.

Definition 4.56
*Let $T : \Omega \to \Omega$ be measure-preserving with invariant σ-algebra $\mathcal{I}$. T is said to be **ergodic** if $\mathcal{I}$ is trivial. A stationary sequnece X_n is **ergodic** if the associated measure-preserving mapping is ergodic.*

Proposition 4.57
Let X_n be independent and identically distributed. Then X_n is ergodic.

Proof. Notice that the invariant σ-algebra $\mathcal{I}$ generated by the shift operator satisfies that $\mathcal{I} \subset \mathcal{T}$, where $\mathcal{T}$ is the tail σ-algebra. Indeed, if $A \in \mathcal{I}$, then $(\theta^n)^{-1}A = \theta^{-1}A = A$ almost surely. This implies that $A \in \sigma(X_n, \ldots)$ for all n. Hence $A \in \mathcal{T}$ and $\mathcal{I} \subset \mathcal{T}$. Now by the Kolmogorov's zero one law, $\mathcal{T}$ is trivial and so is $\mathcal{I}$. ∎

Theorem 4.58
Let X_n be an irreducible Markov chain on a countable space starting with the stationary distribution π. Then X_n is ergodic.

Proof. Suppose that $A \in \mathcal{I}$ and $\mathcal{F}_n = \sigma(X_0, \ldots, X_n)$. Then $\mathbf{1}_A\theta = \mathbf{1}_A$. Now

$$\mathrm{E}_\pi\left[\mathbf{1}_A|\mathcal{F}_n\right] = \mathrm{E}_\pi\left[\mathbf{1}_A\theta^n|\mathcal{F}_n\right] = \mathrm{E}_{X_n}\left[\mathbf{1}_A\right].$$

Denote $h(x) = \mathrm{E}_x\left[\mathbf{1}_A\right]$. By Levy's zero one law, $\mathrm{E}_\pi\left[\mathbf{1}_A|\mathcal{F}_n\right] \to \mathbf{1}_A$ almost surely. Note that $h(X_n)$ is a bounded martingale. By the martingale convergence theorem, $h(X_n)$ converges almost surely to some random variable. If h is not a constant function, there are x, y such that $h(x) \neq h(y)$ and X_n visits x and y infinitely many times by the positive recurrence, contradicting to the fact that $h(X_n)$ converges. Thus h is constant. Hence $h = 0$ or 1. This implies that $\mathrm{P}(A|\mathcal{F}_n) = 0$ or 1 and $\mathrm{P}(A) = 0$ or 1. We conclude that $\mathcal{I}$ is trivial. ∎

Example
Fix $\theta \in [0,1]$. Consider a dynamical system starting from $\omega_0 \in [0,1]$ and $\omega_{n+1} = \omega_n + \theta \mod 1$. Suppose that the measure is the usual Lebesgue measure. Then the process is ergodic if and only if $\theta \notin \mathbb{Q}$.

First suppose that $\theta \in \mathbb{Q}$, then ω_n returns to ω_0 in finite number of time, say $\omega_n = \omega_0$. Then we may find a tiny interval I_i centered at ω_i for each $i \leq n$. The intervals can be selected such that $\mathrm{P}(\cup_{i\leq n} I_i) \in (0,1)$. But $\cup_{i\leq n} I_i$ is invariant. Hence the process is not ergodic.

Next, we show that if $\theta \notin \mathbb{Q}$, then the process is ergodic. For any $A \subset [0,1]$, if A is invariant, then $\mathbf{1}_A(\omega) = \mathbf{1}_A(\omega + \theta)$. By the Fourier expansion,

$$\sum_{n\in\mathbb{Z}} c_n e^{2\pi i\omega n} = \mathbf{1}_A(\omega) = \mathbf{1}_A(\omega+\theta) = \sum_{n\in\mathbb{Z}} c_n e^{2\pi i n(\omega+\theta)} = \sum_{n\in\mathbb{Z}} \left(c_n e^{2in\theta\pi}\right) e^{2\pi i n\omega}$$

where

$$c_n = \int_0^1 \mathbf{1}_A(\omega) e^{-2\pi i n\omega} d\omega.$$

By the uniqueness of Fourier expansion, since θ is irrational, $c_n = 0$ for all $n \neq 0$. Thus $c_0 = \mathbf{1}_A$ and $\mathrm{P}(A) = 0$ or 1.

Example
Let $\Omega = [0,1]$ and $T : \omega \mapsto \{2\omega\}$ where $\{x\}$ denote the fractional part of x. Then T is ergodic with respect to the Lebesgue measure. To see this, note that for all $\omega \in [0,1]$, $\sum_{i=1}^{\infty} 2^{-i}\omega_i$ where $\omega_i \overset{iid}{\sim} Ber(1/2)$ follows the Lebesgue measure. $T : (\omega_1, \ldots) = (\omega_2, \ldots)$ is a shift of independent and identically distributed random variables and hence is ergodic.

Lemma 4.59 (Hopf's Maximal Ergodic Lemma)
Let T be a measure-preserving map and $X \in \mathcal{L}^1$. Suppose $X_k(\omega) = X(T^k\omega)$. Let $S_k = \sum_{j=0}^{k-1} X_j$ and $M_n = \max\{0, S_1 \ldots, S_n\}$. Then

$$\mathrm{E}\left[X\mathbf{1}\left\{M_n > 0\right\}\right] \geq 0.$$

Proof. Write $M_k(T\omega) = \max\{0, X_1, \ldots, X_1 + \cdots + X_k\}$. Then

$$X_0 + M_k(T\omega) = \max\{X_0, \ldots, X_0 + X_1 + \cdots + X_k\} \geq S_j$$

for all $j = 1, \ldots k+1$. This implies that $[X_0 + M_k(T\omega)]\,\mathbf{1}\{M_k > 0\} \geq M_k\mathbf{1}\{M_k > 0\}$ and $X_0\mathbf{1}\{M_k > 0\} \geq (M_k - M_kT)\mathbf{1}\{M_k > 0\}$. Notice that by the stationarity,

$$0 = \mathrm{E}[M_k - M_kT] = \mathrm{E}\left[(M_k - M_kT)\mathbf{1}\{M_k = 0\}\right] + \mathrm{E}\left[(M_k - M_kT)\mathbf{1}\{M_k > 0\}\right].$$

We know that $\mathrm{E}\left[(M_k - M_kT)\mathbf{1}\{M_k = 0\}\right] \leq 0$ and hence $\mathrm{E}\left[(M_k - M_kT)\mathbf{1}\{M_k > 0\}\right] \geq 0$. Thus

$$\mathrm{E}\left[X\mathbf{1}\{M_n > 0\}\right] \geq \mathrm{E}\left[(M_k - M_kT)\mathbf{1}\{M_k > 0\}\right] \geq 0.$$

The proof is complete. ∎

Theorem 4.60 (Birkhoff's Pointwise Ergodic Theorem)
Let $X \in \mathcal{L}^1$ and T be a measure-preserving mapping with the invariant σ-algebra $\mathcal{I}$. Then

$$\frac{1}{n}\sum_{k=0}^{n-1} X(T^k\omega) \to \mathrm{E}\left[X|\mathcal{I}\right]$$

almost surely and in $\mathcal{L}^1$. If T is ergodic, then $\mathrm{E}\left[X|\mathcal{I}\right] = \mathrm{E}\left[X\right]$.

Proof. Since $\mathrm{E}\left[X|\mathcal{I}\right]$ is $\mathcal{I}$-measurable, $X(T^k\omega) - \mathrm{E}\left[X|\mathcal{I}\right]$ is again stationary and hence we may without loss of generality assume that $\mathrm{E}\left[X|\mathcal{I}\right] = 0$. Put

$$\bar{X} = \limsup_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1} X_k$$

where $X_k = XT^k$. Observe that

$$\bar{X}(T\omega) = \limsup_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} X_k = \limsup_{n\to\infty} \frac{n+1}{n}\cdot\frac{1}{n+1}\sum_{k=0}^{n} X_k - \frac{1}{n}X_0 = \bar{X}.$$

Thus $\bar{X}$ is $\mathcal{I}$-measurable. We claim that $\mathrm{P}(\bar{X} > \epsilon) = 0$ for all $\epsilon > 0$. Let $D = \left\{\bar{X} > \epsilon\right\} \in \mathcal{I}$ and $X^* = (X-\epsilon)\mathbf{1}_D$. Set $S_n = X^* + X^*T + \cdots + X^*T^{n-1}$ and $M_n = \max\{0, S_1, \ldots, S_n\}$. Write $F_n = \{M_n > 0\}$ and $F_n \nearrow F \coloneqq \left\{\sup_k k^{-1}S_k > 0\right\}$. Since D is invariant, $X^*T^n = (X_n - \epsilon)\mathbf{1}_D$. By the definition of D, $F = D \cap \left\{\sup_k k^{-1}S_k > 0\right\} = D$. By Hopf's maximal ergodic lemma, $\mathrm{E}\left[X^*\mathbf{1}_{F_n}\right] \geq 0$. Since $|X^*| \leq |X| + \epsilon \in \mathcal{L}^1$, $\mathrm{E}\left[X^*\mathbf{1}_{F_n}\right] \to \mathrm{E}\left[X^*\mathbf{1}_F\right]$ by LDCT. Thus $\mathrm{E}\left[X^*\mathbf{1}_F\right] \geq 0$. Now

$$\mathrm{E}\left[X\mathbf{1}_D\right] - \epsilon\,\mathrm{P}(D) = \mathrm{E}\left[(X-\epsilon)\mathbf{1}_D\right] = \mathrm{E}\left[X^*\mathbf{1}_D\right] \geq 0.$$

But $\mathrm{E}\left[X\mathbf{1}_D\right] = \mathrm{E}\left[\mathrm{E}\left[X\mathbf{1}_D|\mathcal{I}\right]\right] = \mathrm{E}\left[\mathbf{1}_D\,\mathrm{E}\left[X|\mathcal{I}\right]\right] = 0$. Hence $\mathrm{P}(D) = 0$. The claim follows and thus $\limsup_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1} X_k \leq 0$ almost surely.

Similarly, one can show that $\liminf_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1} X_k \geq 0$ almost surely. The almost sure convergence follows. To prove the $\mathcal{L}^1$ convergence, write $X = Y + Z$ where $Y = X\mathbf{1}\{|X| \leq M\}$ and $Z = X\mathbf{1}\{|X| > M\}$. Taking $n \to \infty$ and then $M \to \infty$,

$$\frac{1}{n}\sum_{k=0}^{n-1} YT^k \to \mathrm{E}\left[X\mathbf{1}\{|X| \leq M\}\,|\mathcal{I}\right] \to \mathrm{E}\left[X|\mathcal{I}\right]$$

in $\mathcal{L}^1$ by the bounded convergence theorem and LMCT. Also,

$$\mathrm{E}\left[\frac{1}{n}\sum_{k=0}^{n-1}\left|ZT^k\right|\right] = \mathrm{E}\left[|Z|\right] = \mathrm{E}\left[X\mathbf{1}\{|X| > M\}\right] \to 0$$

as $M \to \infty$ by LDCT. This completes the proof. ∎

Definition 4.61
Let $\mathcal{H}$ be a Hilbert space. A linear operator S on $\mathcal{H}$ is ***isometric*** *if $\langle Sx,\, Sy\rangle = \langle x,\, y\rangle$ for all $x, y \in \mathcal{H}$.*

Theorem 4.62 (Von Neumann's Mean Ergodic Theorem)
Let $\mathcal{H}$ be a Hilbert space, $S : \mathcal{H} \to \mathcal{H}$ be an isometric linear operator and P be the orthogonal projection onto $\ker(S - I)$. *Then*

$$\frac{1}{n}\sum_{k=0}^{n-1} S^k x \to Px$$

in the norm on $\mathcal{H}$ for all $x \in \mathcal{H}$.

Proof. We first claim that $\mathcal{H} = \ker(S-I) \oplus \overline{R(S-I)}$. To prove the claim, it suffices to show that $R(S-I)^\perp = \overline{R(S-I)}^\perp = \ker(S-I)$. Indeed, if $x \in \ker(S-I)$, then $Sx = x$ and for all $y \in R(S-I)$, $y = Sz - z$,

$$\langle x,\, y\rangle = \langle x,\, Sz - z\rangle = \langle x,\, Sz\rangle - \langle x,\, z\rangle = \langle Sx,\, z\rangle - \langle x,\, z\rangle = 0.$$

Hence $x \in R(S-I)^\perp$. Conversely, if $x \in R(S-I)^\perp$,

$$0 = \langle x,\, Sx - x\rangle = \left\langle \frac{1}{2}(x - Sx) + \frac{1}{2}(Sx + x),\, Sx - x \right\rangle = -\frac{1}{2}\|Sx - x\|_{\mathcal{H}}^2 + \frac{1}{2}\Big(\|Sx\|_{\mathcal{H}}^2 - \|x\|_{\mathcal{H}}^2\Big).$$

Since S is isometric, $\|Sx\|_{\mathcal{H}} = \|x\|_{\mathcal{H}}$ and $Sx = x$. Thus $x \in \ker(S-I)$. We see that $R(S-I)^\perp = \ker(S-I)$.

Now let $A_n = \frac{1}{n}\sum_{k=0}^{n-1} S^k$. For all $x \in \mathcal{H}$, write $x = y + z$ where $y = Px \in \ker(S-I)$ and $z \in \overline{R(S-I)}$. $A_n y = \frac{1}{n}\sum_{k=0}^{n-1} S^k y = y$ since $Sy = y$. Also, there is v such that $z = Sv - v$.

$$A_n z = \frac{1}{n}\sum_{k=0}^{n-1} S^k(Sv - v) = \frac{1}{n}(S^n v - v) \to 0$$

since $\|S^n - I\| \le \|S^n\| + 1 = 2$. We conclude that $A_n x \to x$ as $n \to \infty$. ∎

Corollary 4.63
Let $X \in \mathcal{L}^2(\Omega, \mathrm{P})$, $T : \Omega \to \Omega$ be measure-preserving and $X_k(\omega) = X(T^k\omega)$. If $\mathcal{I}$ is the invariant σ-algebra, then

$$\frac{1}{n}\sum_{k=0}^{n-1} X_k \to \mathrm{E}\,[X|\mathcal{I}]$$

in $\mathcal{L}^2$.

Proof. Define $S : X \mapsto XT$ to be a linear operator. Note that

$$\int X(T\omega)Y(T\omega)\,d\,\mathrm{P} = \int XY\,d(\mathrm{P}\circ T^{-1}) = \int XY\,d\,\mathrm{P}\,.$$

Hence S is an isometry. Notice that $\ker(S - I) = \mathcal{L}^2(\mathcal{I})$. By the von Neumann's mean ergodic theorem, $\frac{1}{n}\sum_{k=0}^{n-1} X_k \to PX$ where P is the projection onto $\mathcal{L}^2(\mathcal{I})$. By definition, $PX = \mathrm{E}[X|\mathcal{I}]$. The corollary follows. ∎

Definition 4.64

*A measure-preserving transformation T on $(\Omega, \mathcal{F}, \mathrm{P})$ is called **mixing** if $\lim_{n\to\infty} \mathrm{P}(A \cap T^{-n}B) = \mathrm{P}(A)\,\mathrm{P}(B)$ for all $A, B \in \mathcal{F}$.*

Definition 4.65

*A measure-preserving transformation T on $(\Omega, \mathcal{F}, \mathrm{P})$ is called **average mixing** if*

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=0}^{n-1} \mathrm{P}(A \cap T^{-n}B) = \mathrm{P}(A)\,\mathrm{P}(B).$$

Remark

Clearly mixing implies average mixing. This follows from the fact that a sequence converges, then it converges in Cesaro's sense.

Proposition 4.66

Suppose that T is measure-preserving on $(\Omega, \mathcal{F}, \mathrm{P})$. Show that T is ergodic if and only if T is average mixing.

Proof. Suppose that T is ergodic. By the Birkhoff's ergodic theorem,

$$\frac{1}{n}\sum_{k=0}^{n-1} \mathbf{1}_{T^{-1}B} = \frac{1}{n}\sum_{k=0}^{n-1} \mathbf{1}_B(T\omega) \to \mathrm{E}[\mathbf{1}_B|\mathcal{I}] = \mathrm{P}(B)$$

almost surely. Hence

$$\frac{1}{n}\sum_{k=0}^{n-1} \mathbf{1}_{A\cap T^{-1}B} = \mathbf{1}_A\left[\frac{1}{n}\sum_{k=0}^{n-1} \mathbf{1}_{T^{-1}B}\right] \to \mathbf{1}_A\,\mathrm{P}(B)$$

almost surely. Clearly the left hand side is bounded, the bounded convergence theorem implies that

$$\frac{1}{n}\sum_{k=0}^{n-1} \mathrm{P}(A \cap T^{-1}B) \to \mathrm{E}[\mathbf{1}_A\,\mathrm{P}(B)] = \mathrm{P}(A)\,\mathrm{P}(B).$$

Conversely, let $A \in \mathcal{I}$.

$$\frac{1}{n}\sum_{k=0}^{n-1} \mathrm{P}(A \cap T^{-k}B) = \frac{1}{n}\sum_{k=0}^{n-1} \mathrm{P}(T^{-k}A \cap T^{-k}B) = \frac{1}{n}\sum_{k=0}^{n-1} \mathrm{P}(T^{-k}(A \cap B)) = \mathrm{P}(A \cap B).$$

On the other hand, $\frac{1}{n}\sum_{k=0}^{n-1} \mathrm{P}(A \cap T^{-k}B) \to \mathrm{P}(A)\,\mathrm{P}(B)$ almost surely. Hence $\mathrm{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$. This means that $\mathcal{I}$ is independent of $\mathcal{F}$ and in particular $\mathrm{P}(A) = \mathrm{P}(A)^2$ implies that $\mathrm{P}(A) = 0$ or 1. Thus $\mathcal{I}$ is trivial and T is ergodic. ∎

Remark
Consider $T : \omega \to \{\omega + \theta\}$ with $\theta \notin \mathbb{Q}$. $\{\cdot\}$ denote the fractional part. We have seen that T is ergodic. We claim that it is not mixing.

Consider $A = B = [0, 1/2]$. Observe that there is a subsequence n_k such that $\{n_k\theta\} \to 0$ as $k \to \infty$. To see this, fix $\epsilon > 0$. Let $N \in \mathbb{N}$ with $1/N < \epsilon$. By Dirichlet's theorem, there are integers p, q, $1 \le q \le N$ such that $|q\theta - p| < 1/N < \epsilon$. Hence $\{q\theta\}$ satisfies either $\{q\theta\} < \epsilon$ or $\{q\theta\} > 1 - \epsilon$. In the later case, $\{mq\theta\} = 1 - m(1 - \{q\theta\})$ for m such that $1 - m(1 - \{q\theta\}) > 0$ and we may find m such that $1 - m(1 - \{q\theta\}) < \epsilon$ since $1 - \{q\theta\} < \epsilon$. We conclude that we can always find $r \in \mathbb{N}$ such that $\{r\theta\} < \epsilon$. Now pick $\epsilon_k \searrow 0$, we can find associate subsequence n_k such that $\{n_k\theta\} \to 0$. For such subsequence,

$$\mathrm{P}(A \cap T^{-n_k} B) = \frac{1}{2} - \{n_k\theta\} \to \frac{1}{2} \neq \frac{1}{4} = \mathrm{P}(A)\,\mathrm{P}(B).$$

Thus T is not mixing.

Lemma 4.67 (Fekete)
Let $a_n \in \mathbb{R}$ be such that $a_k + a_l \ge a_{k+l}$. Then

$$\lim_{n\to\infty} \frac{a_n}{n} = \inf_n \frac{a_n}{n}.$$

Proof. Since $\liminf_{n\to\infty} \frac{a_n}{n} \ge \inf_n \frac{a_n}{n}$. It suffices to show that $\limsup_{n\to\infty} \frac{a_n}{n} \le \inf_n \frac{a_n}{n}$. Let $n = km + l$ with $0 \le l \le m - 1$. Then $a_{km} + a_l \ge a_n$ and $ka_m \ge a_{km}$. Thus $ka_m + a_l \ge a_n$. Dividing by n,

$$\frac{km}{n}\frac{a_m}{m} + \frac{a_l}{n} \ge \frac{a_n}{n}.$$

Fix m, l and let $n, k \to \infty$. Then

$$\frac{a_m}{m} \ge \limsup_{n\to\infty} \frac{a_n}{n}$$

for all m. Thus $\inf_m \frac{a_m}{m} \ge \limsup_{n\to\infty} \frac{a_n}{n}$. The lemma follows. ∎

Theorem 4.68 (Subadditive Ergodic Theorem, Liggett)
Suppose that $X_{m,n}$ with $0 \le m < n$ satisfies

(a) $X_{0,m} + X_{m,n} \ge X_{0,n}$ *for* $0 \le m < n$.

(b) Given $k \ge 1$, $\left\{X_{nk,(n+1)k}\right\}$ *is stationary.*

(c) The distribution of $X_{m,m+k}$ *does not depend on* m.

(d) $\mathrm{E}\left[X_{0,1}^+\right] < \infty$ *and* $\mathrm{E}\left[X_{0,n}\right] \ge -cn$ *for some constant* c.

Then

(a) $\gamma := \lim_{n\to\infty} \frac{\mathrm{E}[X_{0,n}]}{n} = \inf_n \frac{\mathrm{E}[X_{0,n}]}{n} > -\infty$.

(b) $\frac{X_{0,n}}{n} \to X$ *almost surely and in* $\mathcal{L}^1$ *for some* $X \in \mathcal{L}^1$ *with* $\mathrm{E}[X] = \gamma$.

(c) If the sequence in condition (b) is ergodic, then $X = \gamma$.

Proof. For (a), notice that by conditions (a), (c), $\mathrm{E}\left[X_{0,m}\right] + \mathrm{E}\left[X_{0,n-m}\right] \geq \mathrm{E}\left[X_{0,n}\right]$. Applying Fekete's lemma shows that

$$\gamma = \lim_{n\to\infty} \frac{\mathrm{E}\left[X_{0,n}\right]}{n} = \inf_n \frac{\mathrm{E}\left[X_{0,n}\right]}{n}$$

Furthermore, by condition (d), $\gamma \geq -c > -\infty$ for some constant c.

Next, define

$$\bar{X} = \limsup_{n\to\infty} \frac{X_{0,n}}{n}, \quad \underline{X} = \liminf_{n\to\infty} \frac{X_{0,n}}{n}.$$

We claim that $\mathrm{E}\left[\bar{X}\right] \leq \gamma$ and if the stationary process in condition (b) is ergodic, $\bar{X} \leq \gamma$ almost surely. To prove this, fix $k \geq 1$. By the condition (a),

$$X_{0,kn} \leq \sum_{j=1}^{n} X_{k(j-1),kj}.$$

Using the Birkhoff's ergodic theorem,

$$\frac{1}{n}\sum_{j=1}^{n} X_{k(j-1),kj} \to X_k$$

almost surely and in $\mathcal{L}^1$ for some random variable X_k. Hence

$$\mathrm{E}\left[\limsup_{n\to\infty} \frac{X_{k,n}}{kn}\right] \leq \frac{\mathrm{E}\left[X_k\right]}{k}.$$

Notice also that $X_{kn+j} \leq X_{kn} + X_{kn,kn+j}$. By condition (c), the distribution of $X_{kn,kn+j}$ only depends on j. By condition (d), $X_{kn,kn+j}$ has finite first moment. By the Borel-Cantelli lemma, one can show that

$$\lim_{n\to\infty} \frac{X_{kn,kn+j}}{n} = 0$$

almost surely for all j. Thus $\mathrm{E}\left[\bar{X}\right] \leq \mathrm{E}\left[X_k\right]/k \to \gamma$. If the process is ergodic, then X_k is deterministic and $\bar{X} \leq \gamma$ almost surely.

Now we shall claim that $\mathrm{E}\left[\underline{X}\right] \geq \gamma$. Let U_n be independent of $X_{k,n}$ and distributed uniformly on $\{1,\ldots,n\}$. Set $Y_k^n = X_{0,k+U_n} - X_{0,k+U_n-1}$. Notice that

$$\mathrm{E}\left[Y_k^n\right] = \frac{1}{n}\sum_{j=1}^{n} \mathrm{E}\left[X_{0,k+j} - X_{0,k+j-1}\right] = \frac{1}{n}\left(\mathrm{E}\left[X_{0,k+n}\right] - \mathrm{E}\left[X_{0,k}\right]\right)$$

and

$$\mathrm{E}\left[(Y_k^n)^+\right] = \frac{1}{n}\sum_{j=1}^{n} \mathrm{E}\left[(X_{0,k+j} - X_{0,k+j-1})^+\right] \leq \frac{1}{n}\sum_{j=1}^{n} \mathrm{E}\left[X_{k+j-1,k+j}^+\right] = \mathrm{E}\left[X_{0,1}^+\right].$$

Hence $\sup_n \mathrm{E}\left[\left|Y_k^n\right|\right] < \infty$ and $\lim_{n\to\infty} \mathrm{E}\left[Y_k^n\right] = \gamma$ for all $k \geq 1$. Now there is a subsequence, along which the joint distribution $\left\{Y_k^{n_i}\right\}$ converges to the joint distribution, say $\{Y_k\}$. For any

$f \in C_b(\mathbb{R}^{\mathbb{N}})$ depending only on finitely many coordinates, we have

$$\mathrm{E}\left[f(Y_1, \ldots)\right] = \lim_{i\to\infty} \frac{1}{n_i} \sum_{j=1}^{n_i} \mathrm{E}\left[f(X_{0,j+1} - X_{0,j}, X_{0,j+2} - X_{0,j+1}, \ldots)\right].$$

Thus $\{Y_k\}$ is a stationary sequence. Using condition (a),

$$Y_1^n = X_{0,U_n+1} - X_{0,U_n} \le X_{U_n,U_n+1} \stackrel{d}{=} X_1.$$

Hence by condition (d), $\left\{(Y_1^n)^+\right\}$ is uniformly integrable. By Fatou's lemma,

$$\mathrm{E}\left[Y_1\right] \ge \limsup_{n\to\infty} \mathrm{E}\left[Y_1^n\right] = \gamma.$$

Using Birkhoff's ergodic theorem again shows that

$$\frac{1}{n}\sum_{k=1}^{n} Y_k \to Y$$

almost surely for some Y with $\mathrm{E}\left[Y\right] = \mathrm{E}\left[Y_1\right] \ge \gamma$. To establish the claim, it remains to show that $\underline{\mathrm{X}}$ is stochastically larger than Y.[1] To see this, note that for every increasing $f \in C_b(\mathbb{R}^{\mathbb{N}})$ depending only on finitely many coordinates,

$$\begin{aligned}\mathrm{E}\left[f(Y_1, Y_1+Y_2, Y_1+Y_2+Y_3, \ldots)\right] &= \lim_{i\to\infty} \frac{1}{n_i} \sum_{j=1}^{n_i} \mathrm{E}\left[f(X_{0,j+1} - X_{0,j}, X_{0,j+2} - X_{0,j+1}, \ldots)\right] \\ &\le \lim_{i\to\infty} \frac{1}{n_i} \sum_{j=1}^{n_i} \mathrm{E}\left[f(X_{j,j+1}, X_{j+1,j+2}, \ldots)\right] = \mathrm{E}\left[f(X_{0,1}, \ldots)\right].\end{aligned}$$

Hence the sequence $X_1, \ldots$ is stochastically larger than $Y_1, Y_1+Y_2, \ldots$. It follows that $\underline{\mathrm{X}}$ is stochastically larger than Y. This proves the almost sure convergence in (b) and (c).

Finally, for the $\mathcal{L}^1$ convergence, let $X = \bar{X} = \underline{\mathrm{X}}$. $\mathrm{E}\left[X\right] = \gamma$. Since

$$X_{0,n} \le \sum_{j=1}^{n} X_{(j-1),j},$$

$\frac{1}{n}X_{0,n}^+$ is uniformly integrable. Thus $\mathrm{E}\left[(n^{-1}X_{0,n} - X)\right]^+ \to 0$. Now

$$0 = \lim_{n\to\infty} \mathrm{E}\left[\frac{1}{n}X_{0,n} - X\right] \le \lim_{n\to\infty} \mathrm{E}\left[\left(\frac{1}{n}X_{0,n} - X\right)^+\right] - \mathrm{E}\left[\left(\frac{1}{n}X_{0,n} - X\right)^-\right].$$

Thus the negative part also converges to 0. This gives the desired $\mathcal{L}^1$ convergence. ∎

[1] A real-valued random variable X is said to be stochastically larger than Y if $\mathrm{P}(X > a) \ge \mathrm{P}(Y > a)$ for all $a \in \mathbb{R}$.

5. Continuous-Time Stochastic Processes

5.1. Brownian Motion

Definition 5.1
A standard Brownian motion is a stochastic process $B_t(\omega)$ with $t \geq 0$ such that

(a) $B_0 = 0$.

(b) For $t_1 < \cdots < t_n$, $B_{t_i} - B_{t_{i-1}}$ are independent.

(c) $B_{t+s} - B_t \sim N(0, s)$ *for* $s > 0$.

(d) $t \mapsto B(t, \omega)$ *is continuous almost surely.*

Remark
The uniqueness fail when we drop the condition (d). For example, one may consider $\tilde{B}_t = B_t \mathbf{1}\{U \neq t\}$ where $U \sim Unif[0,1]$ is independent of B_t. It is clear that condition (a) to (c) are satisfied for $\tilde{B}_t$.

Theorem 5.2 (Kolmogorov's Continuity Theorem)
Let X_t be a stochastic process on T. Suppose that there are $\alpha, \beta, K > 0$ such that

$$\mathrm{E}\left[|X_t - X_s|^\beta\right] \leq K\,|t-s|^{1+\alpha}$$

for all t, s. Then there is a version of X_t such that X_t is locally γ-Hölder continuous for every $0 < \gamma < \alpha/\beta$

Proof. Without loss of generality, consider $T = [0,1]$. Consider the restriction of X_t on $D_n = \{2^{-n} i \mid 0 \leq i \leq 2^n\}$. Put

$$\epsilon_n = \max\left\{|X_t - X_s| \mid |t-s| = 2^{-n},\ t, s \in D_n\right\}.$$

Notice that there are $2^n + 1$ pairs of such s and t. Thus

$$\mathrm{E}\left[\sum_n (2^{\gamma n}\epsilon_n)^\beta\right] = \sum_n 2^{\gamma\beta n}\,\mathrm{E}\left[\epsilon_n^\beta\right] \lesssim \sum_n 2^{\gamma\beta n} \cdot 2^n 2^{-(1+\alpha)n} = \sum_n 2^{(\gamma\beta-\alpha)n} < \infty.$$

Hence $\epsilon_n \lesssim 2^{-\gamma n}$ almost surely. For any $t, s \in \cup_n D_n$ with $|t-s| \leq 2^{-m}$, if $r \in [2^{-m-1}, 2^{-m}]$,

$$\sup_{\substack{t,s \in \cup_n D_n \\ |t-s| \leq r}} |X_t - X_s| \lesssim \sum_{n \geq m} \epsilon_n \lesssim \sum_{n \geq m} 2^{-\gamma n} \lesssim 2^{-\gamma m} \lesssim r^\gamma.$$

The first inequality comes from the fact that we can always take one step in D_n for each $n \geq m$ so that t and s are linked in finite steps. This shows that X_t is γ-Hölder continuous on $\cup_n D_n$. It follows that $X|_{\cup_n D_n}$ can be extend to a γ-Hölder continuous process Y_t. It remains to

show that $Y_t = X_t$ almost surely. Indeed, for any $t \in T$, pick $t_i \in \cup_n D_n$ such that $t_i \to t$. Then $X_{t_i} = Y_{t_i}$ for all i and $Y_{t_i} \to Y_t$ almost surely by the continuity. By the Chebyshev's inequality,

$$\mathrm{P}\left(\left|X_{t_i} - X_t\right| > \epsilon\right) = \mathrm{P}\left(\left|X_{t_i} - X_t\right|^\beta > \epsilon^\beta\right) \leq \frac{1}{\epsilon^\beta}\,\mathrm{E}\left[\left|X_{t_i} - X_t\right|^\beta\right] \leq \frac{1}{\epsilon^\beta}\,|t_i - t|^\beta \to 0$$

for any $\epsilon > 0$. Hence $X_{t_i} \xrightarrow{p} X_t$. It follows that $X_t = Y_t$ almost surely. The proof is complete. ■

Corollary 5.3
B_t is well-defined and is locally γ-Hölder continuous almost surely for every $0 < \gamma < 1/2$.

Proof. It is enough to show that the Brownian motion is well-defined on $[0, 1]$. Let $D_n = \{2^{-n}i \mid 0 \leq i \leq 2^n\}$. Define B_t on D_n recursively. On D_0, $B_0 = 0$ and $B_1 \sim N(0, 1)$. Suppose that B_t has been constructed on D_n. For $d \in D_{n+1} - D_n$, define

$$B(d) = B(d - 2^{-n-1}) + 2^{-(n+1)/2}Z_d + \frac{B(d + 2^{-n-1}) - B(d - 2^{-n-1})}{2}$$

where Z_d are independent standard normal from increments $B(d + 2^{-n-1}) - B(d - 2^{-n-1})$ on D_n. Then

$$B(d + 2^{-n-1}) - B(d) = -2^{-(n+1)/2}Z_d + \frac{B(d + 2^{-n-1}) - B(d - 2^{-n-1})}{2}$$

and one can observe that $\mathrm{Cov}\left[B(d + 2^{-n-1}) - B(d), B(d) - B(d - 2^{-n-1})\right] = 0$. Hence the contructed process has independent increments on D_{n+1}. Repeat the procedure, we obtain a process defined on $\cup_n D_n$ by the Kolmogorov's extension theorem. Now observe that

$$\mathrm{E}\left[|B_t - B_s|^4\right] = 3\,|t - s|^2\,.$$

By the Kolmogorov's continuity theorem, there is a locally γ-Hölder continuous almost surely version of B_t for every $0 < \gamma < 1/2$. ■

Theorem 5.4 (Garsia-Rodemich-Rumsey Inequality)
Let be $f : [0, 1] \to \mathbb{R}$ continuous and $\psi, p : [0, \infty) \to [0, \infty)$ are continuous and strictly increasing with $p(0) = \psi(0) = 0$ and $\psi(x) \to \infty$ as $x \to \infty$. If

$$\int_{[0,1]^2} \psi\left(\frac{|f(x) - f(y)|}{p(|x - y|)}\right)dxdy \leq A < \infty,$$

then for all $x, y \in [0, 1]$,

$$|f(x) - f(y)| \leq 8\int_0^{|x-y|} \psi^{-1}\left(\frac{4A}{u^2}\right)dp(u).$$

Proof. We shall first prove the case $x = 1$ and $y = 0$. Let

$$I(x) = \int_0^1 \psi\left(\frac{|f(x) - f(y)|}{p(|x - y|)}\right) dy.$$

For $x_0 \in (0, 1)$, $I(x_0) \leq A$. Observe that we can pick $x_n \searrow 0$ and $d_n = p^{-1}(p(x_n)/2)$ such that $x_{n+1} \leq d_n$, with

$$I(x_{n+1}) \leq \frac{2A}{d_n}, \quad \text{and} \quad \psi\left(\frac{|f(x_{n+1}) - f(x_n)|}{p(|x_{n+1} - x_n|)}\right) \leq \frac{2I(x_n)}{d_n}.$$

Since $d_{n+1} \leq d_n$, $I(x_{n+1}) \leq \frac{2A}{d_{n+1}}$. Hence,

$$|f(x_{n+1}) - f(x_n)| \leq p(|x_{n+1} - x_n|)\psi^{-1}\left(\frac{2I(x_n)}{d_n}\right) \leq p(|x_{n+1} - x_n|)\psi^{-1}\left(\frac{4A}{d_n^2}\right).$$

Also,

$$p(|x_{n+1} - x_n|) \leq p(x_n) = 2p(d_n).$$

Notice that by construction, $2p(d_{n+1}) \leq p(d_n)$ and thus $2p(d_n) \leq 4(p(d_n) - p(d_{n+1}))$. This shows that

$$p(|x_{n+1} - x_n|) \leq 4(p(d_n) - p(d_{n+1})).$$

Now

$$\begin{aligned} |f(x_0) - f(0)| &\leq 4\sum_{n=0}^{\infty} [p(d_n) - p(d_{n+1})]\, \psi^{-1}\left(\frac{4A}{d_n^2}\right) \leq 4\sum_{n=0}^{\infty} \int_{d_n}^{d_{n+1}} \psi^{-1}\left(\frac{4A}{u^2}\right) dp(u) \\ &\leq 4\int_0^1 \psi^{-1}\left(\frac{4A}{u^2}\right) dp(u). \end{aligned}$$

By a similar argument applying to $f(1 - x)$,

$$|f(x_0) - f(1)| \leq 4\int_0^1 \psi^{-1}\left(\frac{4A}{u^2}\right) dp(u).$$

Hence

$$|f(1) - f(0)| \leq 8\int_0^1 \psi^{-1}\left(\frac{4A}{u^2}\right) dp(u).$$

For the general case, we can consider $\tilde{f}(t) = f(x + t(y - x))$ and $\tilde{p}(u) = p(u\,|x - y|)$. Then

$$|f(x) - f(y)| = \left|\tilde{f}(0) - \tilde{f}(1)\right| \leq 8\int_0^1 \psi^{-1}\left(\frac{4A}{u^2\,|x - y|^2}\right) dp(u\,|x - y|) = 8\int_0^1 \psi^{-1}\left(\frac{4A}{u^2}\right) dp(u).$$

The proof is complete. ∎

Proposition 5.5

Let B_t be a Brownian motion. Then

$$|B_{t+h} - B_t| \leq C(\omega)\sqrt{-h\log(h)}$$

for h small enough.

Proof. Let $\psi(x) = e^{\alpha x^2} - 1$ where $\alpha \in (0, 1/2)$. Set $p(u) = \sqrt{u}$. Then by the Fubini theorem

$$\begin{aligned}\mathrm{E}\left[\int_{[0,1]^2} \psi\left(\frac{|B_s - B_t|}{p(|s-t|)}\right) dsdt\right] &= \int_{[0,1]^2} \mathrm{E}\left[\psi\left(\frac{|B_s - B_t|}{p(|s-t|)}\right)\right] dsdt = \int_{[0,1]^2} \mathrm{E}\left[e^{\alpha Z^2} - 1\right] dsdt \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2} + \alpha x^2} dx - 1 < \infty\end{aligned}$$

where $Z = \frac{B_s - B_t}{\sqrt{|t-s|}} \sim N(0,1)$. Hence

$$\int_{[0,1]^2} \psi\left(\frac{|B_s - B_t|}{p(|s-t|)}\right) dsdt \leq A(\omega) < \infty$$

almost surely. Note that $\psi^{-1}(y) = \sqrt{\log(1+y)/\alpha}$. Apply the Garsia-Rodemich-Rumsey inequality.

$$|B_{t+h} - B_t| \leq 8\frac{1}{\sqrt{\alpha}} \int_0^h \sqrt{\log\left(1 + \frac{4A}{u^2}\right)} \frac{1}{2\sqrt{u}} du \lesssim C(\omega)\sqrt{-h\log(h)}$$

for h small. ∎

Theorem 5.6 (Lévy)
Let B_t be a standrad Brownian motion on $[0,1]$. Then

$$\limsup_{h \to 0} \sup_{0 \leq t \leq 1-h} \frac{|B_{t+h} - B_t|}{\sqrt{-2h\log(h)}} = 1$$

almost surely.

Proof. We first establish the upper bound. Consider the dyadic points of the form $t_{n,k} = k2^{-n}$ with $k = 0, \ldots, n$. By the property of Brownian motion we have

$$\mathrm{P}\left(\left|B_{t_{n,k+1}} - B_{t_{n,k}}\right| \geq x\right) \leq 2e^{-2^n \frac{x^2}{2}}$$

∎

5.2. Poisson Process

5.3. Semigroup Theory

Definition 5.7
*Let $(S, \mathcal{S})$ be a measurable space. A **probability kernel** is a function $\mu : (s, A) \mapsto [0, 1]$ such that*

(a) For every fixed $s \in S$, $A \mapsto \mu(s, A)$ is a probability measure on S.

(b) For every fixed $A \in \mathcal{S}$, $s \mapsto \mu(s, A)$ is measurable.

Definition 5.8
*Let $(S, \mathcal{S})$ be a measurable space and μ be a probability kernel on S. The **transition operator** T associated with μ is an operator acting on*

$$M = \{f : S \to \mathbb{R} \mid f \text{ is measurable and non-negative or bounded}\},$$

and defined by

$$Tf(x) = \int \mu(x, dy) f(y).$$

Remark
For every $f \in M$, Tf is again measurable. Indeed, for any characteristic function $\mathbf{1}_A$, $A \in \mathcal{S}$, $(T\mathbf{1}_A)^{-1}(B) = \{x \in S \mid \mu(x, A) = B\} \in \mathcal{S}$ for all $B \in \mathcal{S}$ by the measurability of $x \mapsto \mu(x, A)$. The result extends by linearity and approximation to all functions in M.

Definition 5.9
*Let $\{T_t\}_{t \geq 0}$ be a family of parametrized operator. T_t is called a **semigroup** if*

(a) For $s, t \geq 0$, $T_s T_t = T_{s+t}$.

(b) $T_0 = I$.

Proposition 5.10
For each $t \geq 0$, consider the probability kernel μ_t and the associated transition operator T_t. Then $T_t T_s = T_{t+s}$ if and only if $\mu_t \mu_s = \mu_{t+s}$.

Proof. Let $B \in \mathcal{S}$. Then

$$T_t T_s \mathbf{1}_B(x) = \int \mu_t(x, dy)(T_s \mathbf{1}_B)(y) = \int \mu_t(x, dy)\mu_s(y, B) = (\mu_t \mu_s)(x, B).$$

Now the result holds for every measurable characteristic functions. The result extends by the linearity and the monotone convergence. ∎

Definition 5.11
*Let S be a locally compact and separable metric space and $(C_0, \|\cdot\|_\infty)$ collect all the functions on S that are continuous and vanishes at the infinities. A semigroup on C_0 is called a **Feller semigroup** if*

(a) $T_t C_0 \subset C_0$, *and*

(b) $T_t f(x) \to f(x)$ *as* $t \to 0$ *for* $f \in C_0$ *and* $x \in S$.